SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering

Somaiya TRUST

FY B Tech (Common to All)
SVU R-2025 Version 3.0
SEM I

Teaching and Credit Scheme

Course Code		Course Category	Name of the Course	Teaching Scheme TH-PR-TUT	Total (hrs.)	Credit Scheme TH-PR-TUT	Total Credits
SVU	KJSSE						
	316U06C101	BS	Applied Mathematics – 1	3 – 0 – 1	4	3 – 0 – 1	4
	316U06C102	BS	Engineering Physics	2 – 0 – 0	2	2 – 0 – 0	2
	316U06C103	BS	Engineering Chemistry	2 – 0 – 0	2	2 – 0 – 0	2
	316U06C104	ES	Basic Electrical Engineering	2 – 0 – 0	2	2 – 0 – 0	2
	316U06C105	ES	Engineering Drawing	2 – 0 – 1	3	2 – 0 – 1	3
	316U06C106	BS	Biology for Engineers	2 – 0 – 0	2	2 – 0 – 0	2
	316U06C107	ES	Structured Programming Methodology	2 – 2 – 0	4	2 – 1 – 0	3
	316U06L101	BS	Applied Science Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
	316U06L102	ES	Basic Electrical Engineering Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
	316U06L103	ES	Engineering Drawing Laboratory	0 – 2 – 0	2	0 – 1 – 0	1
	316U06L104	ES	Maker Space Laboratory – I	0 – 2 – 0	2	0 – 1 – 0	1
			Total	15 – 10 – 2	27	15 – 5 – 2	22

Examination Scheme

Course Code		Course Category	Name of the Course	Lab/Tut CA	CA		ESE		Total
SVU	KJSSE				IA	MSE	PR/OR Exam/ OST/ Project	Theory Exam	
	316U06C101	BS	Applied Mathematics – I	25	20	30	--	50	125
	316U06C102	BS	Engineering Physics	--	20	30	--	50	100
	316U06C103	BS	Engineering Chemistry	--	20	30	--	50	100
	316U06C104	ES	Basic Electrical Engineering	--	20	30	--	50	100
	316U06C105	ES	Engineering Drawing	--	20	30	--	50	100
	316U06C106	BS	Biology for Engineers	--	50	--	--	--	50
	316U06C107	ES	Structured Programming Methodology	50	--	--	50	--	100
	316U06L101	BS	Applied Science Laboratory	50	--	--	--	--	50
	316U06L102	ES	Basic Electrical Engineering Laboratory	50	--	--	--	--	50
	316U06L103	ES	Engineering Drawing Laboratory	50	--	--	--	--	50
	316U06L104	ES	Maker Space Laboratory – I	50	--	--	--	--	50
			Total	275	150	150	50	250	875

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<tr><td>316U06C104</td><td colspan="6">Basic Electrical Engineering</td></tr>
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<tr><td colspan="2">CA</td><td rowspan="2">ESE (Theory/On-Screen)</td><td rowspan="2">PR/OR Exam</td><td rowspan="2">LAB/TUT CA</td><td rowspan="2">Total</td></tr>
<tr><td>IA</td><td>MSE</td></tr>
<tr><td>20</td><td>30</td><td>50</td><td>--</td><td>--</td><td>100</td></tr>
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Course prerequisites:

Knowledge of Basic Electrical parameters: Resistance, Inductance, Capacitance, Frequency, Voltage, Current and Power and Energy, basic laws of magnetism

Course Objectives:

It is difficult to imagine life without electricity. Electricity plays a major role in the working of all minor and major devices used in our day to day life. In this course students acquire fundamental knowledge to understand the electrical systems

Course Outcomes

At the end of the successful completion of the course, the student will be able to

CO1: Analyze resistive DC networks using various network theorems
CO2: Analyze single phase AC circuits
CO3: Analyze three phase AC circuits
CO4: Understand principles and working of Transformer
CO5: Describe principles and working of DC and AC motors

Module No.	Unit No.	Details	Hrs.	CO
1		**DC Circuits**	10	CO1
	1.1	Concept of dependent and independent sources, ideal and practical voltage and current sources, Kirchhoff's Laws, source transformation and network terminology.		
	1.2	Resistive network simplification, Series, parallel connection and Star-Delta transformations		
	1.3	Mesh and nodal analysis, concept of super mesh and super node (Analysis only with independent sources)		
	1.4	Superposition theorem, Thevenin's theorem, Norton's theorem, Maximum power transfer theorem (Analysis only with independent sources)		
2		**Single Phase AC Circuits**	10	CO2
	2.1	Generation of alternating voltage, average value, RMS value, form factor, crest factor, phasor representation in rectangular and polar form.		
	2.2	Steady state behaviour of single phase AC circuits with pure R, L, and C, concept of inductive and capacitive reactance, phasor diagram of impedance, phase relationship in voltage and current.		
	2.3	RL, RC and RLC series and parallel circuits, concept of impedance and admittance, power triangle, power factor, active, reactive and apparent power, concept of power factor improvement.		
	2.4	Series and parallel resonance, Q-factor and bandwidth		
3		**Three Phase AC circuits**	05	CO3
	3.1	Three-phase balanced circuits, voltage and current relations in star and delta connections.		
	3.2	Measurement of power in 3-phase system using two wattmeter method		

4	**Introduction to Electrical Machines**		05	CO4
	4.1	**Single phase transformer :** Construction and principle of working, emf equation of a transformer, losses in transformer, Equivalent circuit of Ideal and practical transformer, and efficiency of transformer, (No numerical expected)		
	4.2	**DC motors** : Construction and working principle of DC motors such as series, shunt and compound, torque-speed characteristics, selection criteria and applications (no derivations and numerical expected)		
	4.3	**Single phase induction motor**: Construction, working principle, double field revolving theory, applications (no derivations and numerical expected)		
	#	**Self-Learning:** BLDC Motors, PMSM and Stepper Motors, SRM Motor, Servo Motor, Switch Fuse Unit (SFU), MCB, ELCB, MCCB ,Types of Wires and Cables, Earthing, Lamps: fluorescent, CFL, LED, Electrical measuring instruments principle and applications-energy meter, megger, tong tester.		
		Total	**30**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	B. L. Thereja	Electrical Technology Vol-1 and Vol-II	S. Chand & Publications, India	25th Edition 2014
2.	Mittle and Mittle	Basic Electrical Engineering	Tata McGraw Hill, India	2nd edition (New) 2001
3.	Singh Ravish R	Basic Electrical Engineering	S. Chand	1st Edition, 2023
4.	B.R. Patil	Basic Electrical Engineering	Oxford University Press	2nd Edition, 2022

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<tr><td colspan="2">316U06L102</td><td colspan="7">Basic Electrical Engineering Laboratory</td></tr>
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Note: List of experiments/tutorials, scheme for continuous assessment and rubrics will be shared by course coordinators from time to time at the beginning of every semester

Basic Electrical Components

Resistors, Capacitors, Inductors (individual and combination boxes) – For circuit design.

Rheostat / Variable Resistor – Used in current control and load variation.

Breadboard and Wires

Measurement Instruments:

1. Digital Multimeter (DMM):
   Used for measuring voltage (AC/DC), current (AC/DC), and resistance.
2. Analog Ammeter and Voltmeter:
   For basic current and voltage measurements and to understand needle deflection.
3. Wattmeter (Dynamometer Type):
   Measures real power in AC circuits.
4. Energy Meter (Single-phase):
   Used for measuring energy consumption over time.

Basic Electrical Components

Power Supplies:

5. DC Power Supply (Variable):

Typically ranges from 0–30V and up to 5A; provides a stable DC source.

6. AC Power Supply (Variac / Auto-transformer):

Variable output up to 20V; used for AC experiments.

7. Function Generator:

Generates sinusoidal, square, and triangular waveforms.

8. Oscilloscope (DSO or CRO):

Used for viewing time-varying signals and measuring phase, amplitude, and frequency.

Machines and Loads:

9. DC Motor / Generator - Used to demonstrate electromechanical energy conversion.

10. Single-phase Transformer (e.g., 230V/12V)

For transformer characteristics and load testing.

11. Resistive Load Bank / Lamp Load

Used to study voltage drop and current flow under loading conditions.

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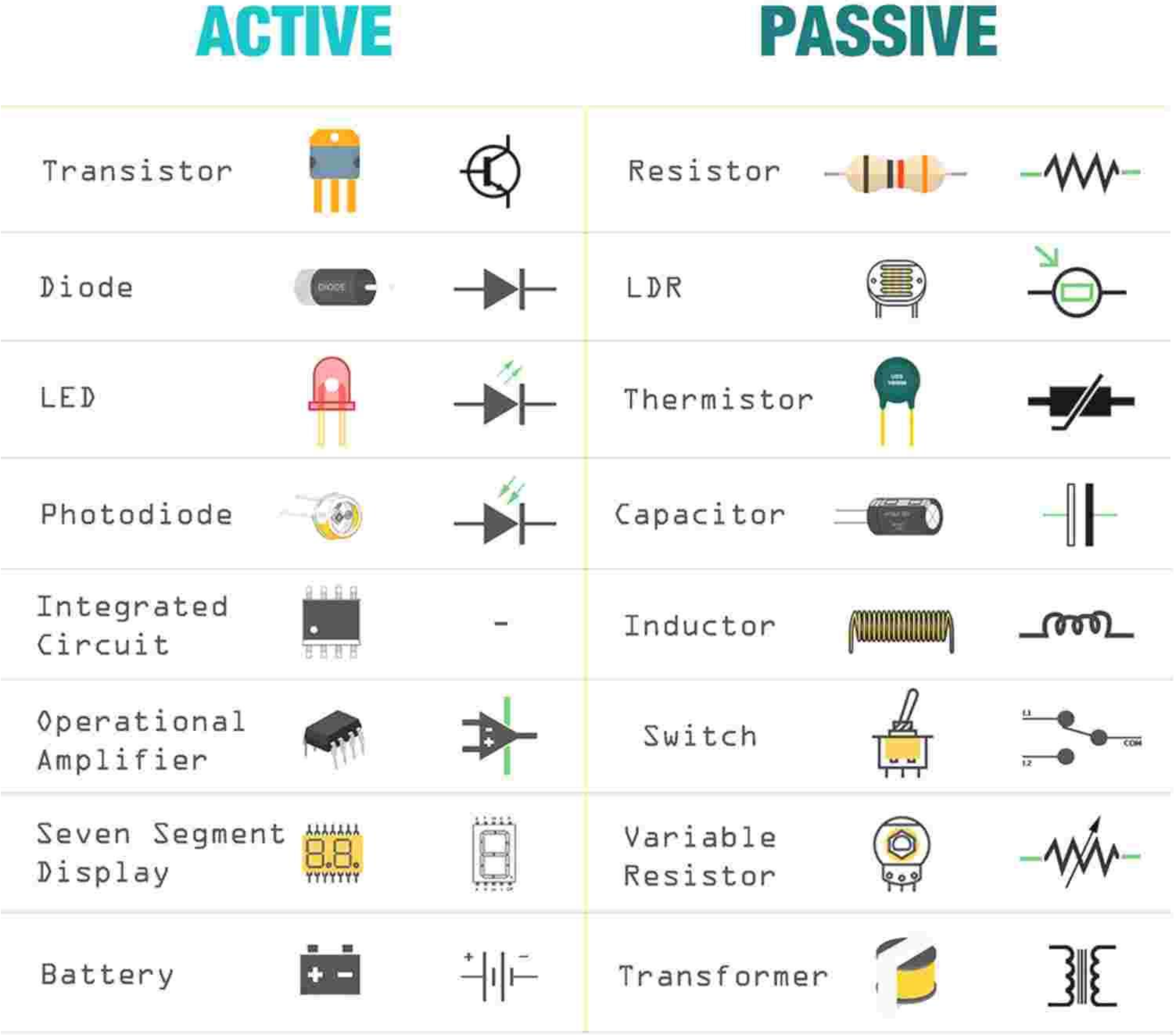
Electrical and Electronics Components



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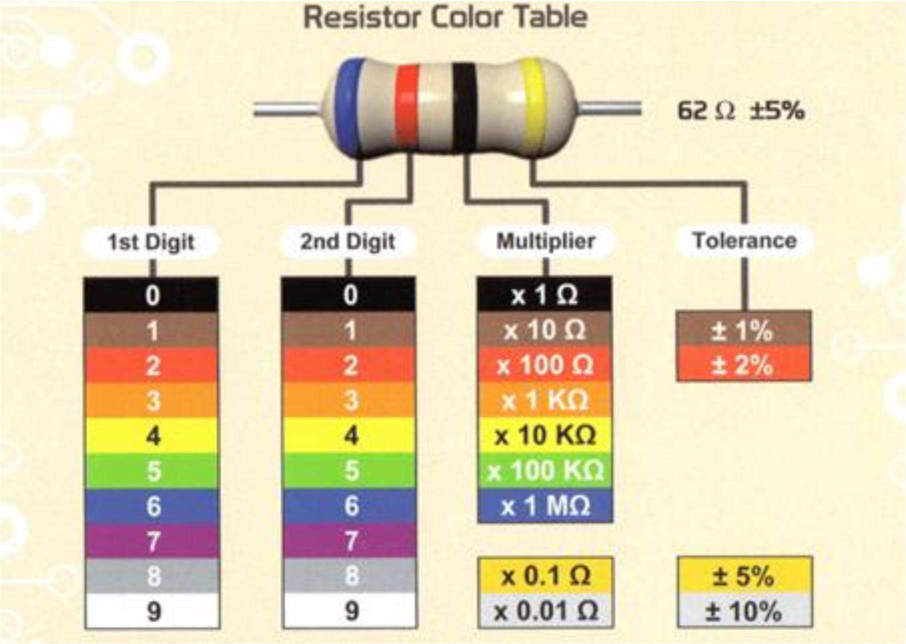
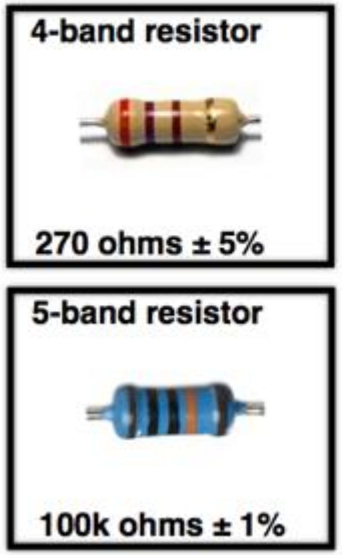
Resistor (Ohm Ω) Colour Codes

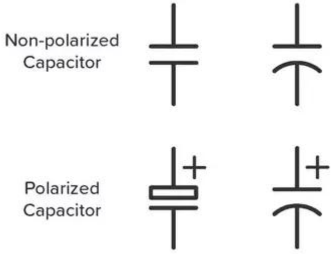
A **resistor** is a two-terminal **passive electrical component** that **opposes the flow of electric current** by converting electrical energy into heat, according to **Ohm's Law**.

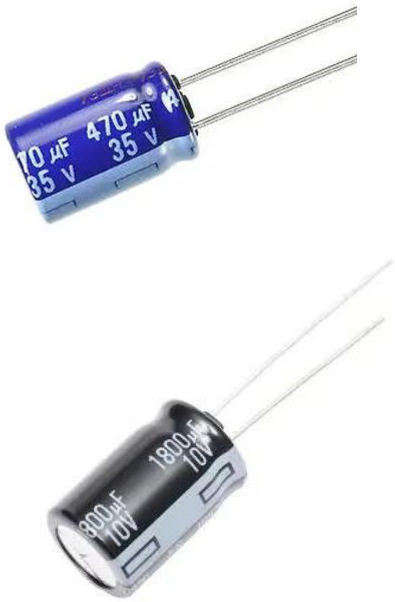


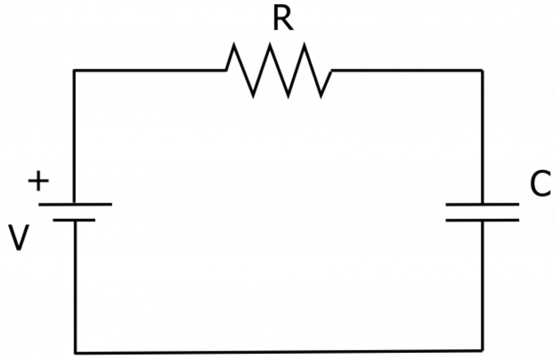


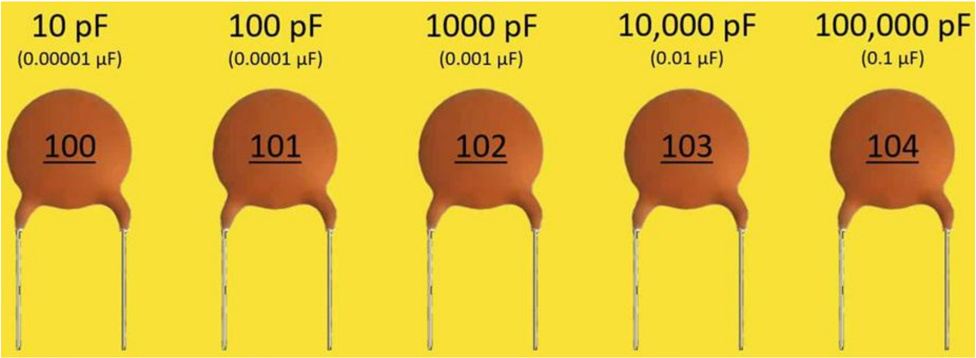
Capacitor Types C (Faraday)

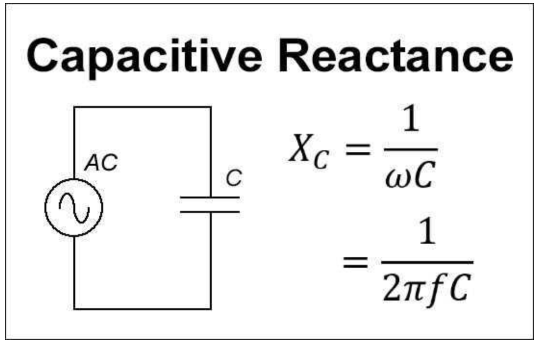
A **capacitor** is a two-terminal **passive electrical component** that **stores energy in an electric field** between its plates due to separation of charge. It **opposes changes in voltage**.

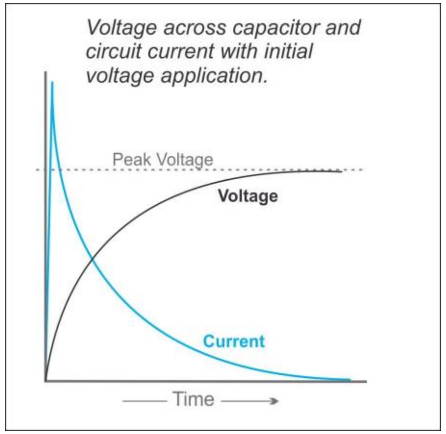






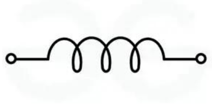






Inductors L (Hendry)

Inductor Symbol



An **inductor** is a two-terminal **passive component** that **stores energy in a magnetic field** when current flows through it. It **opposes changes in current**, not steady current.

Air-Core Inductor

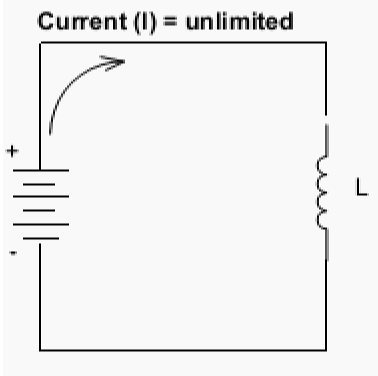
Iron-Core Inductor

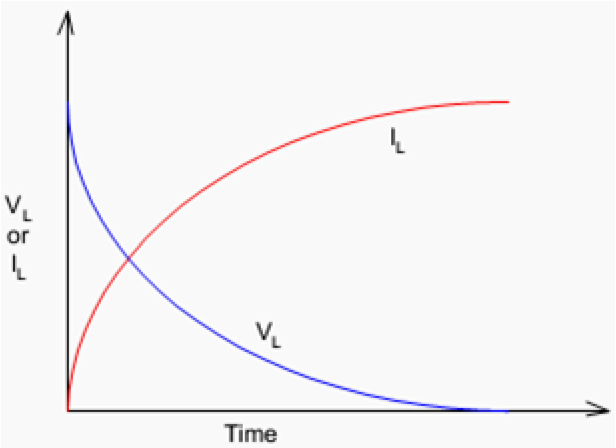
Ferrite-Core Inductor

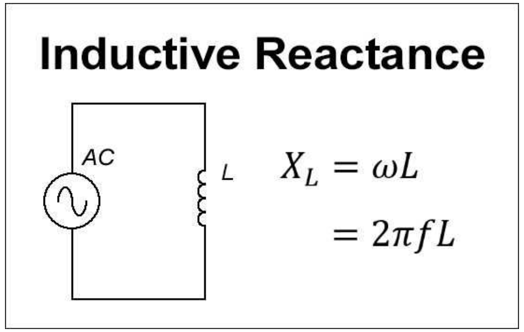
Bobbin Based Inductor

Variable Inductor

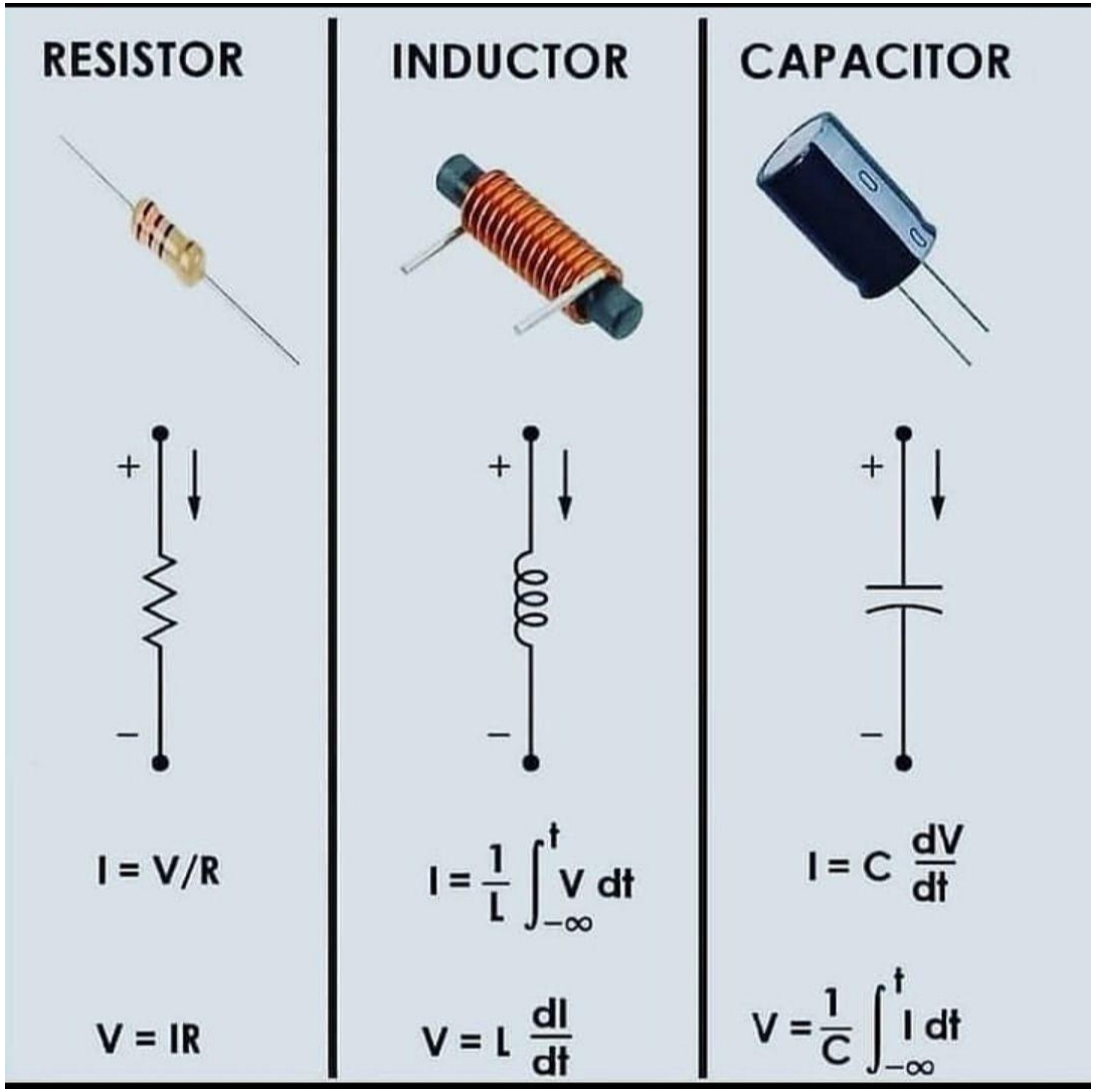
Multilayer Ceramic Inductor



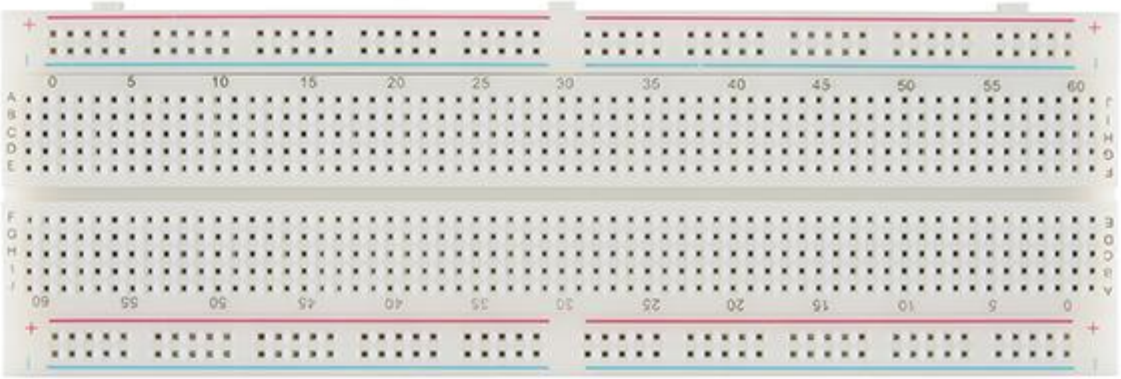


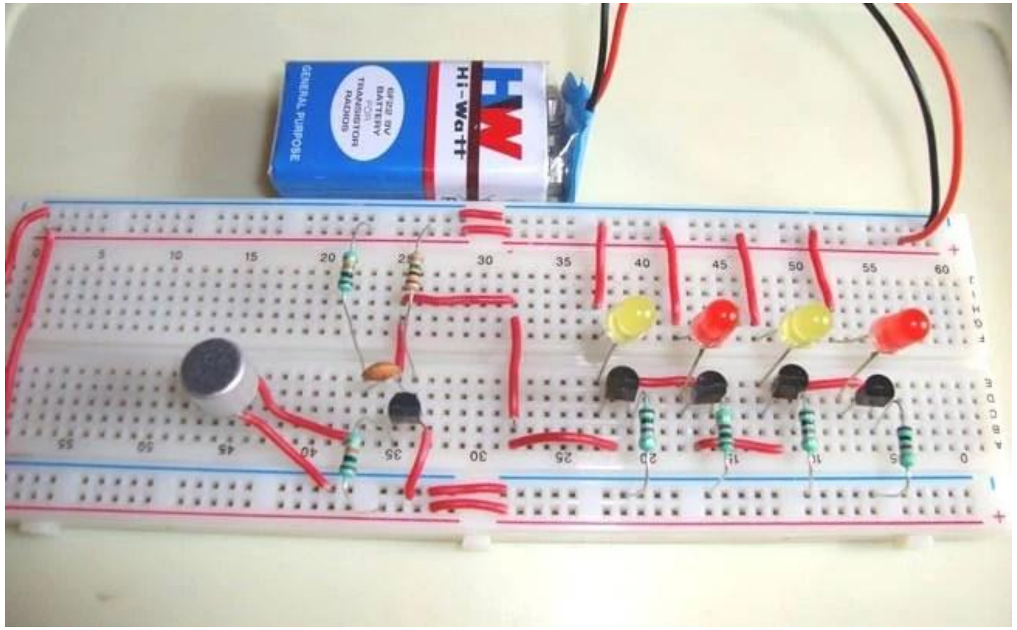


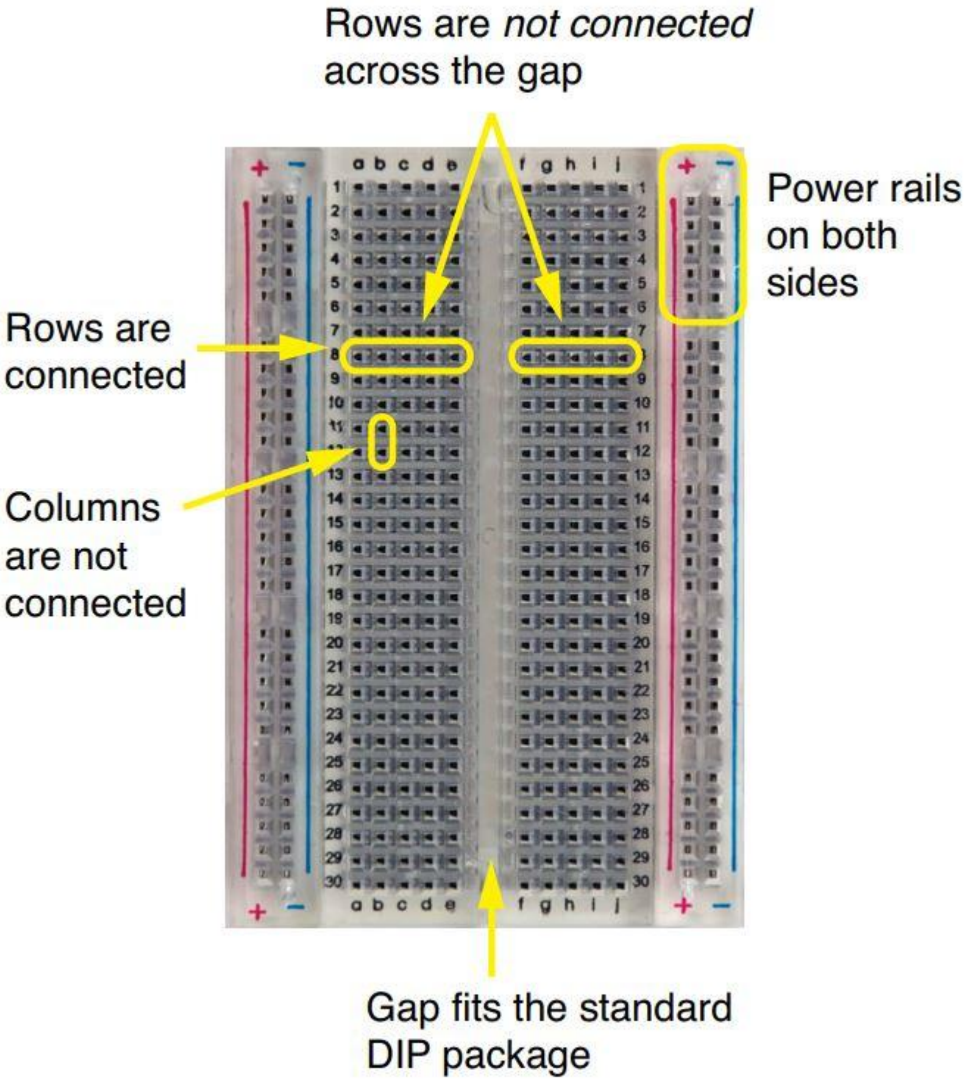
Voltage and current through R, L , C



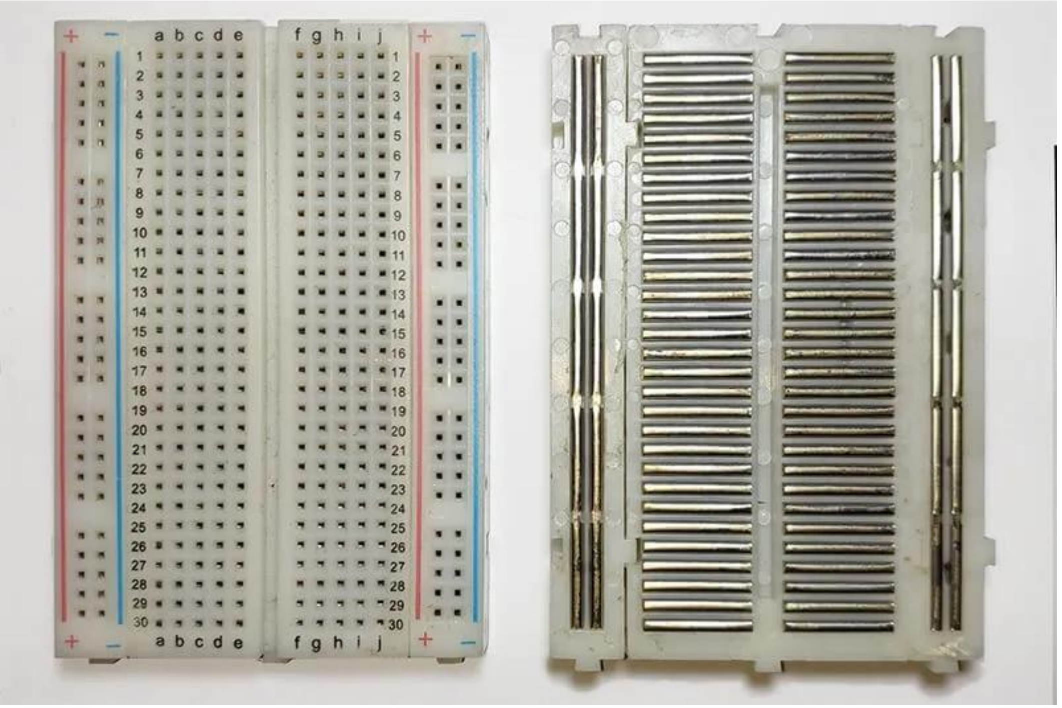
Bread Boards and Wires



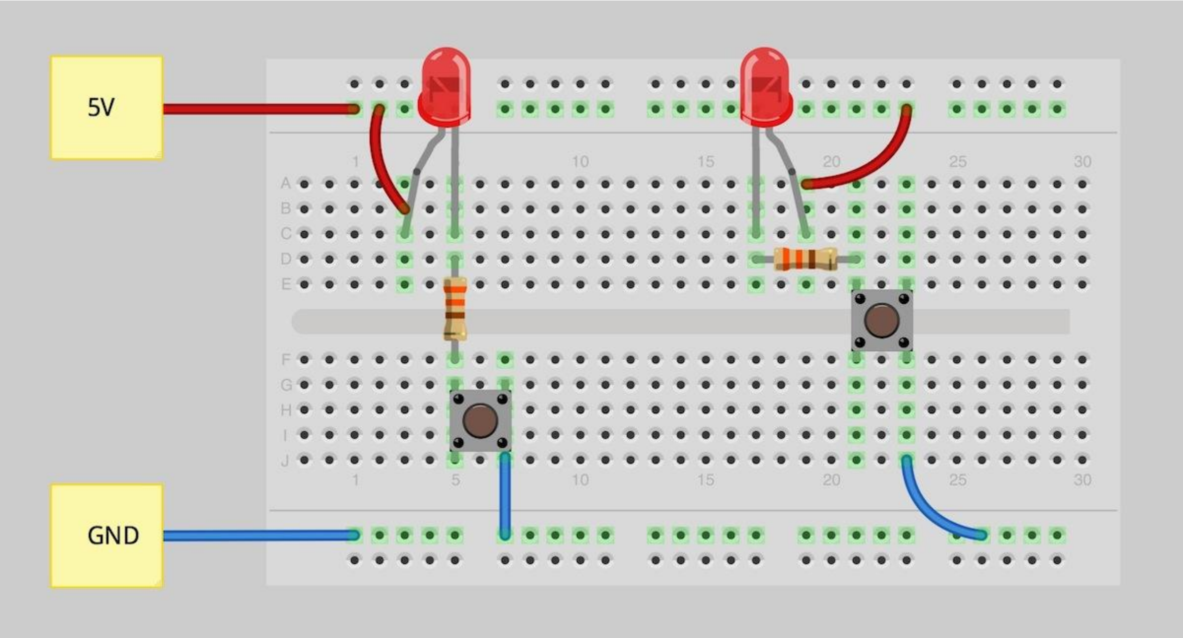




Bread Boards and Wires



Front and Back view of a Breadboard

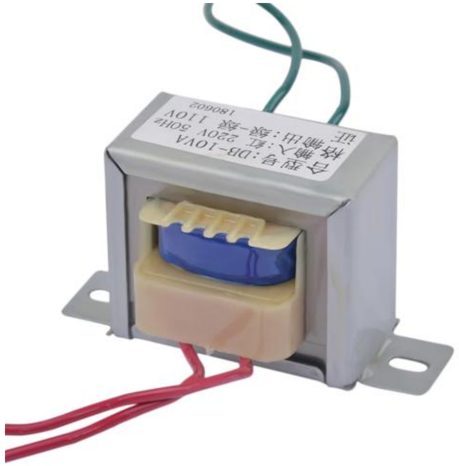


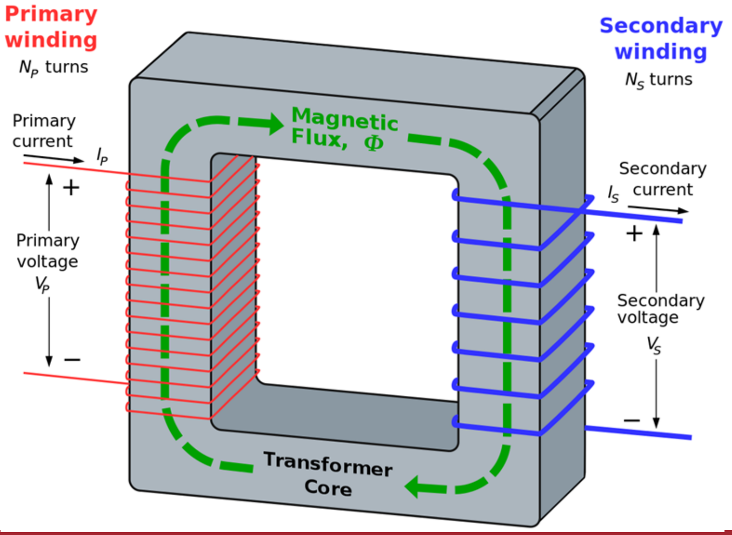
LED and Push Button connection with power supply

Electrical Equipment

- **DC Power Supply (Variable)** – Typically 0–30V, up to 5A.

- **AC Power Supply (Auto-transformer)** – Fixed/Variable output 12-0-12 V , for sinusoidal input.





Electrical Equipment

Function Generator – For generating sinusoidal, square, and triangular waveforms, 0 -20 V, 0.01Hz to 2MHz.

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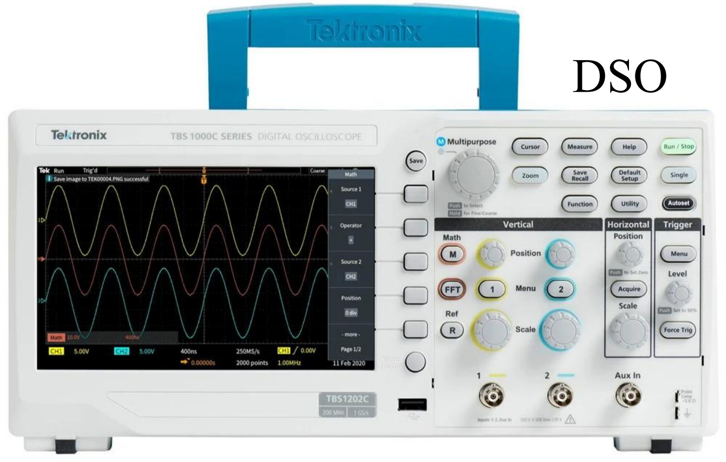
Probe

Oscilloscope (DSO/CRO) – To visualize AC waveforms and verify amplitude, frequency, phase shift. Etc.

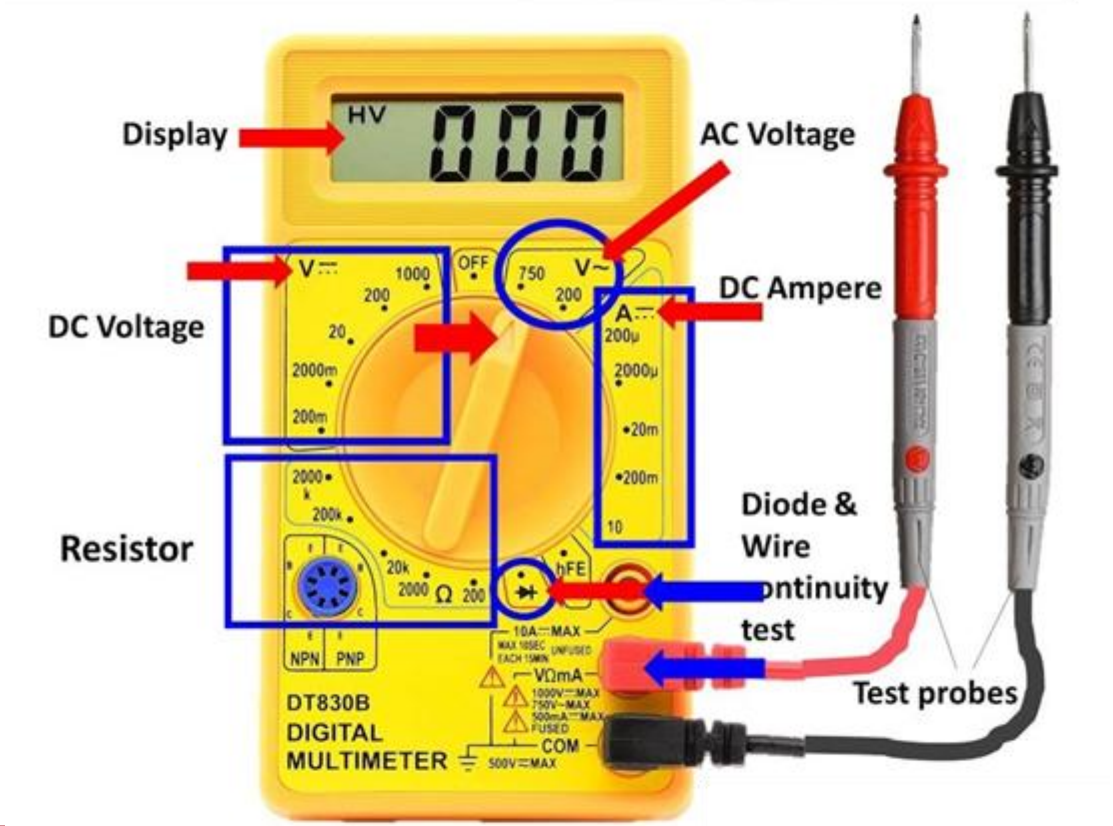
CRO



DSO

Probe

Digital Multimeter



Wattmeter

Sources and Basic Laws

Independent & Dependent Sources:

Independent Sources

These are energy sources whose values (voltage or current) do not depend on any other variable in the circuit. They maintain a fixed value regardless of the load connected.

Independent Voltage Source:

Provides a constant voltage across its terminals.

Example: A 12V battery will always provide 12V regardless of the load.

Independent Current Source:

Provides a constant current through its terminals.

Example: A 3A current source will supply 3A irrespective of the voltage across it.

Dependent (Controlled) Sources

These sources deliver voltage or current that is **dependent** on some other voltage or current in the circuit.

Widely used in electronic circuits, especially for modeling transistors and operational amplifiers.

Types of Controlled sources

Voltage-Controlled Voltage Source (VCVS):

Output voltage is proportional to some voltage in the circuit. $V = kV_x$

Current-Controlled Voltage Source (CCVS):

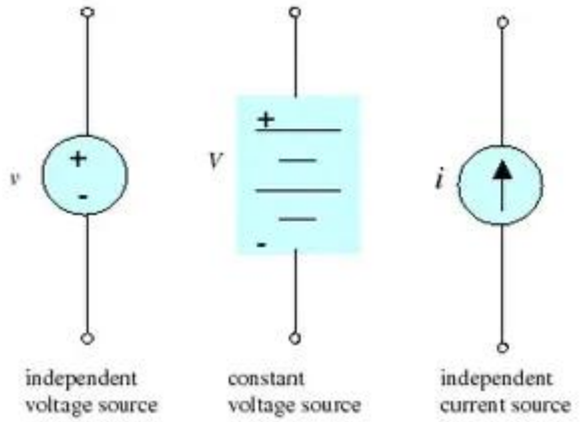
Output voltage is proportional to some current in the circuit. $V = kI_x$

Voltage-Controlled Current Source (VCCS):

Output current is proportional to a voltage elsewhere in the circuit. $I = kV_x$

Current-Controlled Current Source (CCCS):

Output current is proportional to a current elsewhere in the circuit. $I = kI_x$



Importance in Engineering:

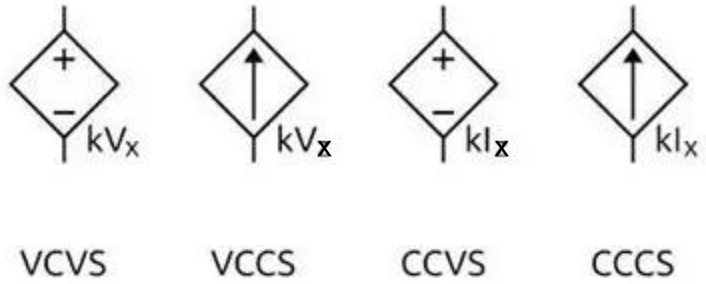
Dependent sources are key for designing amplifiers and modeling behavior of electronic devices.

Simplifies complex systems using equivalent models.

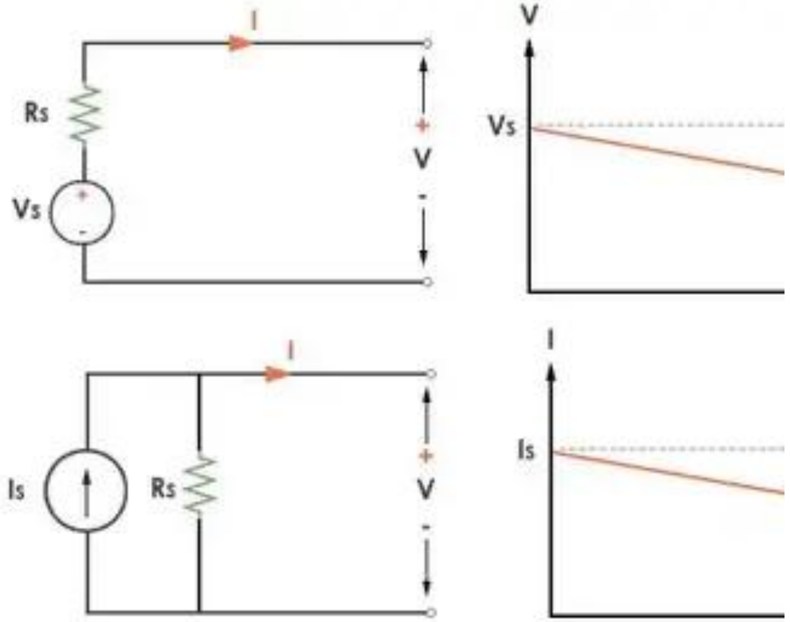
Symbols:

Independent sources are shown with round (voltage) or arrow (current) symbols.

Dependent sources are shown with diamond-shaped symbols.



Ideal and Practical Sources



Ideal Voltage Source:

Maintains the same voltage across its terminals regardless of the current drawn from it.

Internal resistance $R_{int} = 0$

Practical Voltage Source

Real-world sources have some internal resistance which causes the output voltage to drop as the current drawn increases.

Represented by an ideal voltage source in **series** with internal resistance.

Voltage drop: $V_{load} = V_s - IR_{int}$

Ideal Current Source

Maintains constant current regardless of the voltage across it.

Internal resistance $R_{int} = \infty$

Practical Current Source

Real-world current sources are modeled by an ideal current source in **parallel** with a large internal resistance.

Symbols and V-I Characteristics:

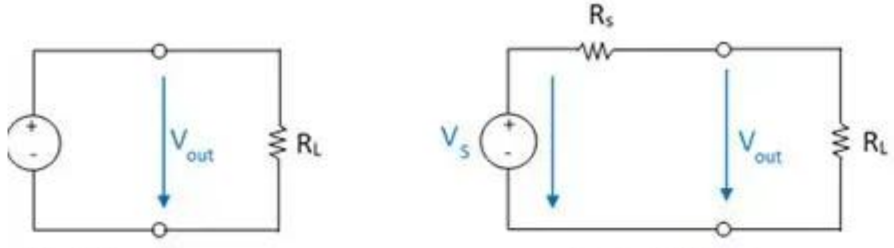
Ideal: perfect horizontal (voltage) or vertical (current) line on a V-I graph.

Practical: sloped lines indicating influence of internal resistance.

Significance:

Recognizing the limitations of real sources allows engineers to make accurate circuit models and performance predictions.

For example, battery voltage drops as it supplies more current due to its internal resistance.



Ideal Voltage Source   *Practical Voltage Source*

Kirchhoff’s Voltage Law (KVL)

States that the **algebraic sum of voltages in any closed loop is zero**.

Based on conservation of energy: net energy gain and drop in a loop must be zero.

$$\sum V_{gains} - \sum V_{drops} = 0$$

Sign Convention for KVL:

Moving from - to + (across source): voltage gain (+)

Moving from + to - (across resistor): voltage drop (-)

Always be consistent.

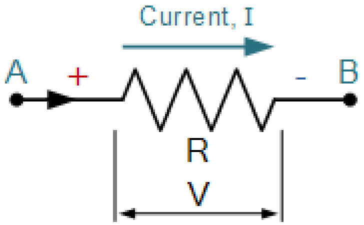
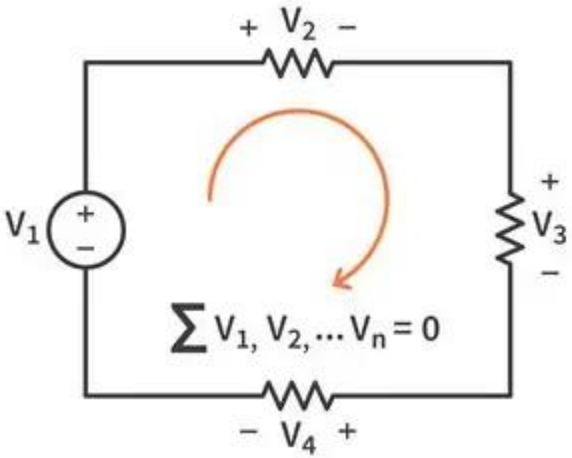
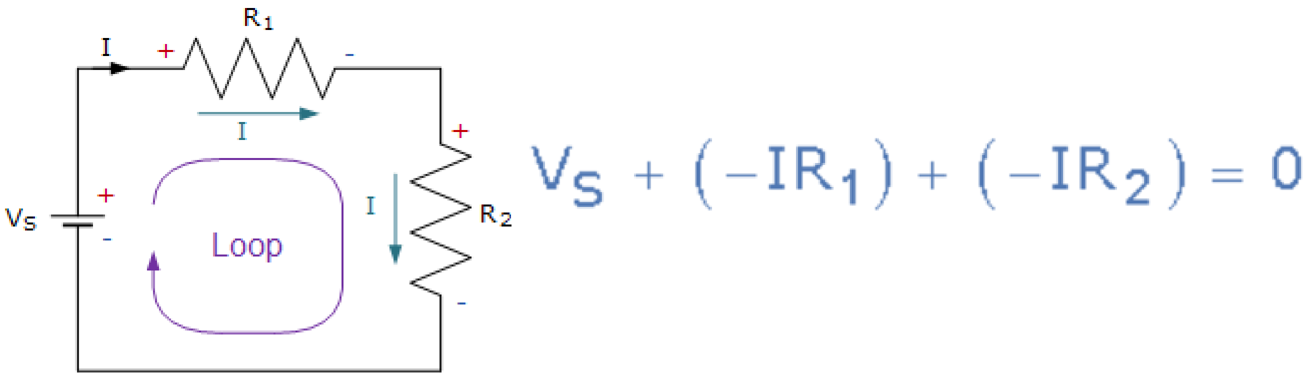


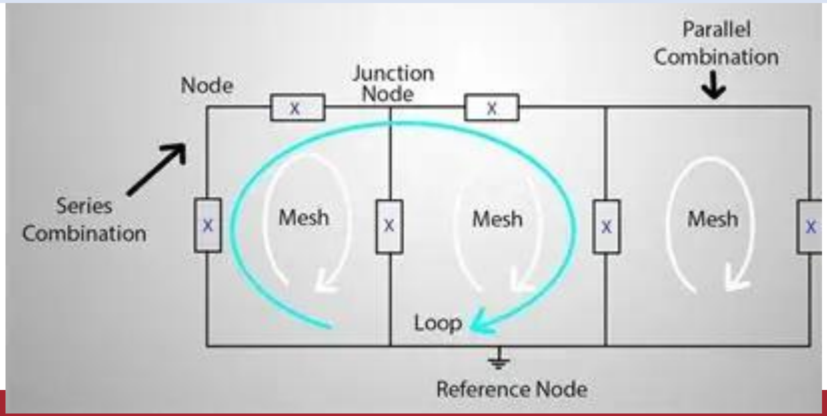
Diagram:





Network Terminology

Term	Explanation
Node	A point where two or more circuit elements are connected.
Branch	A single electrical element such as a resistor or source.
Loop	Any closed path in a circuit.
Mesh	A loop that does not contain any other loop within it.
Circuit	A complete path that allows current to flow.
Network	An interconnection of multiple elements (sources, resistors, etc.).
Linear Network	Elements obey Ohm’s law and superposition.
Nonlinear Network	Elements like diodes, whose V-I relationship is nonlinear.
Active Elements	Can supply energy (e.g., voltage/current sources).
Passive Elements	Cannot supply energy (e.g., resistors, capacitors, inductors).



Example 1: Applying KCL

Problem: At a node, $I_1 = 5A$ and $I_2 = 2A$ are entering. Find I_3, the current leaving.

Solution: By KCL: $I_1 + I_2 = I_3 \Rightarrow I_3 = 5A + 2A = 7A$

Diagram:

Example 2: Applying KVL

Problem: A 10V battery connected in series with 2Ω and 3Ω resistors. Find current and voltage drop across each resistor.

Diagram:

Solution:

- Total resistance $R_{eq} = 2 + 3 = 5\Omega$
- Total current: $I = V/R = 10/5 = 2A$
- Voltage across 2Ω: $V = IR = 2A \times 2\Omega = 4V$
- Voltage across 3Ω: $V = 2A \times 3\Omega = 6V$
- Check: $4V + 6V = 10V$ ✔

Example 3: Real Source Calculation

Problem: A 12V practical voltage source has internal resistance of 1Ω and is supplying 2A. Find voltage across load.

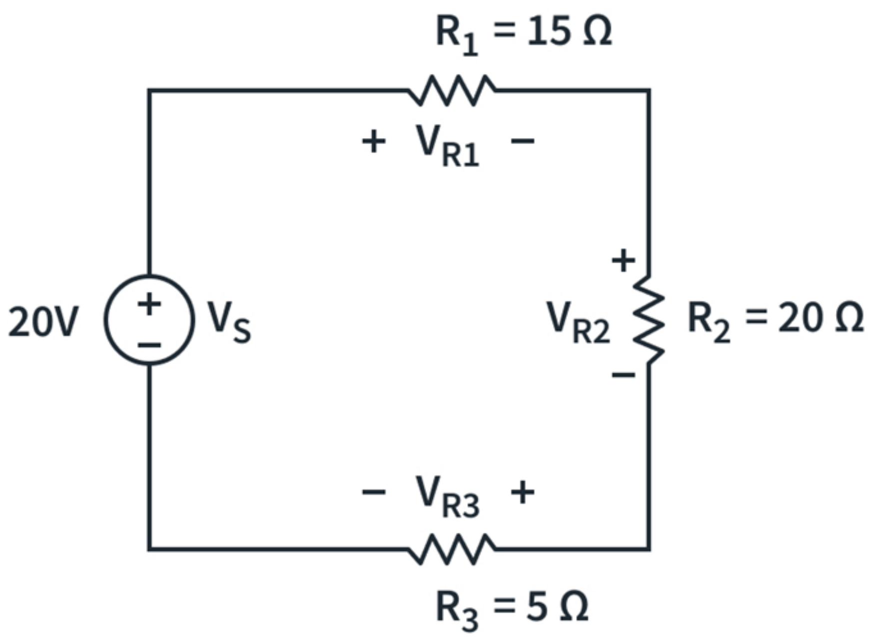
Solution:

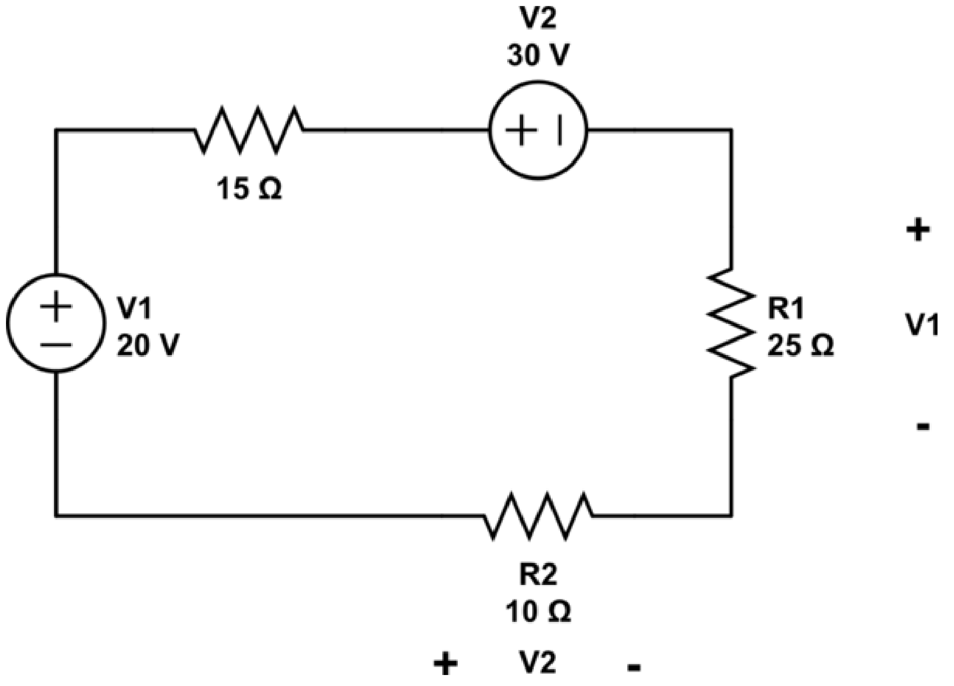
- Voltage drops inside: $V_{int} = 2A \times 1\Omega = 2V$
- Load voltage: $V_L = 12V - 2V = 10V$

Examples

1. **Identify each source as independent or dependent:**
   - A 5V battery → Independent
   - A source controlled by voltage across another resistor → Dependent (VCVS or VCCS)
2. **Use KCL:** At a node, 6A and 4A enter, one branch leaves with 3A. What is the current in the remaining branch?
   - $I_{total\_in} = 10A$, so $I_{out} = 3A + I_x \rightarrow I_x = 7A$
3. **Use KVL:** Circuit with 12V source and two resistors: 4Ω and 2Ω.
   - Total resistance = 6Ω
   - Current = 2A
   - Voltage drop: 8V (4Ω) and 4V (2Ω)

KVL Practice Examples

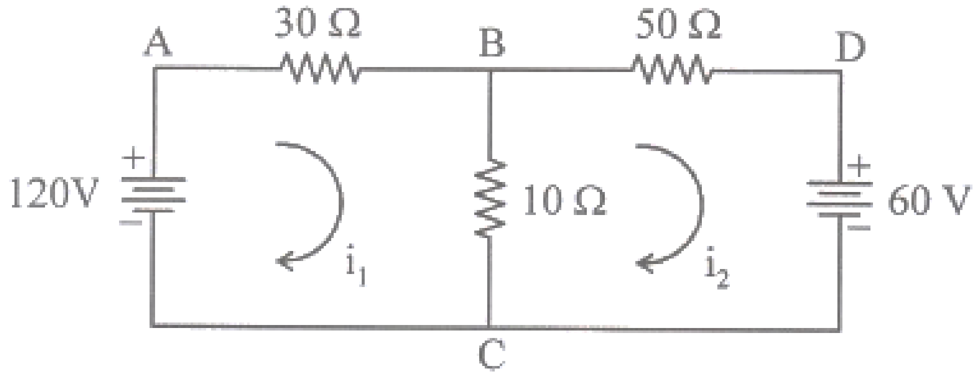




Mesh Analysis (Based on KVL)

It uses Kirchhoff's Voltage Law (KVL) around the meshes (independent loops) of the circuit to find the unknown currents.

Ex. Find mesh currents i1 and i2.



i1 = 114.79/40 = 2869 A

i2 = -120/230 = -0.521 A

Current directions as clockwise mesh ABCA

- 30 i1 - 10(i1 - i2) + 120 = 0

-30 i1 – 10 i1 + 10 i2 + 120 = 0

-40 i1 + 10 i2+ 120 = 0

120 = 40 i1 – 10 i2 …………(1)

Mesh BDCB

-50 i2 - 60 - 10(i2 – i1) = 0

-50 i2 -10 i2 + 10 i1 - 60 = 0

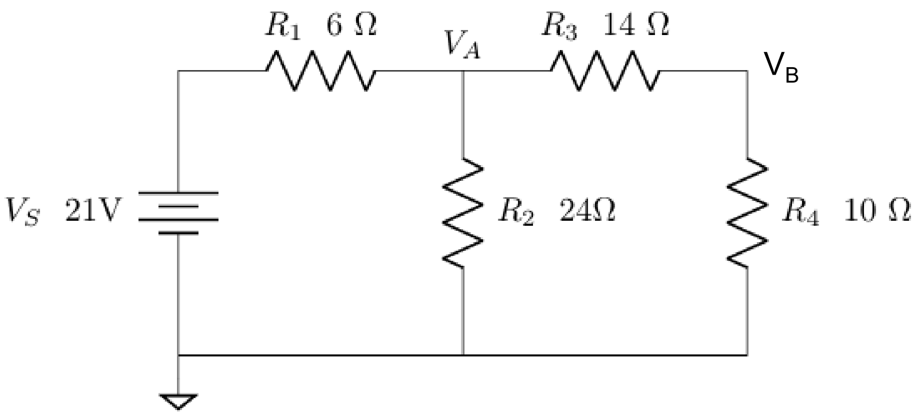
-60 i2 + 10 i1 − 60 = 0

-60 = -10 i1 + 60 i2 …………..(2)

Mesh Analysis (Based on KVL)

EX.

Find V_A using Mesh Analysis



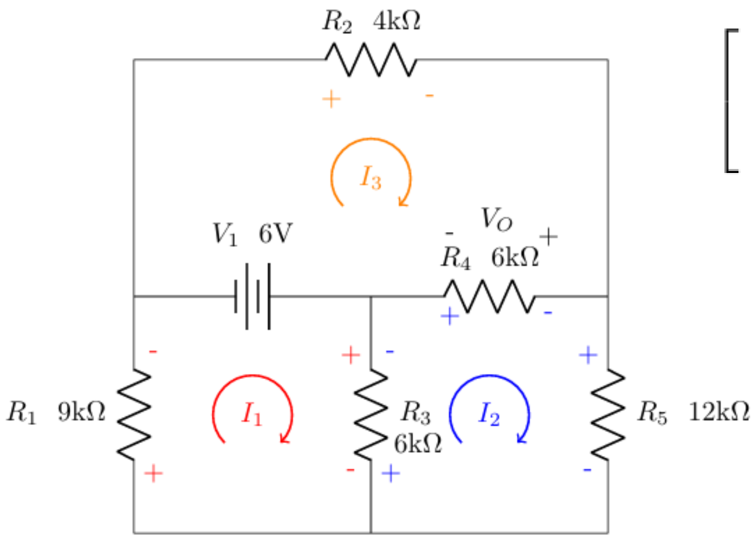
$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1.167 \text{ A} \\ 583.3 \text{ mA} \end{bmatrix}$$

$$V_A = (I_1 - I_2)R_2 = (1.167 \text{ A} - 583.3 \text{ mA})24\ \Omega = 14 \text{ V}$$

$$V_B = I_2 R_4 = (583.3 \text{ mA})(10\ \Omega) = 5.833\text{V}$$

Mesh Analysis (Based on KVL)

Find V_O using mesh analysis.



$$\begin{bmatrix} -15\ \mathrm{k\Omega} & 6\ \mathrm{k\Omega} & 0\ \Omega \\ 6\ \mathrm{k\Omega} & -24\ \mathrm{k\Omega} & 6\ \mathrm{k\Omega} \\ 0\ \Omega & 6\ \mathrm{k\Omega} & -10\ \mathrm{k\Omega} \end{bmatrix}^{-1} \begin{bmatrix} -6\ \mathrm{V} \\ 0\ \mathrm{V} \\ 6\ \mathrm{V} \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 373.33\ \mu\mathrm{A} \\ -66.67\ \mu\mathrm{A} \\ -640.00\ \mu\mathrm{A} \end{bmatrix}$$

$$V_O = (I_3 - I_2)R_4 = (-640.00\ \mu\mathrm{A} + 66.67\ \mu\mathrm{A})6\ \mathrm{k\Omega} = -3.44\mathrm{V}$$

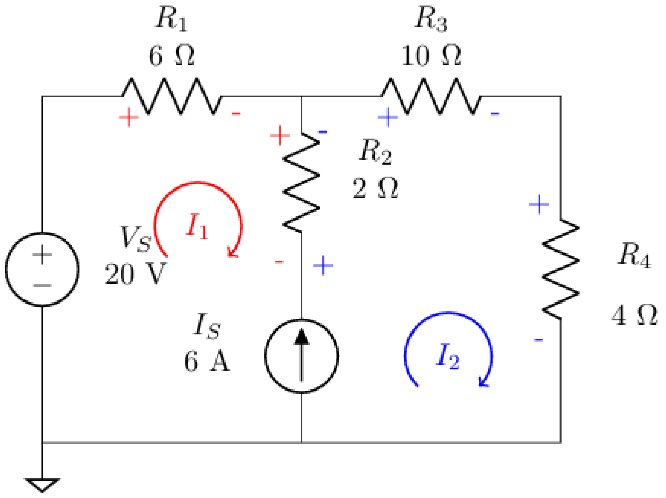
$$-I_1R_1 + V_1 - (I_1 - I_2)R_3 = 0$$
$$-(I_2 - I_1)R_3 - (I_2 - I_3)R_4 - I_2R_5 = 0$$
$$-I_3R_2 - (I_3 - I_2)R_4 - V_1 = 0$$

$$-15k\Omega I_1 + 6k\Omega I_2 = -6\ \mathrm{V}$$
$$6k\Omega I_1 - 24k\Omega I_2 + 6k\Omega I_3 = 0$$
$$6k\Omega I_2 - 10k\Omega I_3 = 6\ \mathrm{V}$$

Super-mesh Analysis

A supermesh is formed when a current source lies between two meshes.
It is obtained by merging the two meshes (excluding the current source branch) and then applying KVL around the combined loop.

Determine the mesh currents



$$-6I_1 - 14I_2 = -20\text{ V}$$

$$\begin{bmatrix} -1 & 1 \\ -6 & -14 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ -20 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -3.2\text{ A} \\ 2.8\text{ A} \end{bmatrix}$$

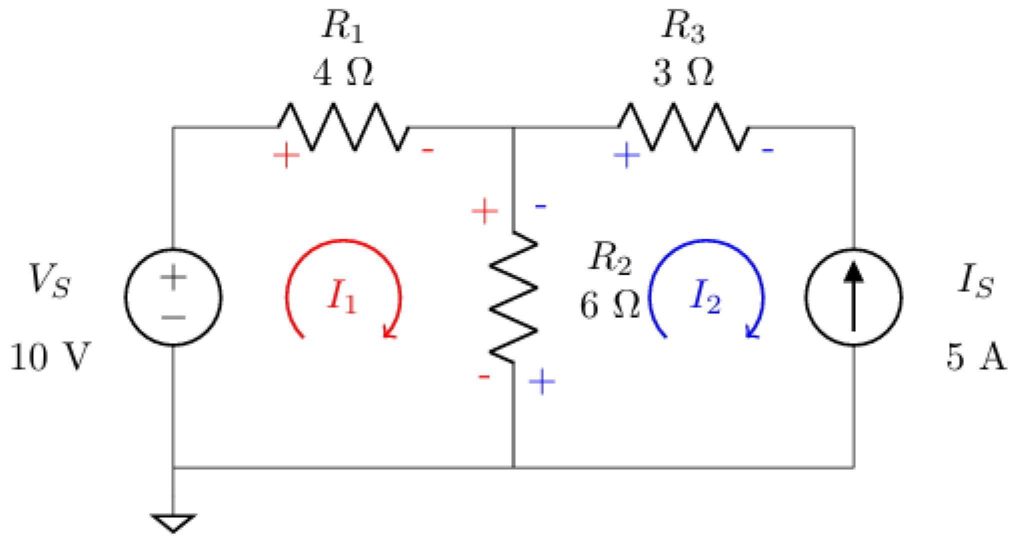
$$-I_1 + I_2 = 6\text{ A}$$

The KVL is then written using the loop around both meshes

$$V_S - V_{R1} - V_{R3} - V_{R4} = 0$$

$$V_S - I_1R_1 - I_2R_3 - I_2R_4 = 0$$

Current Sources in Mesh Current



$$10\ \Omega I_1 - 6\ \Omega I_2 = 10\ \text{V}$$

$$-I_2 = 5\ \text{A}$$

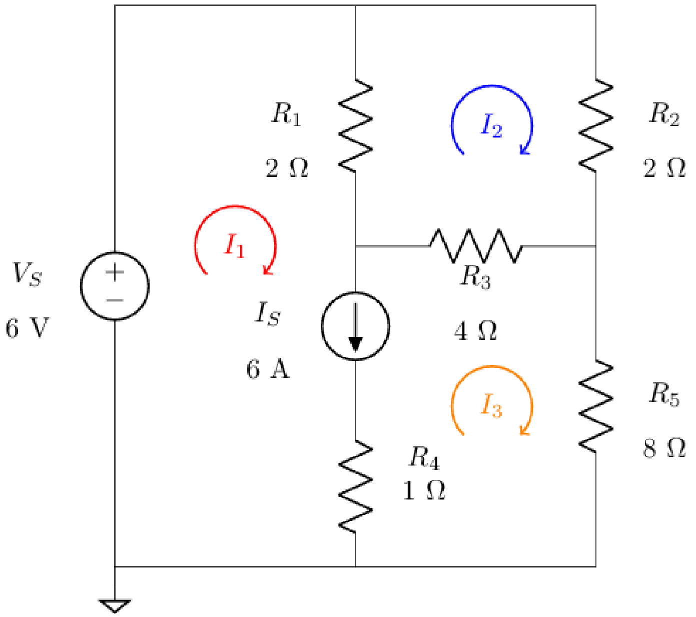
$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -2\ \text{A} \\ -5\ \text{A} \end{bmatrix}$$

$$I_2 = -5\ \text{A}$$

$$-I_2 = 5\ \text{A}$$

Practice Problems

Determine the mesh currents:



$$I_1 + I_3 = 6\ \text{A}$$

The KVL is then written using the loop around both meshes. Note the loop through the circuit by examining which components are included in the KVL

$$V_S - V_{R1} - V_{R3} - V_{R4} = 0$$
$$V_S - I_1 R_1 - I_2 R_3 - I_2 R_4 = 0$$

Kirchhoff's Laws

Kirchhoff's Current Law (KCL):

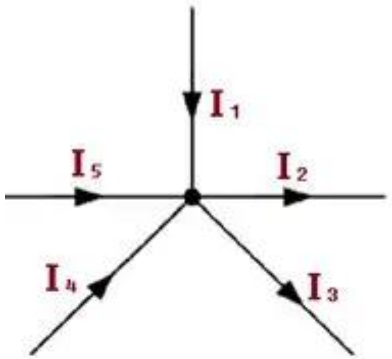
States that the **algebraic sum of all currents entering and leaving a node is zero**.

Based on conservation of charge: current entering a node must equal current leaving.

$$\sum I_{in} - \sum I_{out} = 0$$

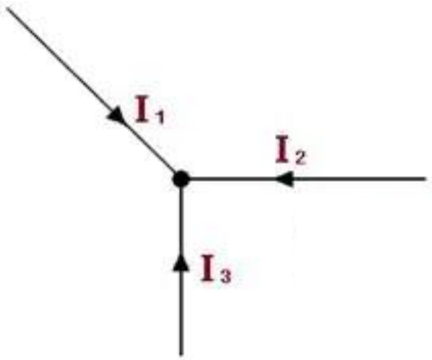
Example: If 3A and 2A enter a node, the total current leaving must be 5A.

Diagram:



$I_1-I_2-I_3+I_4+I_5=0$

(a)



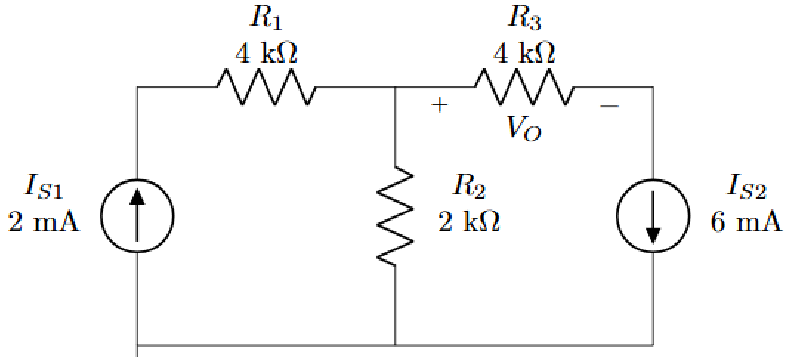
$I_1+I_2+I_3=0$

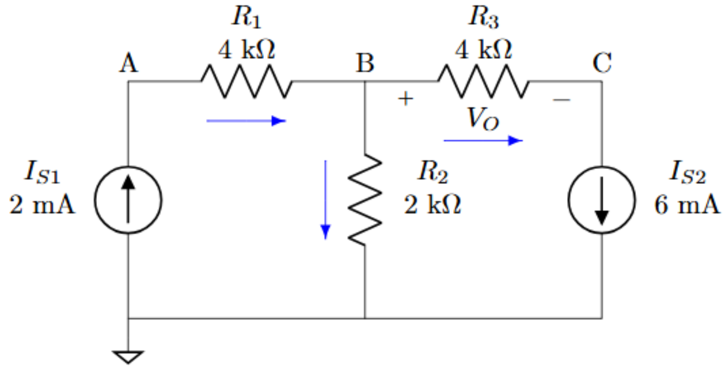
(b)

Node Analysis

- Nodal Analysis is another methodical application of KVL, KCL, and Ohm's law that allow use to analyze any circuit.

Find V_O using Nodal Analysis.





KCL for Node A

$$I_{S1} - I_{R1} = 0$$

and use Ohm's law where we can to rewrite the currents in terms of the node voltages

$$I_{S1} - \frac{V_A - V_B}{R_1} = 0$$

KCL for Node B

$$I_{R1} - I_{R2} - I_{R3} = 0$$

and use Ohm's law where we can to rewrite the currents in terms of the node voltages

$$\frac{V_A - V_B}{R_1} - \frac{V_B - 0}{R_2} - \frac{V_B - V_C}{R_3} = 0$$

KCL for Node C

$$I_{R3} - I_{S2} = 0$$

and use Ohm's law where we can to rewrite the currents in terms of the node voltages

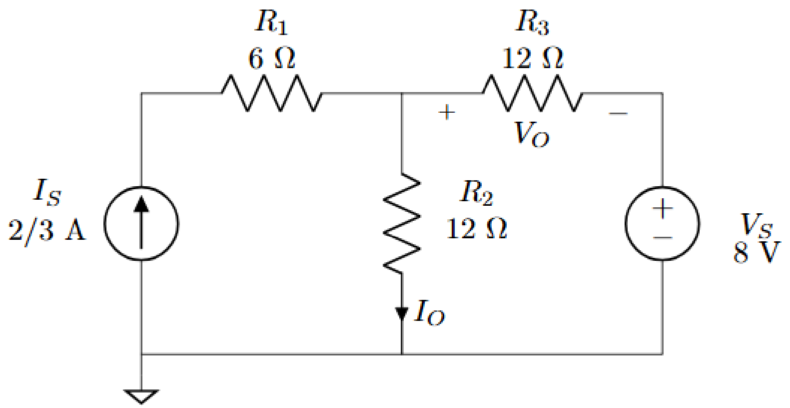
$$\frac{V_B - V_C}{R_3} - I_{S2} = 0$$

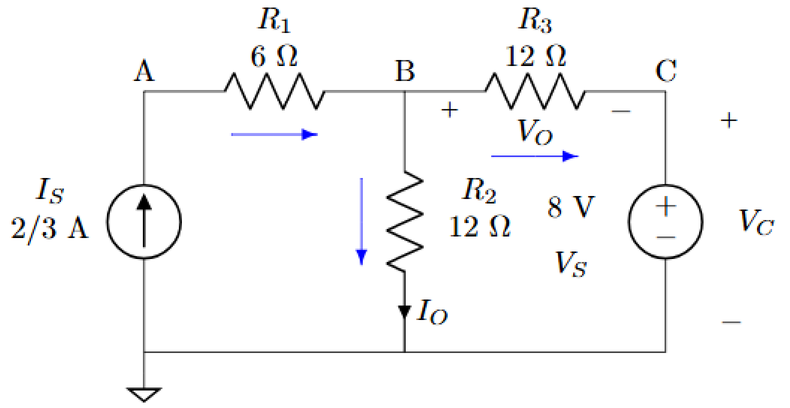
$$\begin{bmatrix} V_A \\ V_B \\ V_C \end{bmatrix} = \begin{bmatrix} 0 \text{ V} \\ -8 \text{ V} \\ -32 \text{ V} \end{bmatrix}$$

$$V_O = V_B - V_C = -8 - (-32) = 24 \text{ V}$$

Examples

Find V_O and I_O using Nodal Analysis.





$$V_C = 8$$

KCL for Node A This KCL is straightforward if you are comfortable with the previous examples of nodal analysis.

$$I_S - I_{R1} = 0$$

and use Ohm's law where we can to rewrite the currents in terms of the node voltages

$$I_{S1} - \frac{V_A - V_B}{R_1} = 0$$

KCL for Node B

$$I_{R1} - I_{R2} - I_{R3} = 0$$

and use Ohm's law where we can to rewrite the currents in terms of the node voltages

$$\frac{V_A - V_B}{R_1} - \frac{V_B - 0}{R_2} - \frac{V_B - V_C}{R_3} = 0$$

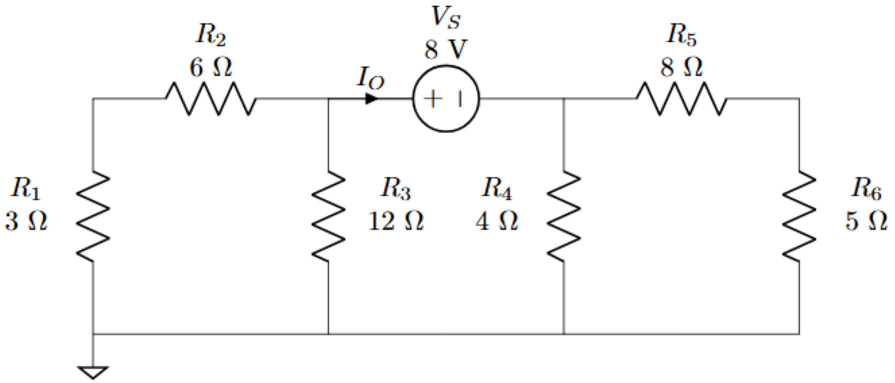
$$\begin{bmatrix} V_A \\ V_B \\ V_C \end{bmatrix} = \begin{bmatrix} 12\ \text{V} \\ 8\ \text{V} \\ 8\ \text{V} \end{bmatrix}$$

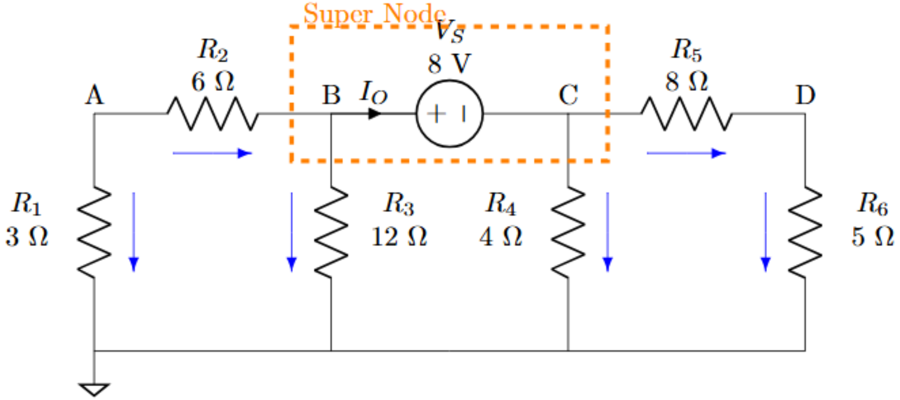
$$V_O = V_B - V_C = 8 - 8 = 0\ \text{V}$$

$$I_O = \frac{V_B - 0}{12} = {}^2/_3\ \text{A}$$

Voltage Sources Connected to Two Non-ground Nodes (Supernode analysis)

- Find Io using node Analysis





$$V_B - V_C = 8$$

KCL for Super-node (B and C)

$$I_{R2} - I_{R3} - I_{R4} - I_{R5} = 0$$

we can to rewrite the currents in terms of the node voltages

$$\frac{V_A - V_B}{R_2} - \frac{V_B - 0}{R_3} - \frac{V_C - 0}{R_4} - \frac{V_C - V_D}{R_5} = 0$$

KCL for Node A

$$-I_{R1} - I_{R2} = 0$$

we can to rewrite the currents in terms of the node voltages

$$-\frac{V_A - 0}{R_1} - \frac{V_A - V_B}{R_2} = 0$$

KCL for Node D

$$I_{R5} - I_{R6} = 0$$

we can to rewrite the currents in terms of the node voltages

$$\frac{V_C - V_D}{R_5} - \frac{V_D - 0}{R_6} = 0$$

put them in matrix form

$$\begin{bmatrix} 0 & 1 & -1 & 0 \\ \frac{1}{6} & -\frac{1}{4} & -\frac{3}{8} & \frac{1}{8} \\ -\frac{1}{2} & \frac{1}{6} & 0 & 0 \\ 0 & 0 & \frac{1}{8} & -\frac{13}{40} \end{bmatrix}^{-1} \begin{bmatrix} 8 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} V_A \\ V_B \\ V_C \\ V_D \end{bmatrix} = \begin{bmatrix} 1.67\text{ V} \\ 5.02\text{ V} \\ -2.98\text{ V} \\ -1.15\text{ V} \end{bmatrix}$$

To find Io, Apply KCL at node 'C' :

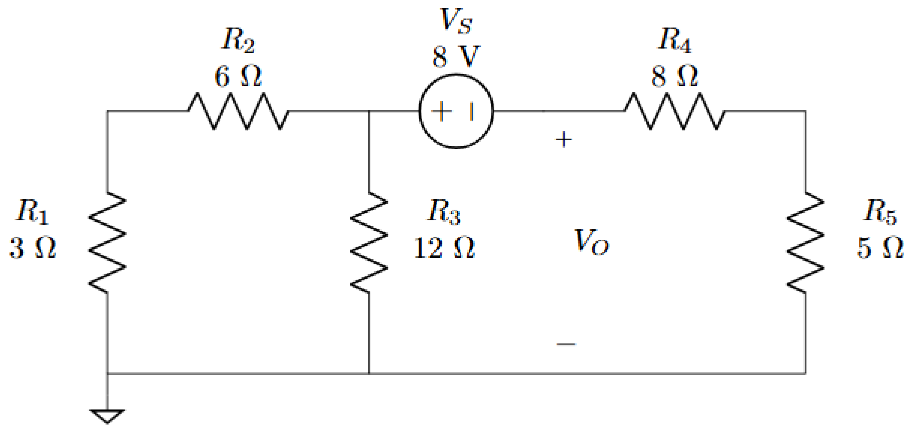
$$I_O - I_{R4} - I_{R5} = 0$$

$$I_O - \frac{V_C - 0}{R_4} - \frac{V_C - V_D}{R_5} = 0$$

substitute the known values

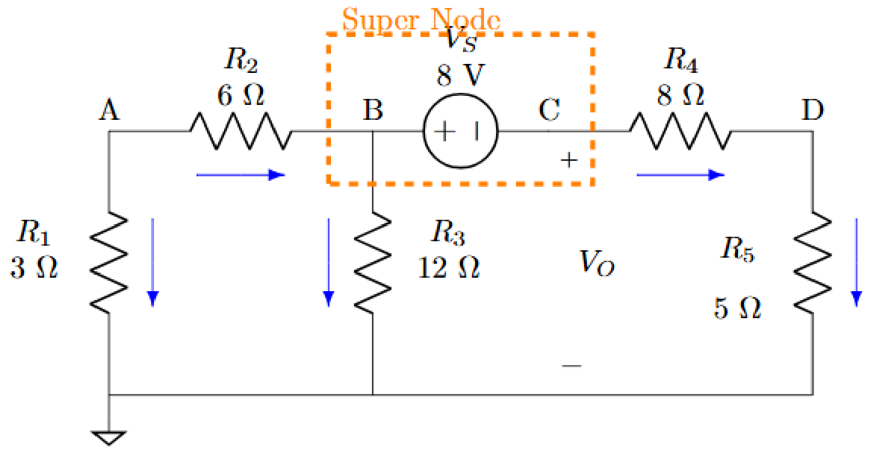
$$I_O = \frac{-2.98}{4} + \frac{-2.98 - (-1.15)}{8} = -975.4\text{ mA}$$

Find V_O using nodal analysis



$$\begin{bmatrix} V_A \\ V_B \\ V_C \\ V_D \end{bmatrix} = \begin{bmatrix} 755.9 \text{ mV} \\ 2.268 \text{ V} \\ -5.732 \text{ V} \\ -2.205 \text{ V} \end{bmatrix}$$

$$V_O = V_C - 0 = -5.732 - 0 = -5.732 \text{ V}$$



Source Transformation

Definition: Source transformation is an analytical technique that allows the conversion between equivalent voltage and current source models for linear bilateral networks. It is based on the principle that two networks are equivalent if they exhibit identical voltage-current characteristics across the terminals.

Voltage Source to Current Source:

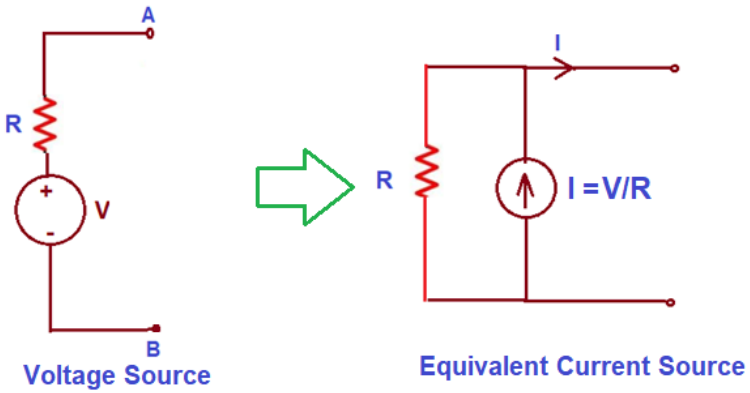
A voltage source in series with a resistor can be transformed into a current source in parallel with the same resistor .

Current Source to Voltage Source:

A current source in parallel with a resistor can be converted into a voltage source in series with the same resistor .

Theoretical Basis:

This transformation is valid under the assumption of linearity and bilateral behavior of the resistor and applies only to passive resistive circuits.



Example:

A 12V source in series with a 4Ω resistor is equivalent to a 3A current source in parallel with a 4Ω resistor, as calculated by Ohm's Law ($I = 12V / 4\Omega = 3A$).

Series and Parallel Resistor Combinations

Series Combination:

- Total or equivalent resistance: $R_{eq} = R_1 + R_2 + \cdots + R_n$
- Same current flows through all resistors.
- Voltage divides according to resistance values (Voltage Division Rule).

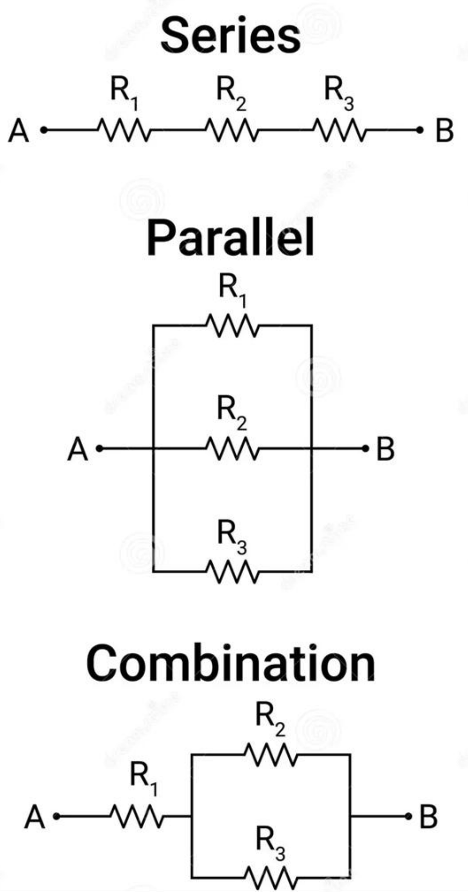
Parallel Combination:

- Equivalent resistance: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$
- Voltage across each resistor is the same.
- Currents divide inversely proportional to resistances (Current Division Rule).

Example:

Calculate equivalent resistance for:

- 4Ω and 6Ω in series → $R_{eq} = 4 + 6 = 10\Omega$
- 4Ω and 6Ω in parallel → $R_{eq} = \frac{1}{\frac{1}{4}+\frac{1}{6}} = 2.4\Omega$

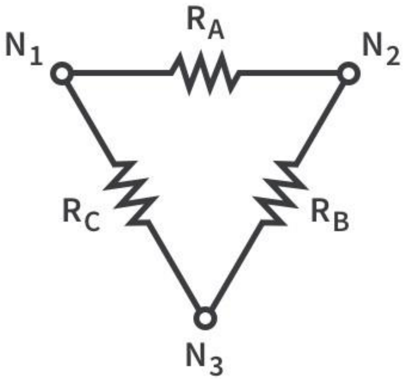


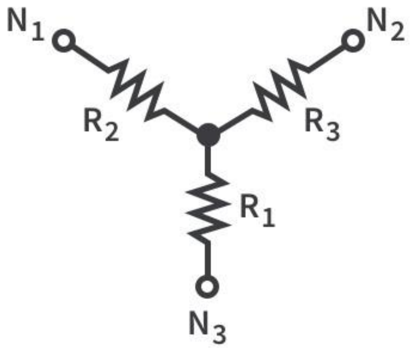
Star-Delta (Y-Δ) Transformation

Necessity:

In networks that are neither purely series nor parallel, especially bridge networks, Y-Δ transformation allows simplification for analysis.

Delta (Δ) to Star (Y) Conversion:

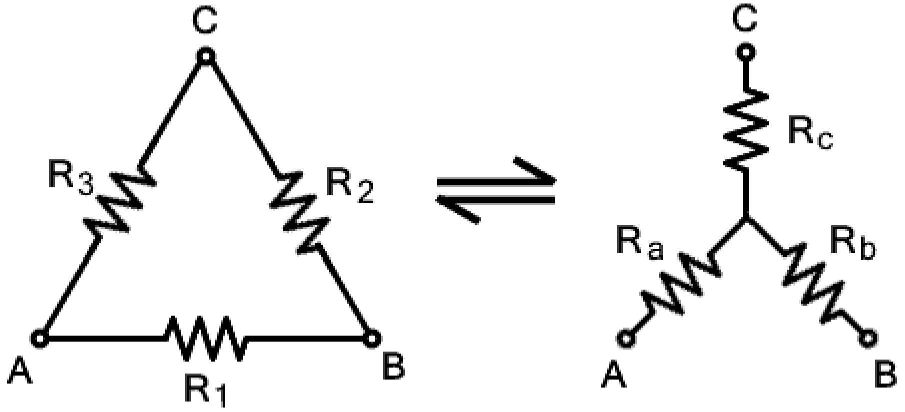




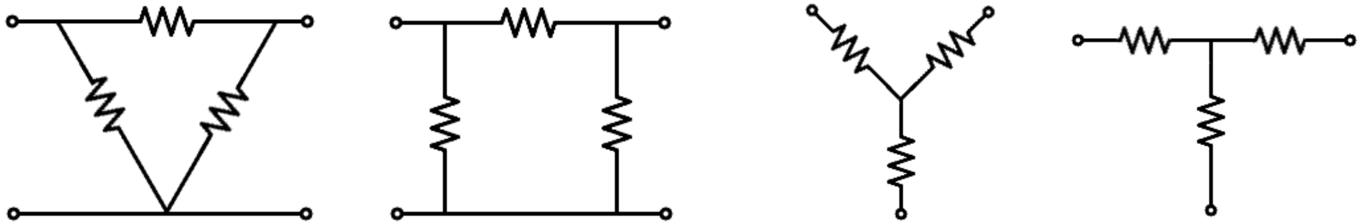
$$R_A = \frac{R_1R_2+R_2R_3+R_3R_1}{R_1} \qquad R_1 = \frac{R_BR_C}{R_A+R_B+R_C}$$

$$R_B = \frac{R_1R_2+R_2R_3+R_3R_1}{R_2} \qquad R_2 = \frac{R_CR_A}{R_A+R_B+R_C}$$

$$R_C = \frac{R_1R_2+R_2R_3+R_3R_1}{R_3} \qquad R_3 = \frac{R_AR_B}{R_A+R_B+R_C}$$



Star Delta Conversion



Delta (Δ) Network

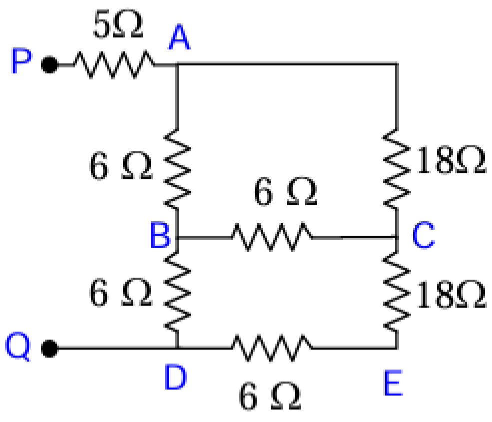
Delta Pi (π) Network

Star Wye(Y) Network

Star (T) Network

Examples

Determine the Input resistance between PQ using star-delta transformation



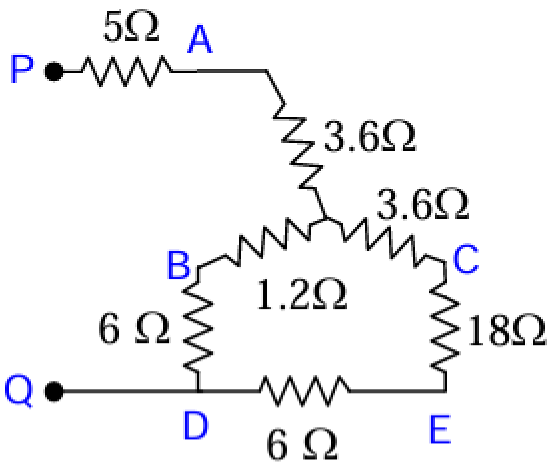
In the given circuit 6 Ω, 6 Ω and 18 connected between A,B and C are in delta connection, convert this into star network. The details are as shown in

$$R_A = \frac{R_{AB} \times R_{AC}}{\sum R_{AB}}$$

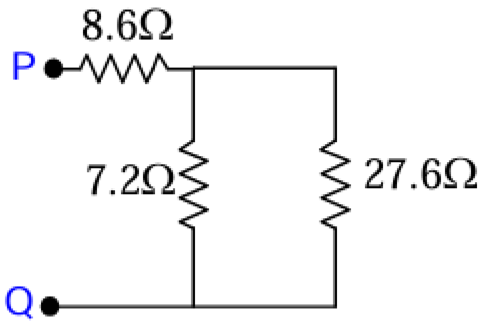
$$R_A = \frac{6 \times 18}{30} = 3.6\Omega$$

$$R_B = \frac{6 \times 6}{30} = 1.2\Omega$$

$$R_C = \frac{6 \times 18}{30} = 3.6\Omega$$



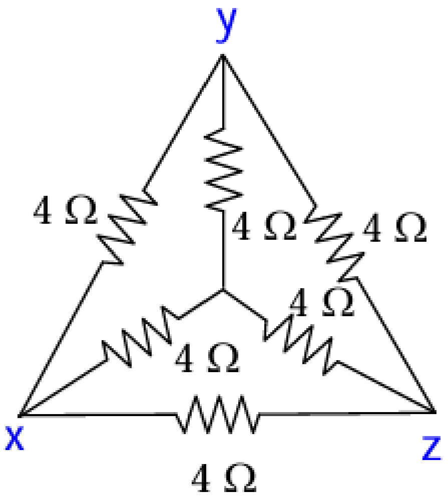
In the circuit 5 Ω and 3.6 Ω are in series, 6 Ω and 1.2 Ω are in series and 3.6 Ω 18 Ω and 6 Ω are in series.



7.2 Ω 27.6 Ω are in parallel, which is in series with 8.6 Ω. The net resistance is.

$$
\begin{aligned}
R_{PQ} &= 8.6 + \frac{7.2 \times 27.6}{7.2 + 27.6} \\
&= 8.6 + 5.71 \\
&= 14.31\Omega
\end{aligned}
$$

For the circuit shown in Figure shown in Figure determine the equivalent resistance between any two terminals.



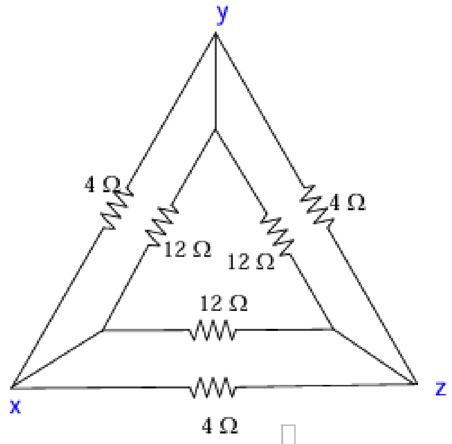
In the given circuit, 4 4 4 Ω at centre are in Star convert them to Delta

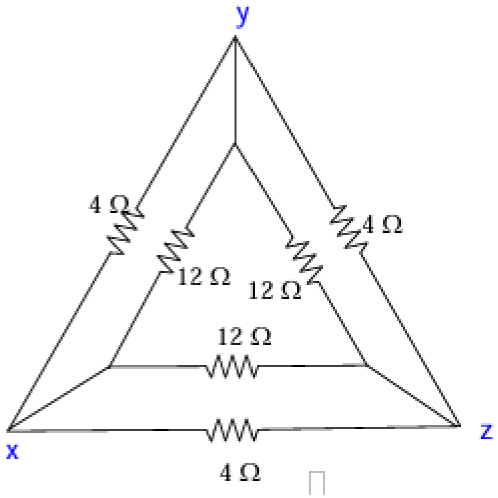
$$R_{xy} = R_x + R_y + \frac{R_x \times R_y}{R_z} = 8 + 4 = 12\Omega$$

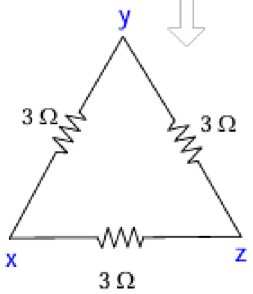
$$R_{xy} = 4 + 4 + \frac{4 \times 4}{4} = 8 + 4 = 12\Omega$$

$$R_{yz} = 4 + 4 + \frac{4 \times 4}{4} = 8 + 4 = 12\Omega$$

$$R_{zx} = 4 + 4 + \frac{4 \times 4}{4} = 8 + 4 = 12\Omega$$







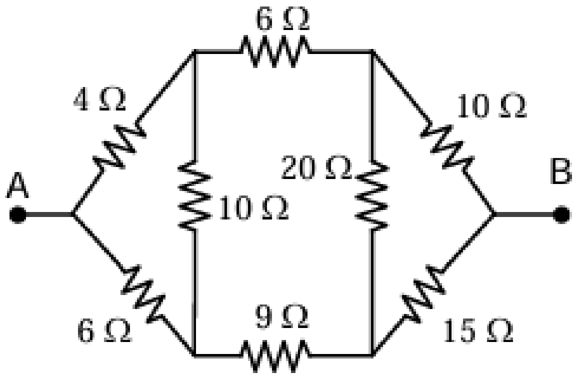
12 Ω is parallel with 4 Ω

$$= \frac{12 \times 4}{12 + 4} = 3\Omega$$

By looking from XZ terminal the 3 Ω resistance is in parallel with series connection of 3 Ω and 3 Ω resistance.

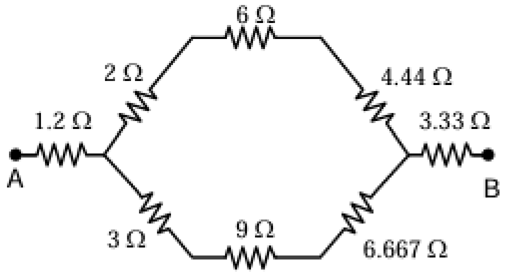
$$R_{XZ} = \frac{3 \times 6}{3 + 6} = 2\Omega$$

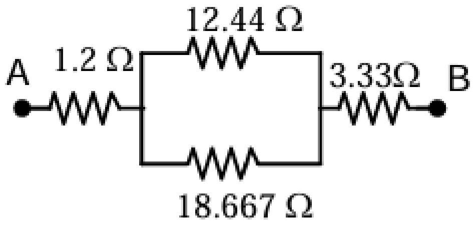
Determine the equivalent resistance between terminals A and B.



$$
\begin{aligned}
R_A &= \frac{4 \times 6}{20} = 1.2\Omega \\
R_B &= \frac{4 \times 10}{20} = 2\Omega \\
R_C &= \frac{6 \times 10}{20} = 3\Omega
\end{aligned}
$$

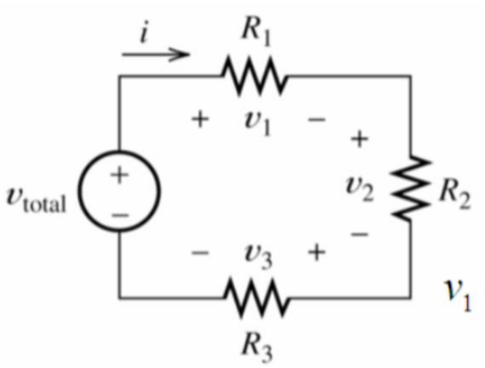
$$
\begin{aligned}
R_A &= \frac{10 \times 15}{45} = 3.33\Omega \\
R_B &= \frac{10 \times 20}{45} = 4.44\Omega \\
R_C &= \frac{20 \times 15}{45} = 6.667\Omega
\end{aligned}
$$





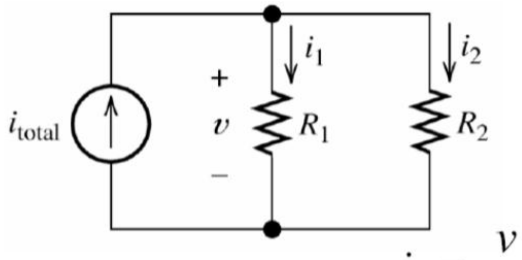
$$
\begin{aligned}
R_{AB} &= 1.2 + \frac{12.44 \times 18.667}{12.44 + 18.667} + 3.3\Omega \\
&= 1.2 + \frac{232.217}{12.44 + 31.187} + 3.3\Omega \\
&= 1.2 + 7.446 + 3.3 = 11.94\Omega
\end{aligned}
$$

Voltage and Current Division Rule



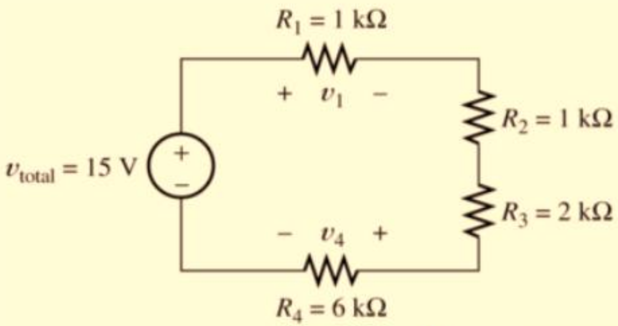
$$v_1 = R_1 i = \frac{R_1}{R_1 + R_2 + R_3} v_{\text{total}}$$

$$v_2 = R_2 i = \frac{R_2}{R_1 + R_2 + R_3} v_{\text{total}}$$



$$i_1 = \frac{v}{R_1} = \frac{R_2}{R_1 + R_2} i_{\text{total}}$$

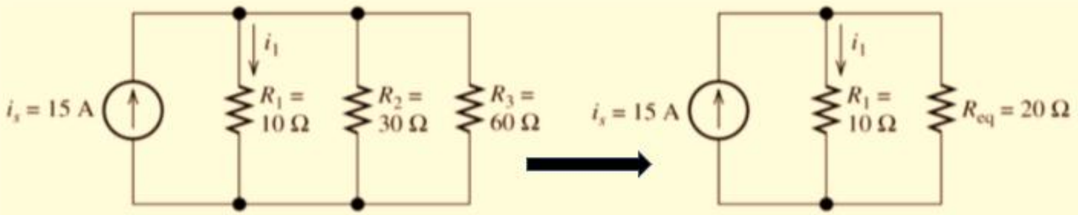
$$i_2 = \frac{v}{R_2} = \frac{R_1}{R_1 + R_2} i_{\text{total}}$$



$$v_1 = \frac{R_1}{R_1 + R_2 + R_3 + R_4} v_{\text{total}}$$

$$= \frac{1000}{1000 + 1000 + 2000 + 6000} \times 15$$

$$= 1.5\text{V}$$



$$R_{eq} = \frac{R_2 R_3}{R_2 + R_3} = \frac{30 \times 60}{30 + 60} = 20\Omega$$

$$i_1 = \frac{R_{eq}}{R_1 + R_{eq}} i_s = \frac{20}{10 + 20} 15 = 10\text{A}$$

Superposition Theorem

In any linear, bilateral circuit having multiple independent sources (voltage or current sources), the response (current or voltage across any element) is equal to the algebraic sum of the responses caused by each independent source acting alone, while replacing all other independent sources with their internal impedances.

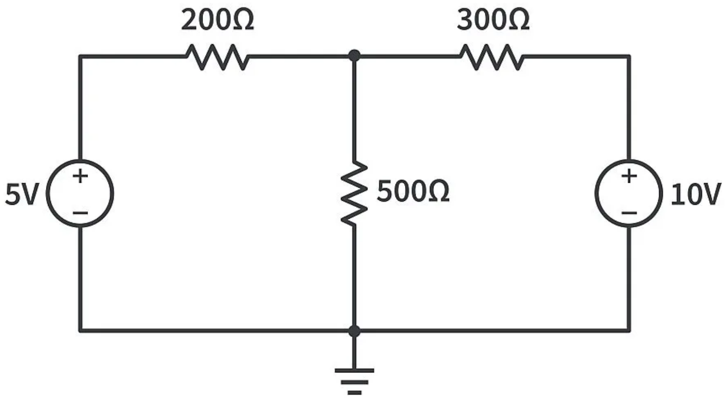
Steps to Apply Superposition Theorem:

- Identify all independent sources (voltage and current).
- Consider one source at a time:
  - Replace all other voltage sources with **short circuits** (since internal resistance of ideal voltage source is zero).
  - Replace all other current sources with **open circuits** (since internal resistance of ideal current source is infinite).
- Solve the circuit and find the contribution of the considered source to the required voltage/current.
- Repeat for each source.
- **Add algebraically all the contributions to get the final result**.

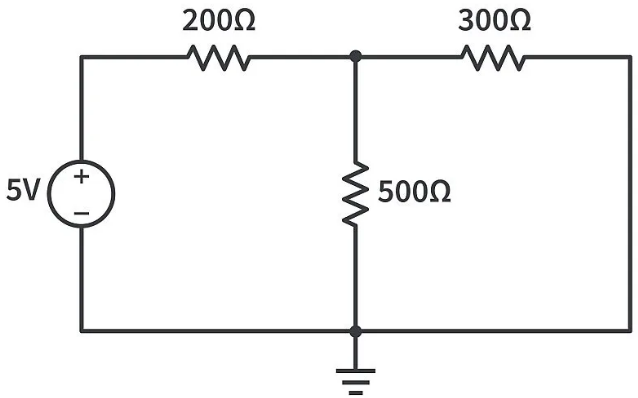
Superposition Theorem

Examples:

Find current through 200 ohm resistor using Superposition theorem



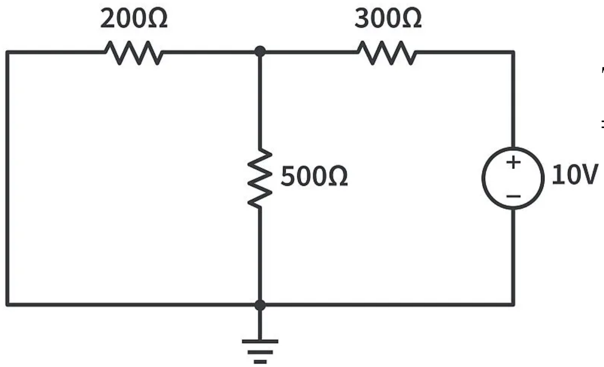
Replace 10 V supply with short circuit



200 Ω in series with 300 Ω || 500 Ω (an equivalent resistance of 187.5 Ω), so the total current through the series circuits will be:

$$\frac{5\text{V}}{200\Omega + 187.5\Omega} = 12.9\text{mA}$$

Now consider the 10V source only:



Total current theough 200Ω resistor
= 12.9 – 16.2 **= -3.3mA**

200Ω resistor: 16.2mA

Thevenin’s Theorem

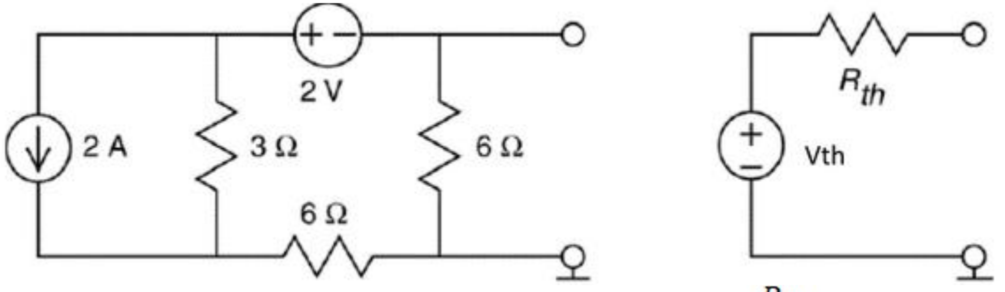
Statement:

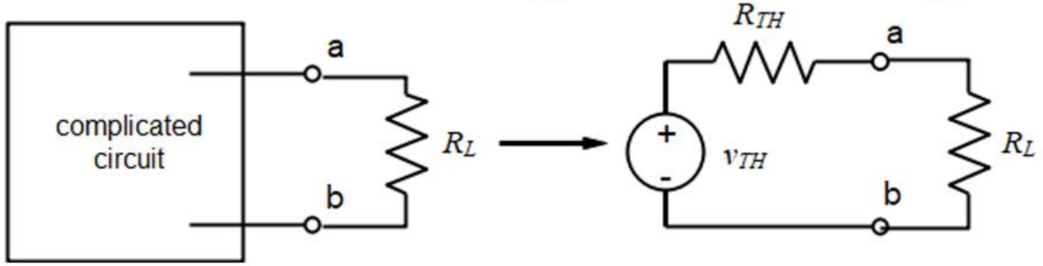
Any linear two-terminal network consisting of voltage sources, current sources, and resistors can be replaced by an equivalent circuit consisting of:

A single **Thevenin’s equivalent voltage source** V_{th} in series with

A single **Thevenin’s equivalent resistance** R_{th},

connected across the load.





Steps to Apply Thevenin’s Theorem

1. Remove the load resistor R_L
2. Find Thevenin Voltage V_{th}:
   Calculate the open-circuit voltage across the load terminals.
3. Find Thevenin Resistance R_{th}:
   Replace voltage sources with a short circuit.
   Replace current sources with an open circuit.
   Calculate the equivalent resistance seen from the load terminals.
4. Draw the Thevenin Equivalent Circuit:
   Voltage source V_{th} in series with resistance R_{th} ,connected to the load R_L.
5. Reconnect the Load R_L:
   Now, the circuit becomes very simple (just one source, one resistance, and the load).

Norton's Theorem

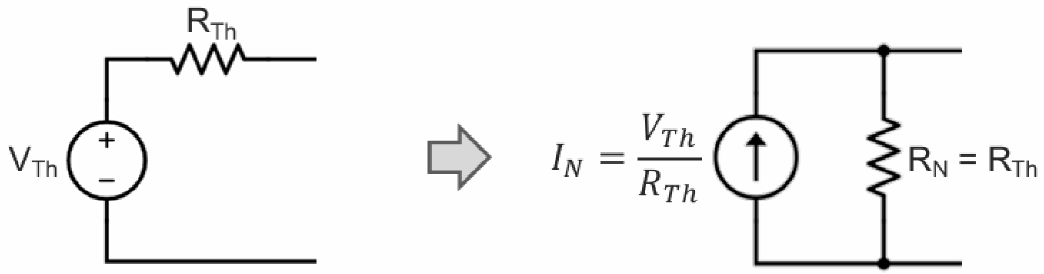
Statement:

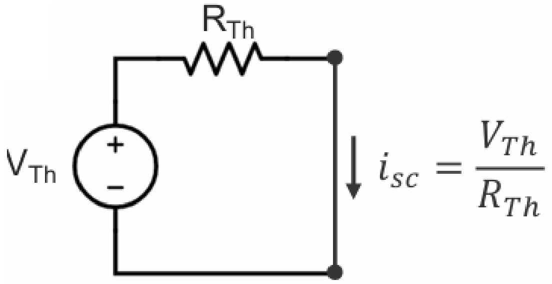
Any linear two-terminal network of sources and resistors can be replaced by an equivalent circuit consisting of:

A single Norton current source I_N in parallel with

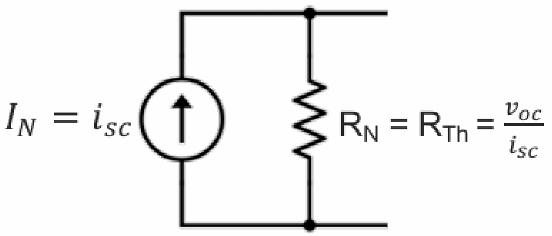
A single Norton resistance R_N,

connected across the load.



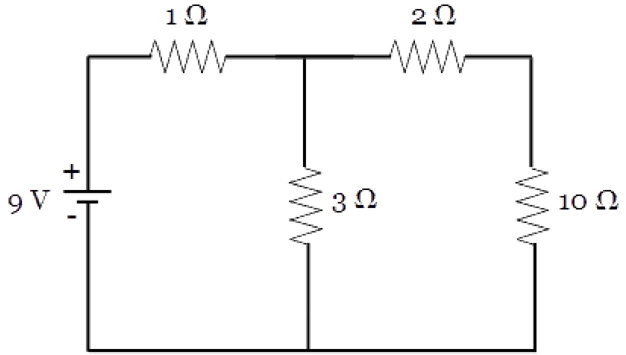


Since, $\frac{V_{Th}}{R_{Th}}$ is the short-circuit current, then, the Norton current equals the short-circuit current



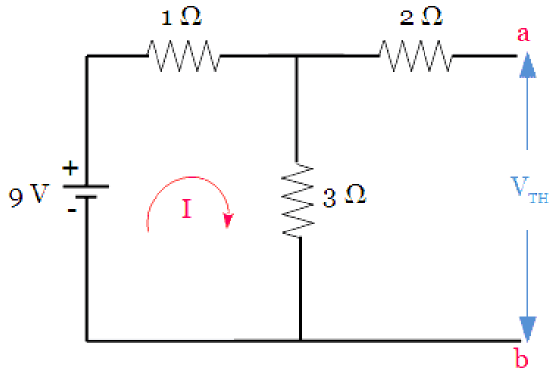
Thevenin's Examples

Find the current through 10 Ω using Thevenin's Theorem.



let us consider 10 Ω as the load resistor.

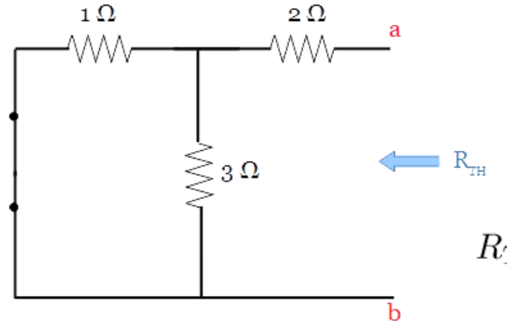
Find Thevenin's voltage, Remove the load resistor(10 Ω)



$$I = \frac{V}{R} = \frac{9}{4} = 2.25A$$

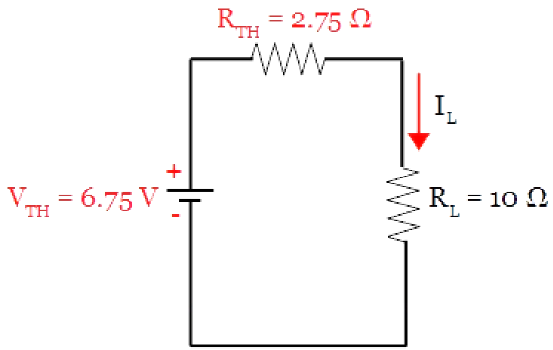
$$V_{TH} = I * R = 2.25 * 3 = 6.75V$$

Find Thevenin's resistance, Remove the load resistor, short the voltage source



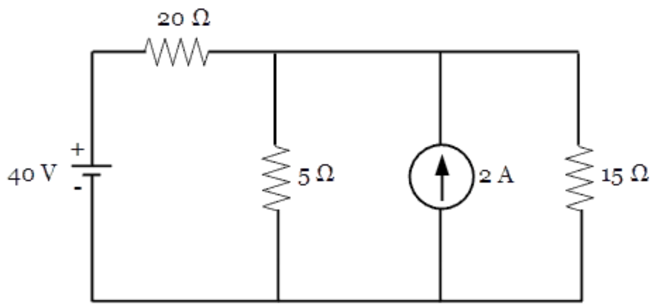
$$R_{TH} = 2 + \frac{1 * 3}{1 + 3} = 2.75\Omega$$

Thevenin's Equivalent Circuit

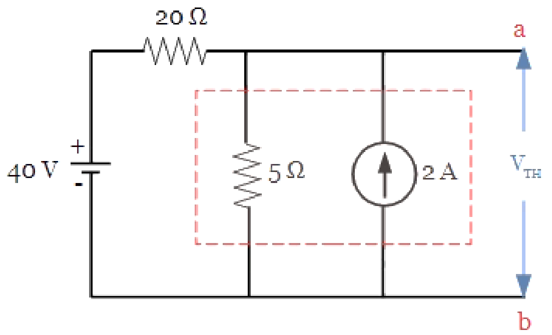


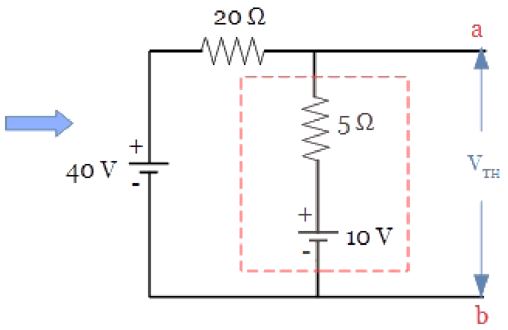
$$I_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{6.75}{2.75 + 10} = 0.529A$$

Find the current through 15 Ω using Thevenin's Theorem

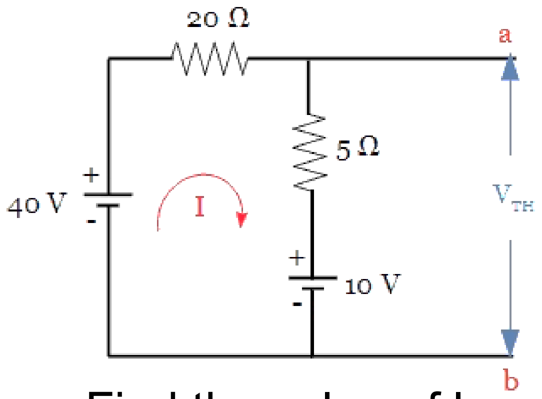


Find Thevenin's voltage





V = IR = 2 x 5 = 10 V



Find the value of loop current.

Applying KVL,

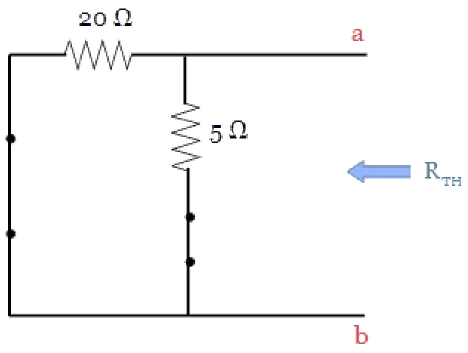
$20I + 5I + 10 - 40 = 0$

$25I = 30$

$I = 1.2\ A$

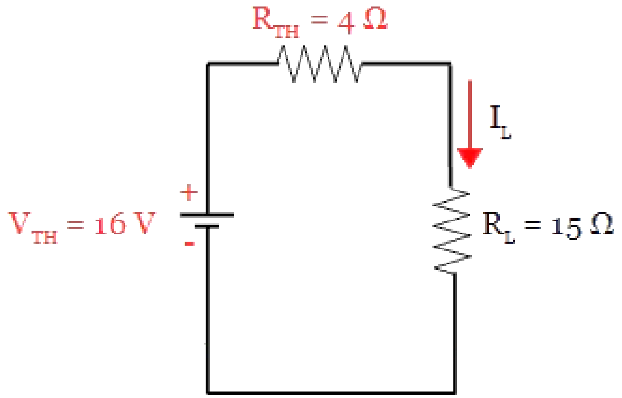
VTH = 10 + (5 * 1.2) = 16 v

Find Thevenin's resistance, Remove the load resistor, short the voltage source.



$$R_{TH} = \frac{5 * 20}{5 + 20} = 4\Omega$$

Final Thevenin's Equivalent Circuit.



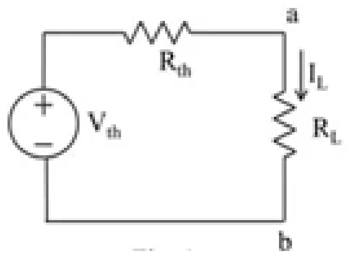
$$I_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{16}{4 + 15} = 0.842A$$

Maximum Power Transfer Theorem

The Maximum Power Transfer Theorem states:

A load will receive maximum power from a network when the load resistance R_L is equal to the Thevenin's resistance R_{th} of the network as seen from the load terminals.

Mathematically: $R_L = R_{th}$



Proof

1. Load Current

$$I = \frac{V_{th}}{R_{th} + R_L}$$

2. Power delivered to load

$$P_L = I^2 R_L = \left(\frac{V_{th}}{R_{th} + R_L}\right)^2 R_L$$

3. To maximize P_L, differentiate with respect to R_L and equate to zero:

$$\frac{dP_L}{dR_L} = 0$$

After solving, we get:

$$\boxed{R_L = R_{th}}$$

$$P_L = \frac{V^2 \cdot R_L}{(R_{th}+R_L)^2}$$

First, we compute:

$$\frac{dP}{dR_L} = \frac{V^2 \cdot \left[(R_{th}+R_L)^2 \cdot 1 - R_L \cdot 2(R_{th}+R_L) \cdot 1\right]}{(R_{th}+R_L)^4}.$$

Simplified:

$$\frac{dP}{dR_L} = \frac{V^2 \cdot (R_{th}-R_L)}{(R_{th}+R_L)^3}.$$

To find the maximum, set this to zero, and you get $R_L = R_{th}$.

Let's also check the second derivative to confirm the maximum.

We start by setting $R_L = R_{th}$, which satisfies the first derivative.

Maximum Power

Substituting $R_L = R_{th}$:

$$\boxed{P_{max} = \frac{V_{th}^2}{4R_{th}}}$$

Maximum power transfer occurs when:

$$R_L = R_{th}$$

Efficiency under maximum power transfer is only **50%**, because half the power is dissipated in R_{th}.

In practical systems (like power distribution), maximum efficiency is more important than maximum power transfer, so R_L is usually made much larger than R_{th}.

However, in communication systems, maximum power transfer is crucial for signal strength.

Numerical Example

A source has Thevenin's equivalent:

$$V_{th} = 20\,V, \quad R_{th} = 10\,\Omega$$

Find R_L for maximum power transfer and the maximum power.

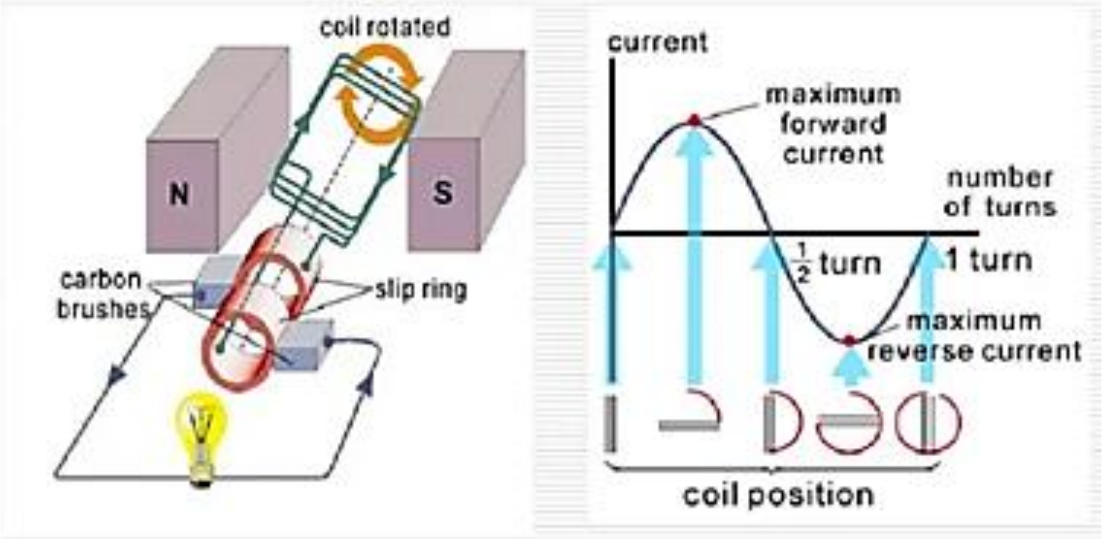
Condition: $R_L = R_{th} = 10\,\Omega$.

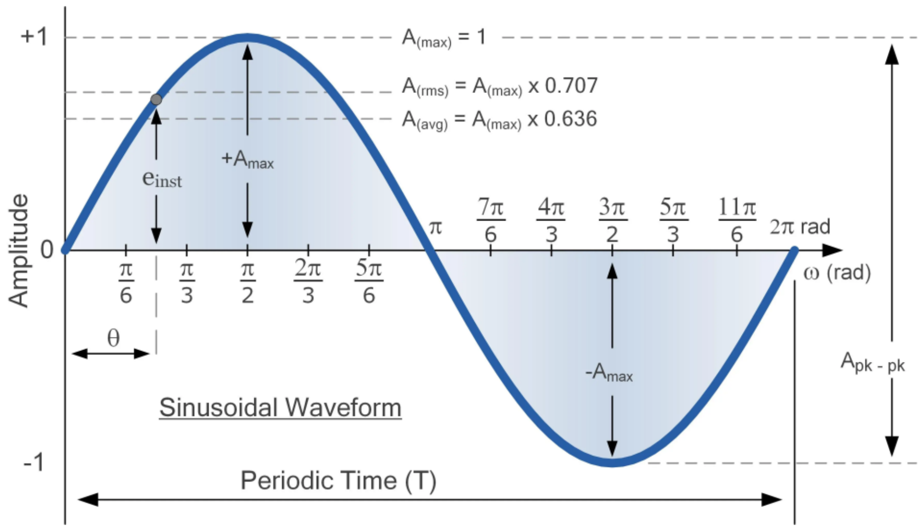
Maximum Power:

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{20^2}{4 \times 10} = \frac{400}{40} = 10\,W$$

AC Circuits

Single Phase AC Circuits	
2.1	Generation of alternating voltage, average value,RMS value, form factor, crest factor, phasor representation in rectangular and polar form.





Generation of Alternating Voltage:

Principle: Alternating voltage is generated when a coil rotates in a uniform magnetic field, based on Faraday's Law of Electromagnetic Induction.

Faraday's Law: $e = -N\dfrac{d\Phi}{dt}$

N= number of turns,
Φ= magnetic flux linkage.

Suppose a coil of N turns and area A rotates at an angular velocity ω in a magnetic field of flux density B. The flux linked with the coil at time t is:

$$\Phi(t) = NBA\cos(\omega t)$$

The induced emf is:

$$e(t) = -\frac{d\Phi}{dt} = NBA\omega\sin(\omega t)$$

Thus,

$$\boxed{e(t) = E_m\sin(\omega t)}$$

where $E_m = NBA\omega$ is the maximum value of emf.

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Average Value of AC

The average value over one half-cycle is:

$$V_{avg} = \frac{1}{\pi} \int_0^\pi V_m \sin(\theta)\, d\theta$$

$$V_{avg} = \frac{V_m}{\pi} [-\cos(\theta)]_0^\pi = \frac{2V_m}{\pi}$$

Therefore,

$$\boxed{V_{avg} = 0.637 V_m}$$

RMS (Root Mean Square) Value

The RMS value is defined as:

$$V_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (V_m \sin(\theta))^2 d\theta}$$

$$V_{rms} = V_m \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2(\theta) d\theta}$$

Using $\sin^2(\theta) = \frac{1}{2}(1 - \cos 2\theta)$,

$$V_{rms} = V_m \sqrt{\frac{1}{2\pi} \cdot \pi} = \frac{V_m}{\sqrt{2}}$$

Thus,

$$\boxed{V_{rms} = 0.707 V_m}$$

Form Factor

$$\text{Form Factor} = \frac{V_{rms}}{V_{avg}} = \frac{0.707V_m}{0.637V_m} = 1.11$$

Crest Factor

$$\text{Crest Factor} = \frac{V_m}{V_{rms}} = \frac{V_m}{0.707V_m} = 1.414$$

Phasor Representation of AC Signal

A sinusoidal quantity can be expressed as:

$$v(t) = V_m \sin(\omega t + \phi)$$

Here, V_m is the peak value, ωt represents angular position, and ϕ is the phase angle.

To simplify the analysis of AC circuits, sinusoidal quantities are represented by **phasors**. A phasor is a vector rotating counter-clockwise at angular velocity ω in the complex plane. The projection of this phasor on the vertical axis gives the instantaneous sinusoidal value.

Rectangular Form:

$$V = V_x + jV_y$$

where $V_x = |V|\cos\theta$ and $V_y = |V|\sin\theta$.

Polar Form:

$$V = |V|\angle\theta$$

where $|V| = \sqrt{V_x^2 + V_y^2}$ and $\theta = \tan^{-1}(V_y/V_x)$.

Exponential Form:

Using Euler's identity, a phasor can also be written as:

$$V = |V|e^{j\theta}$$

This representation is extremely useful for AC circuit analysis since differentiation and integration in the time doma[illegible]pond to simple multiplication/division by $j\omega$ in the phasor domain.

Examples

Q: A sinusoidal voltage has a maximum value of 200 V. Determine its RMS value, average value, form factor, and crest factor.

Solution:

Given $V_m = 200\ \text{V}$.

RMS value:

$$V_{rms} = \frac{200}{\sqrt{2}} = 141.4\ \text{V}$$

Average value:

$$V_{avg} = 0.637 \times 200 = 127.4\ \text{V}$$

Form Factor:

$$FF = \frac{141.4}{127.4} = 1.11$$

Crest Factor:

$$CF = \frac{200}{141.4} = 1.414$$

Steady-State Behavior of Single-Phase AC Circuits
Pure R, L, and C Elements

When a sinusoidal alternating voltage is applied to a circuit, the current also becomes sinusoidal.

Let the applied voltage be: $v(t) = V_m \sin(\omega t)$ where V_m is the peak voltage and $\omega = 2\pi f$ is the angular frequency.

The corresponding RMS (root-mean-square) voltage is: $V_{rms} = \frac{V_m}{\sqrt{2}}$

Similarly, for current: $i(t) = I_m \sin(\omega t + \phi)$ $I_{rms} = \frac{I_m}{\sqrt{2}}$

Purely Resistive Circuit (R)

Circuit: A resistor R connected to an AC voltage source.

Voltage and Current Relation:

$$v(t) = V_m \sin(\omega t)$$

$$i(t) = \frac{v(t)}{R} = \frac{V_m}{R}\sin(\omega t)$$

$$i(t) = I_m \sin(\omega t)$$

where $I_m = \frac{V_m}{R}$

Hence, **voltage and current are in phase.**

Phasor Representation:

$$V = IR$$

Phasor diagram → Voltage and current vectors are **in the same direction.**

Instantaneous Power:

$$p(t) = v(t)\cdot i(t) = V_m I_m \sin^2(\omega t)$$

$$p(t) = \frac{V_m I_m}{2}[1 - \cos(2\omega t)]$$

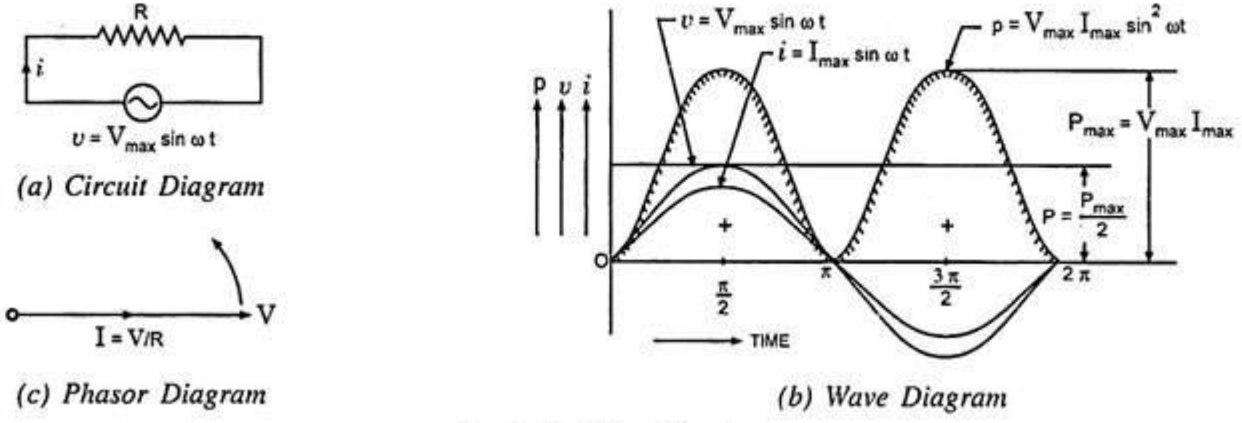
The power is **always positive**, meaning energy is continuously dissipated as heat.

Average Power:

$$P_{avg} = \frac{V_m I_m}{2} = V_{rms} I_{rms}$$

and

Power Factor (pf) = 1 (i.e., unity power factor).



Purely Resistive Circuit

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Inductors L (Hendry)

Somaiya TRUST

An **inductor** is a two-terminal **passive component** that **stores energy in a magnetic field** when current flows through it. It **opposes changes in current**, not steady current.

Air-Core Inductor

Iron-Core Inductor

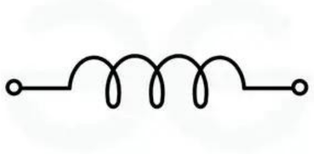
Ferrite-Core Inductor

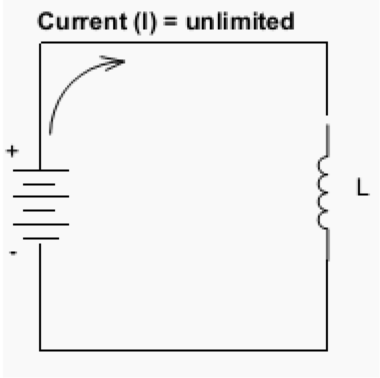
Bobbin Based Inductor

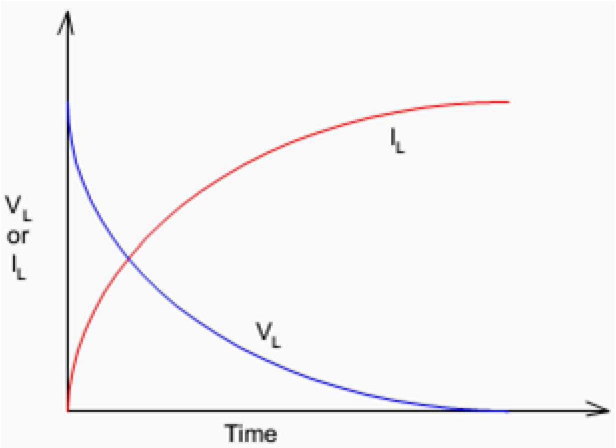
Variable Inductor

Multilayer Ceramic Inductor

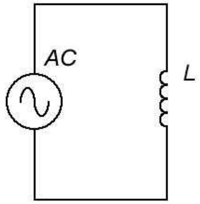
Inductor Symbol







Inductive Reactance



$$X_L = \omega L$$
$$= 2\pi f L$$

Purely Inductive Circuit (L only)

Circuit: A pure inductor L connected to AC supply.

Voltage:

$$v(t) = V_m \sin(\omega t)$$

Current:

For inductor,

$$v(t) = L\frac{di}{dt}$$

$$\Rightarrow i(t) = \frac{1}{L}\int v(t)dt = -\frac{V_m}{\omega L}\cos(\omega t)$$

$$i(t) = \frac{V_m}{\omega L}\sin(\omega t - 90^\circ)$$

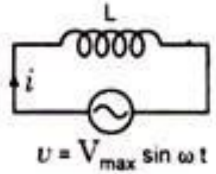
$$i(t) = I_m \sin(\omega t - 90^\circ)$$

Thus, **current lags voltage by 90°**.

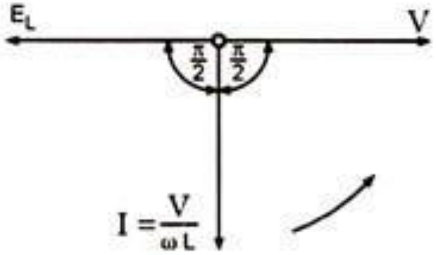
Inductive Reactance:

$$X_L = \omega L$$

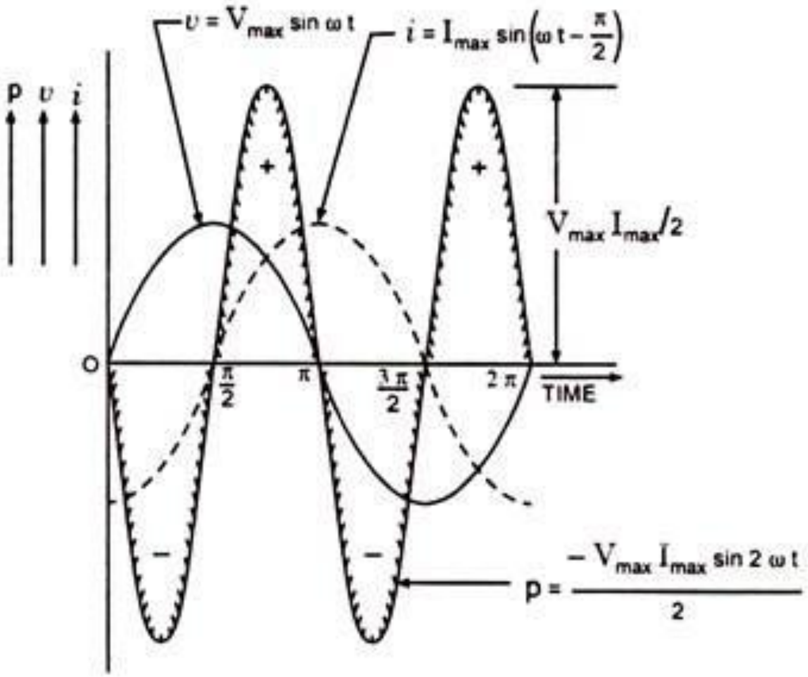
Unit: ohm (Ω)



(a) Circuit Diagram



(c) Phasor Diagram



(b) Wave Diagram

Purely Inductive Circuit (L only)

Phasor Diagram:

Voltage vector **leads** the current vector by 90°.

Instantaneous Power:

$$p(t) = v(t) \cdot i(t) = V_m I_m \sin(\omega t) \sin(\omega t - 90^\circ)$$

$$p(t) = \frac{V_m I_m}{2} [\cos(90^\circ) - \cos(2\omega t - 90^\circ)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\boxed{p(t) = -\frac{V_m I_m}{2} \sin(2\omega t)}$$

Power alternates equally between positive and negative, so **average power = 0**.

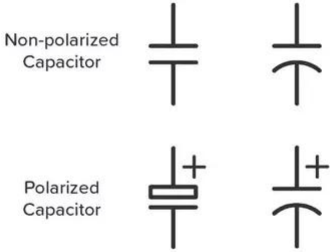
Power Factor:

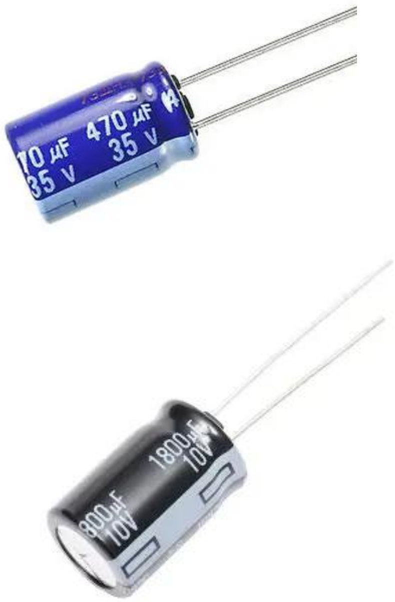
$$\cos\phi = \cos(90^\circ) = 0$$

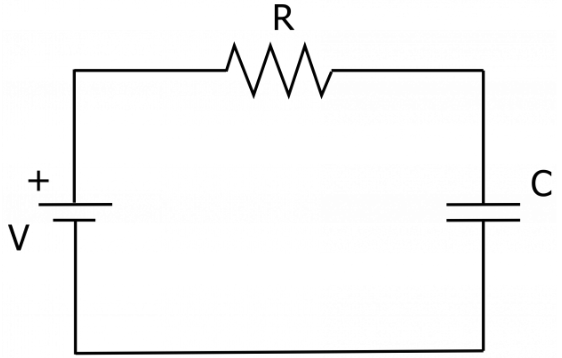
Hence, **no real power is consumed**, only reactive power oscillates.

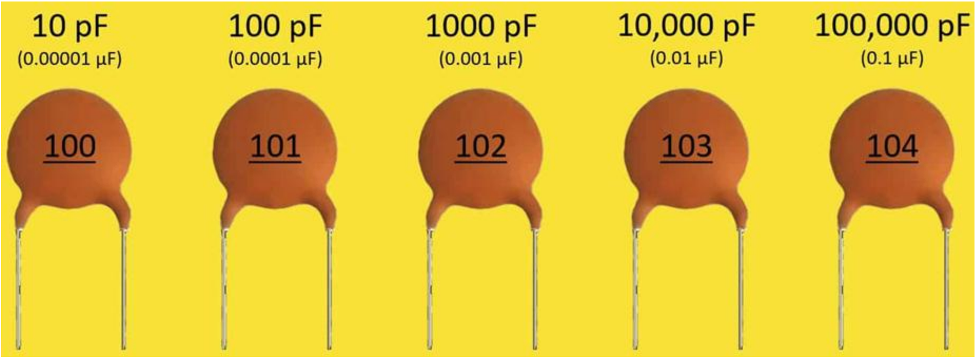
Capacitor Types C (Faraday)

A **capacitor** is a two-terminal **passive electrical component** that **stores energy in an electric field** between its plates due to separation of charge. It **opposes changes in voltage**.

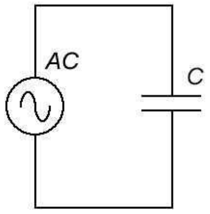




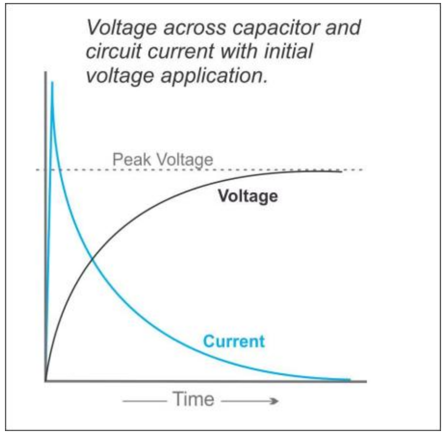




Capacitive Reactance



$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

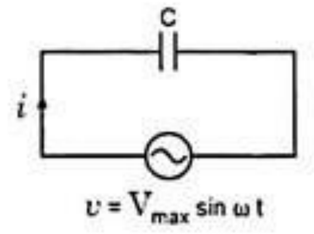


Purely Capacitive Circuit (C only)

Circuit: A pure capacitor C connected to AC supply.

Voltage:

$$v(t) = V_m \sin(\omega t)$$



(a) Circuit Diagram

Current:

For capacitor,

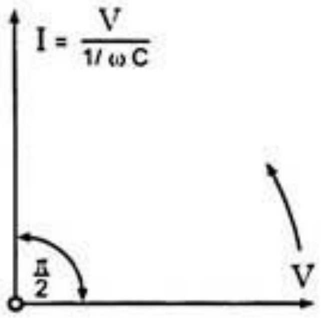
$$i(t) = C\frac{dv}{dt}$$

$$i(t) = C\omega V_m \cos(\omega t)$$

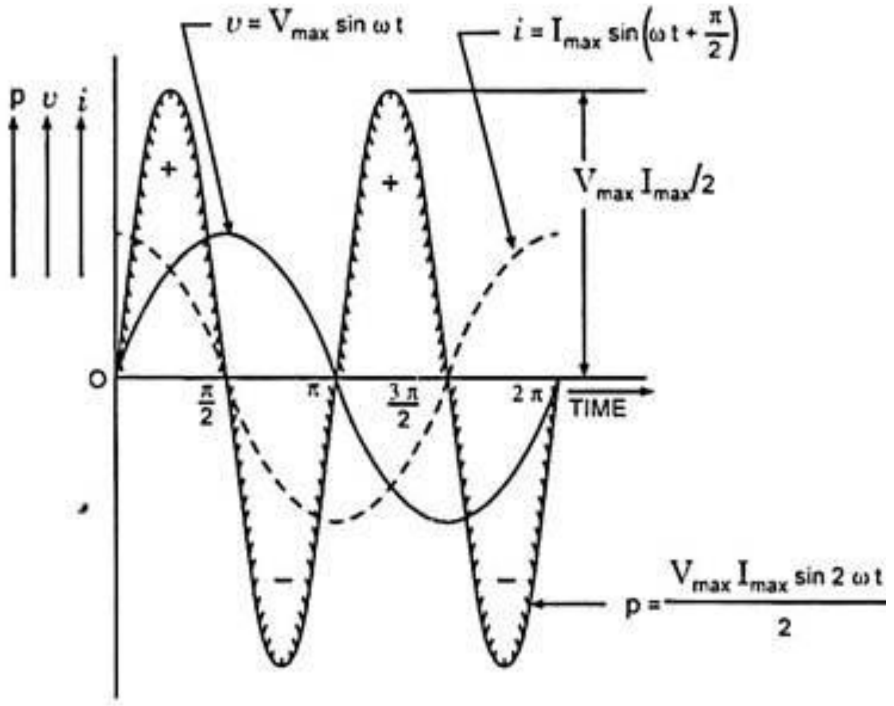
$$i(t) = I_m \sin(\omega t + 90^\circ)$$

where $I_m = \omega C V_m$

Thus, **current leads voltage by 90°**.



(c) Phasor Diagram



(b) Wave Diagram

Capacitive Reactance:

$$X_C = \frac{1}{\omega C}$$

Unit: ohm (Ω)

Purely Capacitive Circuit (C only)

Phasor Diagram:

Current vector **leads** the voltage vector by 90°.

Instantaneous Power:

$$p(t) = v(t) \cdot i(t) = V_m I_m \sin(\omega t) \sin(\omega t + 90^\circ)$$

$$p(t) = \frac{V_m I_m}{2} [\cos(-90^\circ) - \cos(2\omega t + 90^\circ)]$$

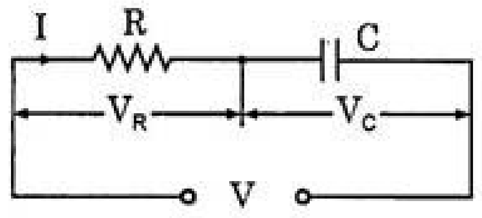
$$\boxed{p(t) = \frac{V_m I_m}{2} \sin(2\omega t)}$$

Power alternates equally in positive and negative halves, so **average power = 0**.

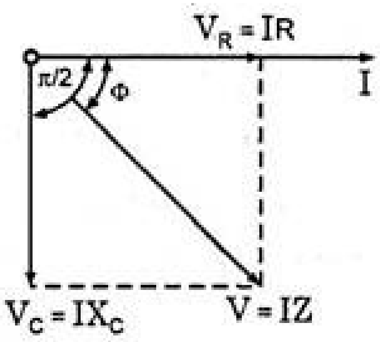
Power Factor:

$$\cos\phi = \cos(90^\circ) = 0$$

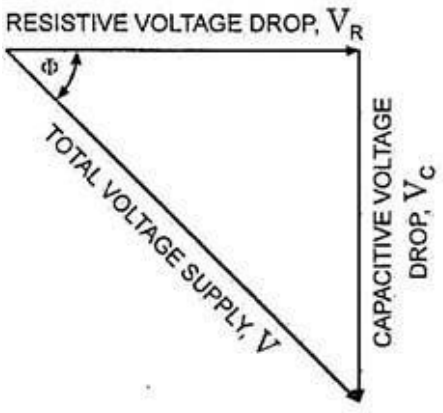
Resistance - Capacitance (R-C) Series Circuit:



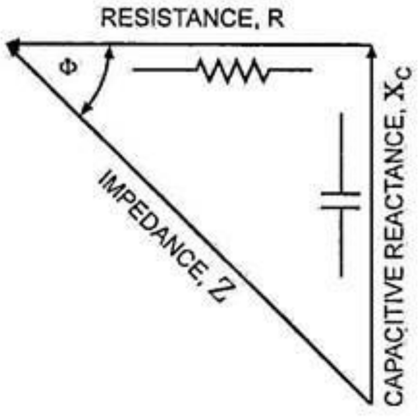
(a) Circuit Diagram



(b) Phasor Diagram



(a) Voltage Triangle



(b) Impedance Triangle

$$V = \sqrt{(V_R)^2 + (V_C)^2} = \sqrt{(IR)^2 + (IX_C)^2} = I\sqrt{R^2 + X_C^2} = IZ$$

where $Z^2 = R^2 + X_C^2$

$$X_C = \frac{1}{\omega C}$$

where $\tan\Phi = \frac{V_C}{V_R} = \frac{IX_C}{IR} = \frac{X_C}{R} = \frac{1}{\omega RC}$ or $\Phi = \mathrm{Tan}^{-1}\ \frac{1}{R\omega C}$

Power factor, $\cos\Phi = \frac{R}{Z}$

Apparent Power, True Power, Reactive Power and Power Factor:

If instantaneous voltage is represented by:

$v = V_{max} \sin \omega t$

Then instantaneous current will be expressed as:

$i = I_{max} \sin (\omega t + \phi)$

And power consumed by the circuit is given by:

$P = VI \cos \phi$

Apparent power, $S = VI$ volt-amperes

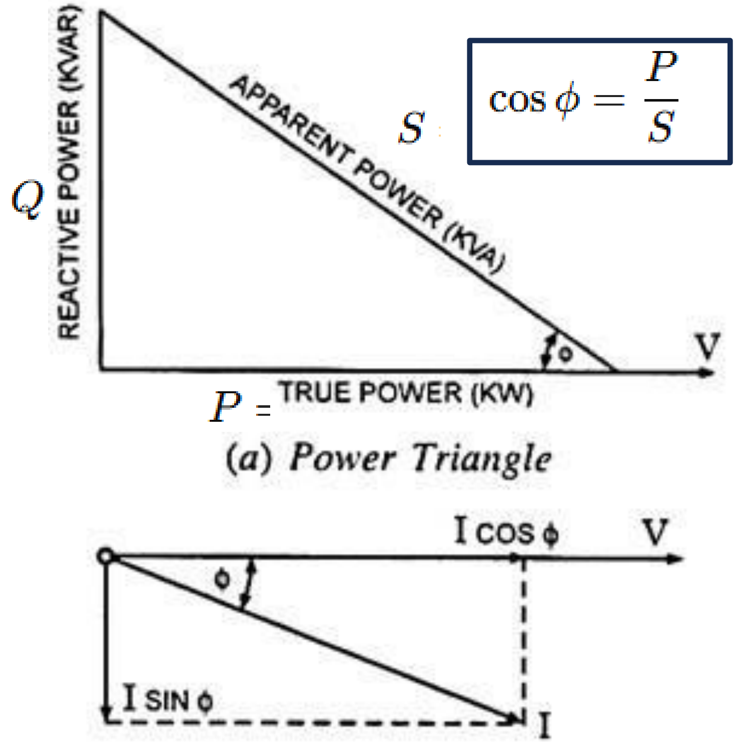
True power, $P = VI \cos \Phi$ watts

Reactive power, $Q = VI \sin \Phi$ VAR

and $kVA = \sqrt{(kW)^2 + (kVAR)^2}$

$$S^2 = P^2 + Q^2$$

Apparent Power, True Power, Reactive Power and Power Factor:



$$\boxed{\cos\phi = \frac{P}{S}}$$

(a) *Power Triangle*

(b)
Current Triangle

Power factor may be defined as:

(i) Cosine of the phase angle between voltage and current,

(ii) The ratio of the resistance to impedance, or

(iii) The ratio of true power to apparent power.

The power factor can never be greater than unity. The power factor is expressed either as fraction or as a percentage. It is usual practice to attach the word ‘lagging’ or ‘ leading’ with the numerical value of power factor to signify whether the current lags behind or leads the voltage.

R–L Series Circuit

A resistor R and inductor L connected in series to an AC source

$$v(t) = V_m \sin(\omega t)$$

Current

$$i(t) = I_m \sin(\omega t - \phi)$$

where

$$\boxed{\tan\phi = \frac{X_L}{R},} \quad \boxed{X_L = \omega L}$$

Hence, current **lags** the voltage by angle ϕ.

Impedance

$$\boxed{Z = \sqrt{R^2 + X_L^2}}$$

Phasor Diagram

Voltage V is the vector sum of:

- $V_R = IR$ (in phase with I)
- $V_L = IX_L$ (leads I by 90°)

Resultant voltage:

$$\boxed{V = \sqrt{V_R^2 + V_L^2}}$$

Power

$$\boxed{P = VI\cos\phi}$$

- $\cos\phi$ → **Power Factor (lagging)**
- Active power (W) = $VI\cos\phi$
- Reactive power (VAR) = $VI\sin\phi$
- Apparent power (VA) = VI

Concept of Impedance (Z) and Phasor Relationship

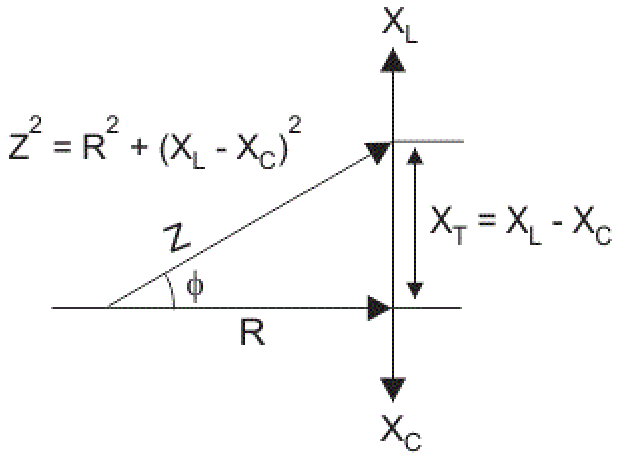
For a general R–L–C series circuit,

$$Z = R + j(X_L - X_C)$$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$$

and

$$\boxed{\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)}$$



Voltage and Current Relationship:

$$\boxed{V = IZ}$$

Current lags voltage if $X_L > X_C$ (inductive circuit),

Current leads voltage if $X_C > X_L$ (capacitive circuit).

Phasor Diagram:

- I is taken as reference.
- $V_R = IR$ (in phase with I)
- $V_L = IX_L$ (leads I by 90°)
- $V_C = IX_C$ (lags I by 90°)
- Resultant voltage V is the phasor sum of these voltages.

Resonance in AC Circuits

Resonance can occur in two types of RLC circuits:

1. **Series Resonance** — when R, L, and C are connected **in series**.
2. **Parallel Resonance** — when L and C are connected **in parallel**.

Resonance in an AC circuit occurs when the inductive reactance and capacitive reactance are **equal in magnitude** but **opposite in phase**, resulting in a purely resistive circuit.

In this condition, the **reactive effects cancel each other**, and the **circuit current and supply voltage are in phase**.

Mathematically:

$$X_L = X_C$$

or

$$\boxed{\omega L = \frac{1}{\omega C}}$$

At this condition, the **net reactance = 0**, and the **impedance** of the circuit is minimum (in series) or maximum (in parallel).

Resonance in AC Circuits

Series Resonance:

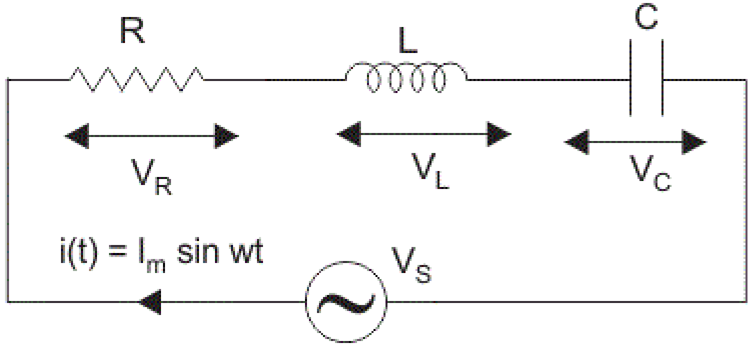
Series resonance occurs in an R–L–C series circuit when the **inductive reactance (X_l)** equals the **capacitive reactance (X_c)**, causing the **impedance to be minimum** and the **circuit current to be maximum**.

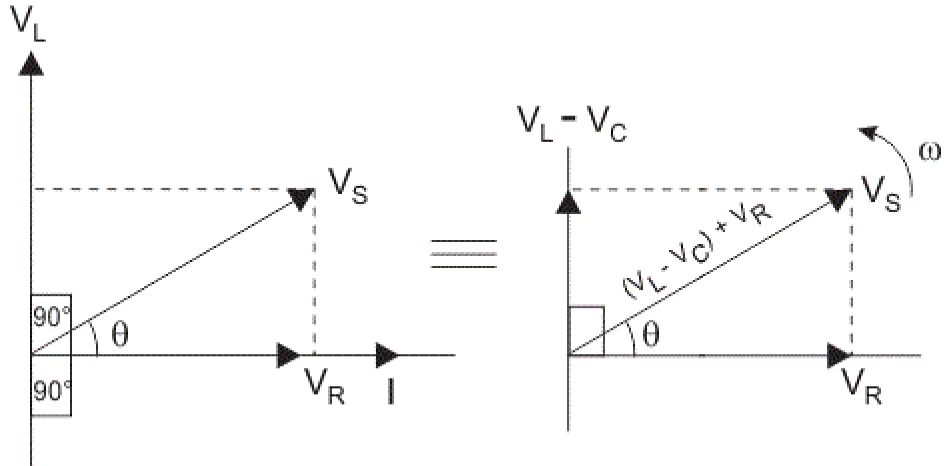
$$\boxed{X_L = X_C}$$

$$\omega_r L = \frac{1}{\omega_r C}$$

$$\therefore \omega_r = \frac{1}{\sqrt{LC}}$$

$$\boxed{f_r = \frac{1}{2\pi\sqrt{LC}}}$$





$$V_S = I\sqrt{R^2 + (X_L - X_C)^2}$$

At Resonance:

- Impedance $Z = R$ (minimum)
- Current $I = \frac{V}{R}$ (maximum)
- Voltage and current are in phase
- Power factor = 1 (unity)
- Reactive powers of L and C cancel each other

Resonance in AC Circuits

Q-Factor (Quality Factor):

Q-factor represents the **sharpness of resonance** or the **selectivity** of the circuit.
It is the ratio of **reactive power** to **real power** at resonance.

$$\boxed{Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r C R}}$$

Higher Q → Sharper resonance.

Bandwidth (BW):

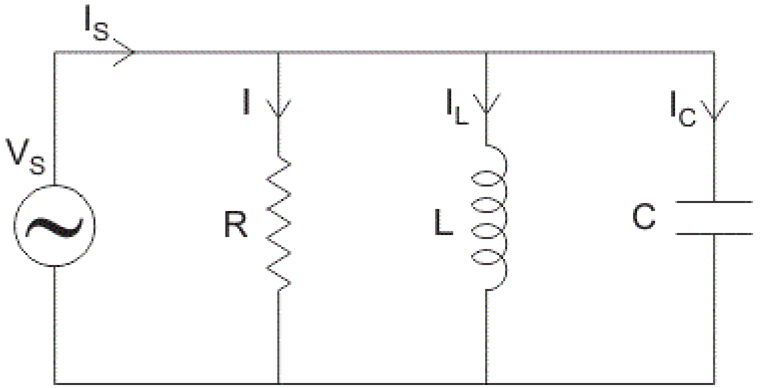
$$\boxed{BW = f_2 - f_1 = \frac{f_r}{Q}}$$

where f_1 and f_2 are the frequencies at which the current drops to 70.7% (1/√2) of the maximum value.

Resonance in AC Circuits

Parallel Resonance:

Parallel resonance occurs in an R–L–C parallel circuit when the **inductive current** and **capacitive current** are equal and opposite, making the **net current minimum** and **circuit impedance maximum**



Resonant Condition:

For practical circuit (L has internal R):

$$\omega_r = \frac{1}{\sqrt{LC}}$$

Q-Factor:

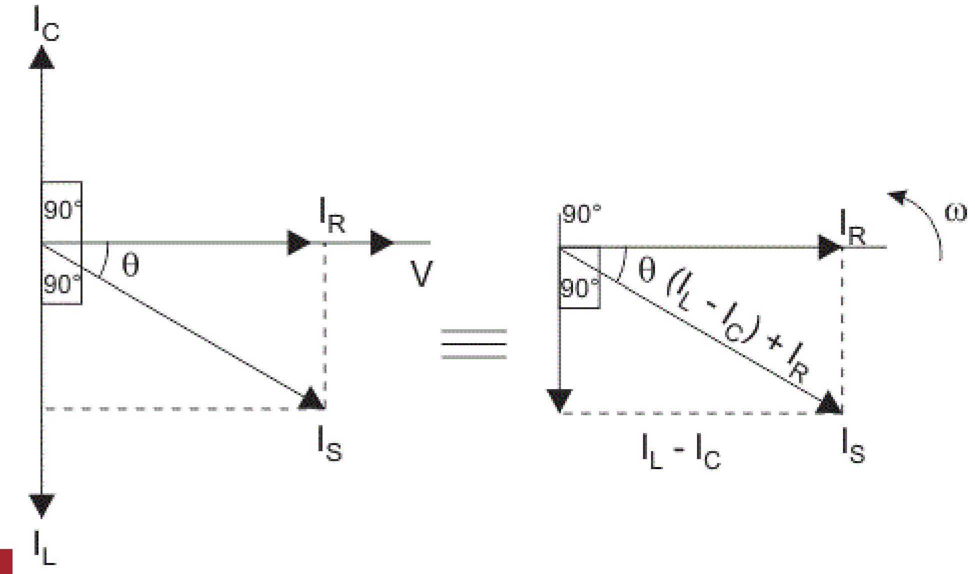
$$Q = R\sqrt{\frac{C}{L}}$$

Bandwidth:

$$BW = \frac{f_r}{Q}$$

At Resonance:

Line current is minimum
Impedance is maximum
Power factor = 1 (unity)
Current through L and C are large and opposite, canceling each other



Resonance in AC Circuits

Comparison of Series and Parallel Resonance

Parameter	Series Resonance Circuit	Parallel Resonance Circuit
1. Definition	Resonance occurs when inductive reactance equals capacitive reactance in a **series R–L–C circuit.**	Resonance occurs when inductive current equals capacitive current in a **parallel R–L–C circuit.**
2. Resonant Condition	$X_L = X_C \Rightarrow \omega_r L = \frac{1}{\omega_r C}$	$I_L = I_C \Rightarrow \omega_r = \frac{1}{\sqrt{LC}}$ (approximately)
3. Impedance (Z)	Minimum, $Z = R$	Maximum, $Z = \frac{L}{CR}$ (approx.)
4. Current at Resonance	Maximum $I = \frac{V}{R}$	Minimum (only small line current flows)
5. Circuit Nature at Resonance	Purely resistive	Purely resistive
6. Phase Relation	Voltage and current are **in phase**	Supply voltage and current are **in phase**
7. Power Factor	Unity (1.0)	Unity (1.0)

8. Behavior with Frequency	Current is **maximum** at resonant frequency	Current is **minimum** at resonant frequency
9. Voltage Across L and C	Large and opposite in phase; cancel each other	Large circulating currents flow through L and C, canceling each other
10. Q-Factor (Quality Factor)	$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r CR}$	$Q = R\sqrt{\frac{C}{L}}$
11. Bandwidth	$BW = \frac{f_r}{Q} = f_2 - f_1$	$BW = \frac{f_r}{Q} = f_2 - f_1$
12. Impedance–Frequency Curve	Minimum impedance at resonance	Maximum impedance at resonance
13. Current–Frequency Curve	Peak (maximum current) at resonance	Dip (minimum current) at resonance
14. Nature of Circuit Above Resonance	Inductive ($X_L > X_C$)	Capacitive
15. Nature of Circuit Below Resonance	Capacitive ($X_C > X_L$)	Inductive
16. Applications	Used in **tuned amplifiers**, **filters**, and **radio receivers** (to select frequency).	Used in **oscillators**, **impedance matching networks**, and **filter circuits**.

Problem 1 — R–L Series Circuit

Problem statement

A series circuit has $R = 50\ \Omega$ and $L = 0.100\ \mathrm{H}$. It is fed from a sinusoidal source of $V_{rms} = 230\ \mathrm{V}$ at frequency $f = 50\ \mathrm{Hz}$.

Find: $X_L,\ Z,\ I_{rms},\ \phi,\ P,\ Q,\ S$ (active, reactive and apparent powers). State the phase relation between voltage and current.

Solution

1. Angular frequency:

$$\omega = 2\pi f = 2\pi(50) = 100\pi\ \mathrm{rad/s} \approx 314.15927\ \mathrm{rad/s}.$$

2. Inductive reactance:

$$X_L = \omega L = 314.15927 \times 0.100 = 31.415927\ \Omega.$$

3. Impedance magnitude (series):

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{50^2 + 31.415927^2} = \sqrt{2500 + 985.96044} = \sqrt{3485.96044} \approx 59.050491\ \Omega.$$

4. RMS current:

$$I_{rms} = \frac{V_{rms}}{Z} = \frac{230}{59.050491} \approx 3.894972 \text{ A}.$$

5. Phase angle (voltage leads current for inductive circuit):

$$\phi = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\left(\frac{31.415927}{50}\right) = \tan^{-1}(0.6283185) \approx 32.1419^\circ.$$

So **current lags voltage by $\mathbf{32.14}^\circ$**.

6. Active (real) power:

   Use $P = I_{rms}^2 R$ for accuracy:

$$P = (3.894972)^2 \times 50 \approx 15.1719 \times 50 = 758.540 \text{ W}.$$

7. Apparent power:

$$S = V_{rms} I_{rms} = 230 \times 3.894972 \approx 895.844 \text{ VA}.$$

8. Reactive power:

$$Q = S \sin\phi = 895.844 \times \sin(32.1419^\circ)$$

$$\sin(32.1419^\circ) \approx 0.53198 \quad \Rightarrow \quad Q \approx 895.844 \times 0.53198 \approx 476.605 \text{ VAR}.$$

Problem 2 — R–C Series Circuit

Problem statement

A series circuit has $R = 200\ \Omega$ and $C = 10.0\ \mu\text{F}$. It is supplied from $V_{rms} = 230\ \text{V}$ at $f = 50\ \text{Hz}$. Find: $X_C,\ Z,\ I_{rms},\ V_C,\ \phi$ (phase between current and supply voltage), $P,\ Q,\ S$.

1. Angular frequency:

$$\omega = 2\pi f = 2\pi(50) = 100\pi \approx 314.159\ \text{rad/s}.$$

2. Capacitive reactance:

$$X_C = \frac{1}{\omega C} = \frac{1}{314.159 \times 10 \times 10^{-6}} = \frac{1}{0.00314159} \approx 318.310\ \Omega.$$

3. Impedance magnitude (series):
   Impedance in phasor form $Z = R - jX_C$.

$$|Z| = \sqrt{R^2 + X_C^2} = \sqrt{200^2 + 318.31^2} = \sqrt{40000 + 101321.18} \approx 375.941\ \Omega.$$

4. RMS current:

$$I_{rms} = \frac{V_{rms}}{|Z|} = \frac{230}{375.941} \approx 0.6123\ \text{A}.$$

5. Voltage across capacitor (rms):

$$V_C = I_{rms}\, X_C \approx 0.6123 \times 318.310 \approx 194.86\ \mathrm{V}.$$

6. Phase angle:

For series RC, impedance angle $\theta_Z = -\tan^{-1}(\frac{X_C}{R})$. Current leads supply voltage by $\phi = |\theta_Z|$ where

$$\phi = \tan^{-1}\left(\frac{X_C}{R}\right) = \tan^{-1}\left(\frac{318.31}{200}\right) \approx 57.75^\circ.$$

So **current leads voltage by** 57.75°.

Powers:

Apparent power: $S = V_{rms} I_{rms} = 230 \times 0.6123 \approx 140.83\ \mathrm{VA}.$

Active power: $P = I_{rms}^2 R = 0.6123^2 \times 200 \approx 74.98\ \mathrm{W}.$

Reactive power (capacitive, sign convention negative if desired): $Q = I_{rms}^2 X_C \approx 0.6123^2 \times 318.31 \approx 119.31\ \mathrm{VAR}.$

Problem on Series Resonance in AC Circuits

Given: A series R–L–C circuit with

$R = 20\ \Omega,\ L = 50.0\ \mathrm{mH},\ C = 10.0\ \mu\mathrm{F}.$

Supply: $V_{rms} = 120\ \mathrm{V}.$

Find: Resonant frequency f_r, quality factor Q, bandwidth BW, line current at resonance I_r, impedance at resonance, voltages across L and C at resonance, active/reactive/apparent powers, and the half-power frequencies f_1, f_2.

1. Convert units:

$$L = 50 \times 10^{-3}\ \mathrm{H}, \qquad C = 10 \times 10^{-6}\ \mathrm{F}.$$

2. Resonant angular frequency and frequency:

$$\omega_r = \frac{1}{\sqrt{LC}}, \qquad f_r = \frac{1}{2\pi\sqrt{LC}}.$$

3. Q-factor for series RLC:

$$Q = \frac{\omega_r L}{R}.$$

$$Q \approx \frac{1414.214 \times 0.05}{20} \approx 3.5355.$$

Resonance in AC Circuits

4. Bandwidth:

$$BW = \frac{f_r}{Q} \approx \frac{225.079}{3.5355} \approx 63.662 \text{ Hz.}$$

5. At resonance (series):

- Net reactance = 0 so impedance $Z = R$ (minimum).
- RMS line current:

$$I_r = \frac{V_{rms}}{Z} = \frac{120}{20} = 6.0 \text{ A.}$$

6. Reactances of L and C at ω_r:

$$X_L = \omega_r L \approx 70.711 \; \Omega, \qquad X_C = \frac{1}{\omega_r C} \approx 70.711 \; \Omega$$

7. Voltages across L and C (rms):

$$V_L = I_r X_L \approx 6 \times 70.711 \approx 424.264 \text{ V},$$

$$V_C = I_r X_C \approx 424.264 \text{ V}.$$

8. Powers:

Active (real) power absorbed by R:

$$P = I_r^2 R = 6^2 \times 20 = 720 \text{ W}.$$

Apparent power:

$$S = V_{rms} I_r = 120 \times 6 = 720 \text{ VA}.$$

Net reactive power of the whole series circuit at resonance:

$$Q_{\text{net}} = 0 \quad (\text{because } X_L = X_C).$$

9. Half-power frequencies (f_1, f_2):

$$f_{1,2} = f_r \mp \frac{BW}{2}$$

Compute:

$$f_1 \approx 193.248 \text{ Hz}, \qquad f_2 \approx 256.910 \text{ Hz}.$$

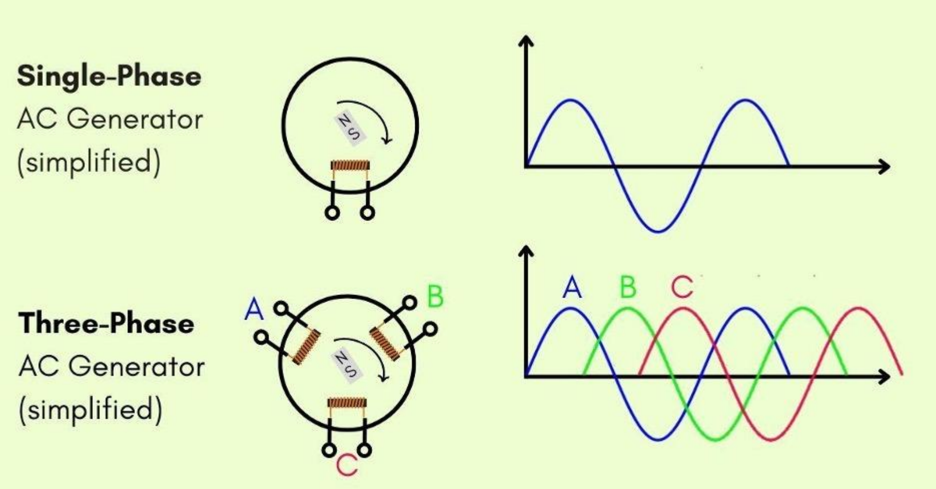
3 - Phase AC Circuits

A **three-phase system** consists of **three alternating voltages (or EMFs)** of **equal magnitude** and **frequency**, but each displaced in **phase by 120°** from the others.

These voltages are generated simultaneously by a **three-phase alternator**, which has three coils spaced 120° apart.

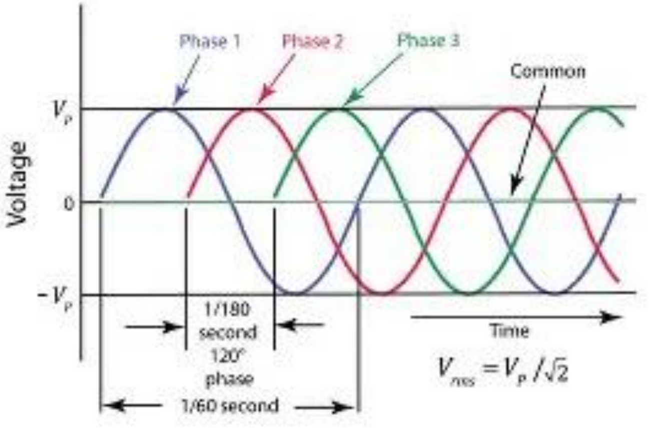
Let the three phase voltages be:

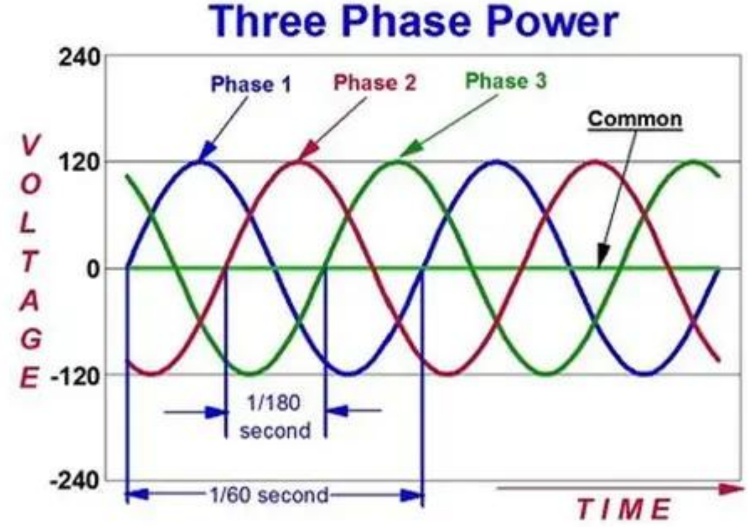
$$v_R = V_m \sin(\omega t)$$
$$v_Y = V_m \sin(\omega t - 120^\circ)$$
$$v_B = V_m \sin(\omega t - 240^\circ)$$



Advantages of Three-Phase System

1. Power delivered is **constant** (not pulsating as in single-phase).
2. For same power, **conductor material** required is less.
3. **Higher efficiency** and **smaller machine size**.
4. **Self-starting torque** in motors.
5. Easy to generate and transmit.





3 - Phase AC Circuits

Types of 3-Phase Connections:

Two common types of connections:

1. Star (Y) connection

2. Delta (Δ) connection

Star (Y) Connection:

In a star connection, one end of each of the three phase windings is connected to a **common neutral point (N)**.

Phase and Line Quantities

Let:

V_{ph} =phase voltage (voltage across each phase winding)

V_L =line voltage (voltage between any two lines)

I_{ph} =phase current

I_L =line current

Relations in Star (Y):

$$V_L = \sqrt{3}V_{ph}$$

$$I_L = I_{ph}$$

and **line voltages are 120° apart.**

Phasor Derivation

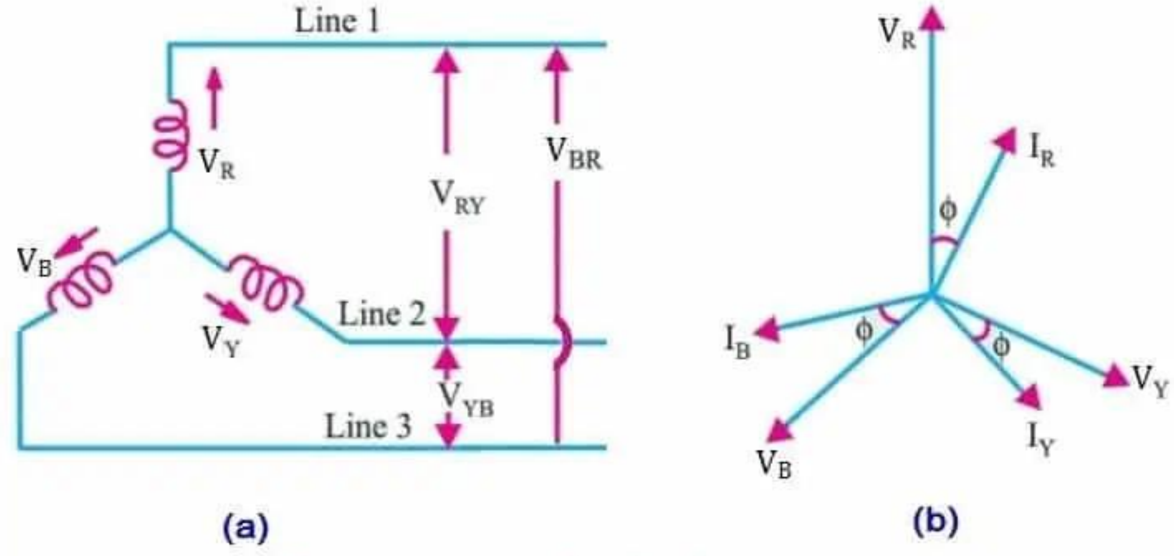
Each phase voltage is 120° apart.

Using vector subtraction:

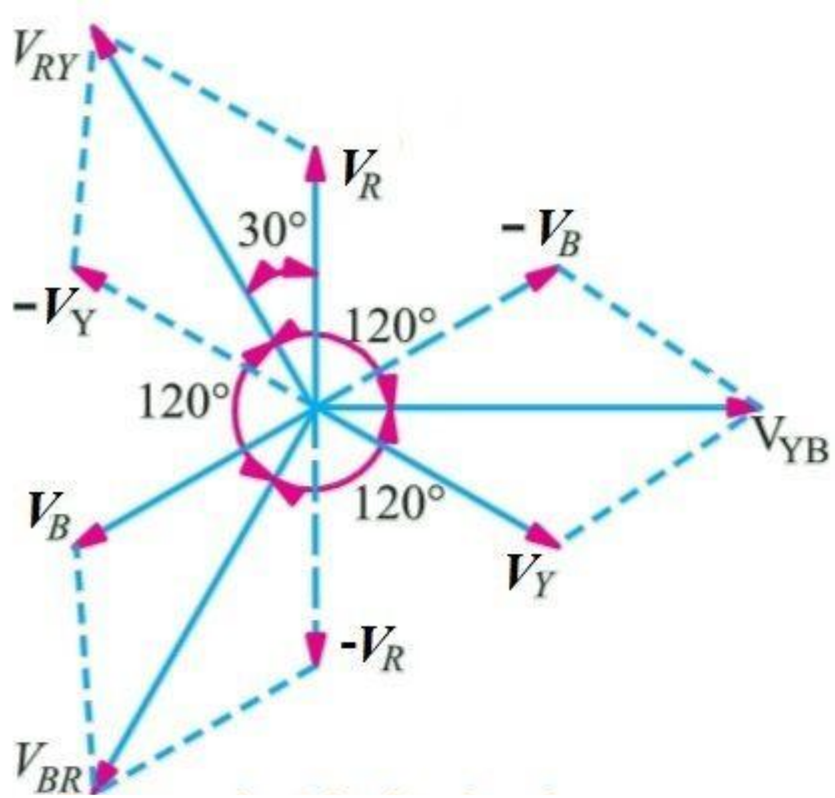
$$V_{RY} = V_R - V_Y$$

$$|V_{RY}| = \sqrt{3}V_{ph}$$

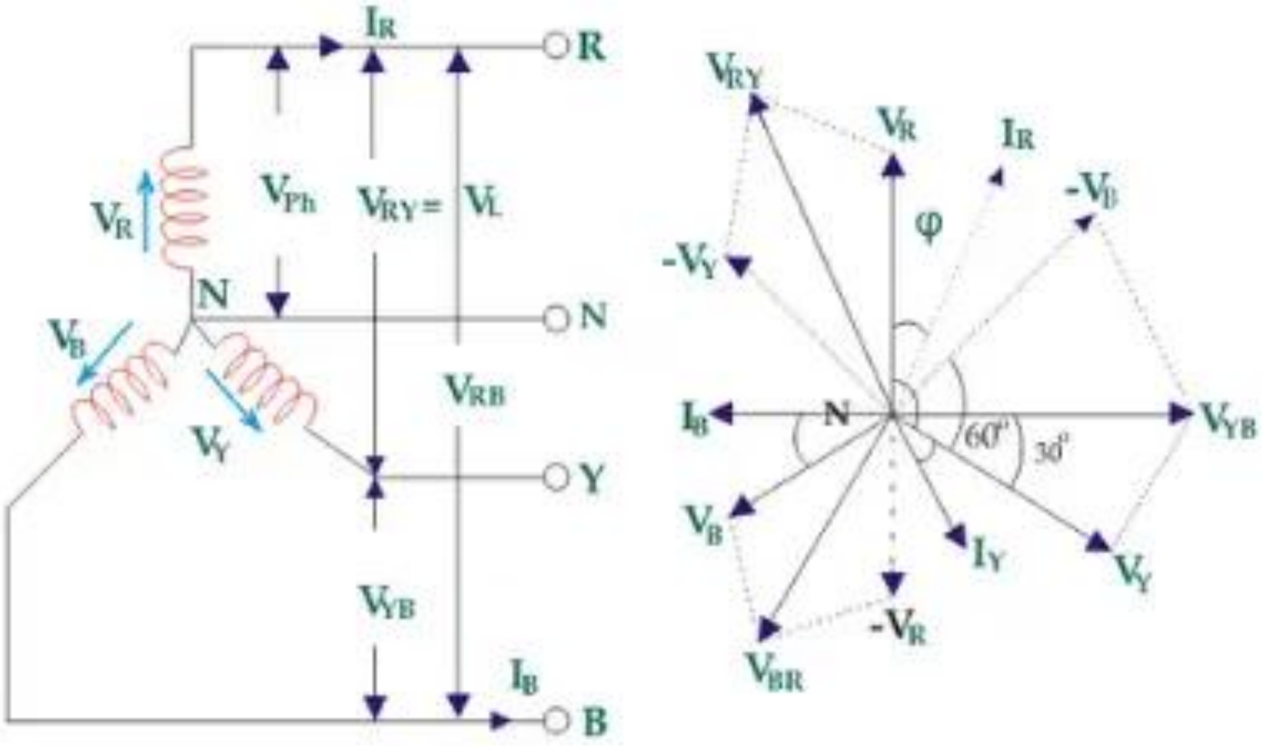
and V_{RY} leads V_R by 30°.

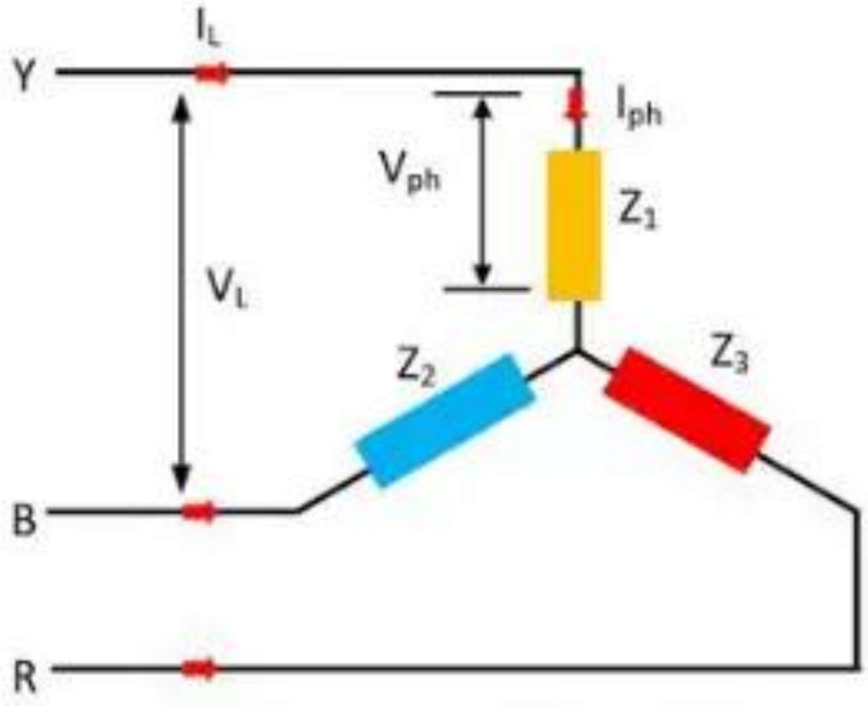


Star (Y) Connection: Phasor Diagram



Star (Y) Connection: Phasor Diagram:





3 - Phase AC Circuits

Delta (Δ) Connection

In delta connection, the **end of one winding** is connected to the **start of the next**, forming a closed loop (Δ shape). The three junctions form the line terminals.

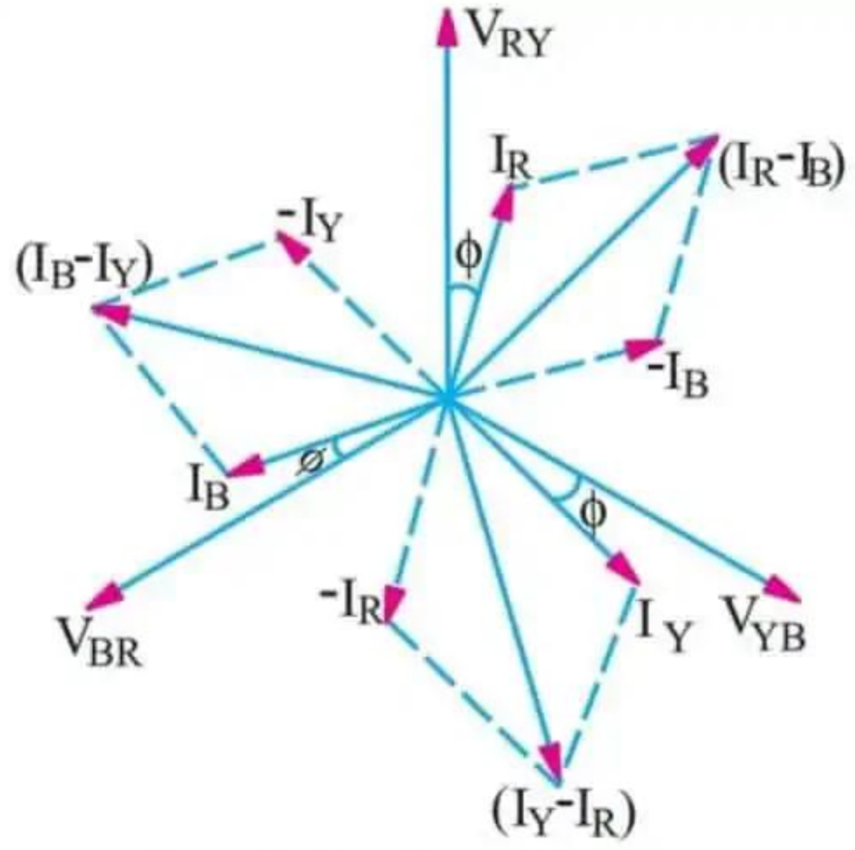
The line current leads its phase current by 30∘30∘

Relations in Delta (Δ):

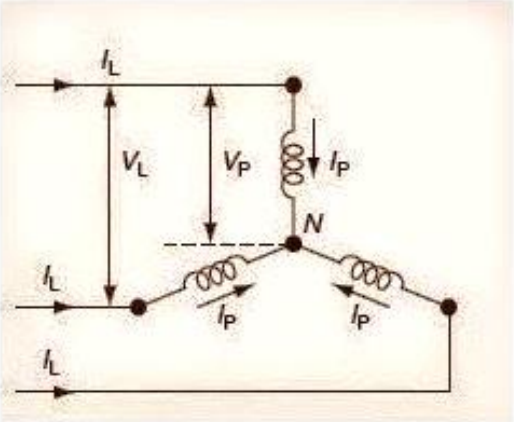
$$V_L = V_{ph}$$

$$I_L = \sqrt{3} I_{ph}$$

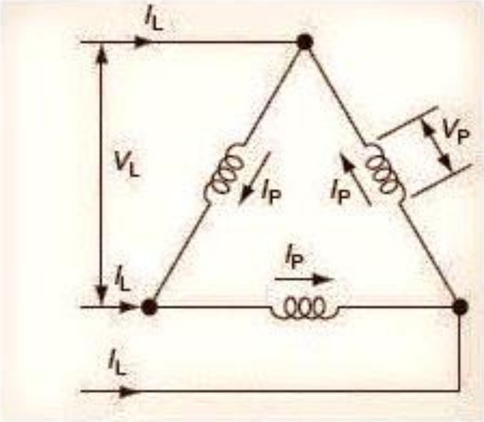
and **line currents are 120° apart**.



Star Connection



Delta Connection



Connection	Phase Voltage	Line Voltage	Phase Current	Line Current
Star	$V_P = V_L \div \sqrt{3}$	$V_L = \sqrt{3} \times V_P$	$I_P = I_L$	$I_L = I_P$
Delta	$V_P = V_L$	$V_L = V_P$	$I_P = I_L \div \sqrt{3}$	$I_L = \sqrt{3} \times I_P$

3 - Phase AC Circuits

Comparison of Star and Delta Connections

Parameter	Star (Y)	Delta (Δ)
Line Voltage & Phase Voltage	$V_L = \sqrt{3}V_{ph}$	$V_L = V_{ph}$
Line Current & Phase Current	$I_L = I_{ph}$	$I_L = \sqrt{3}I_{ph}$
Neutral Point	Exists	Does not exist
Insulation required	Less (phase voltage = $V_L/\sqrt{3}$)	More (phase voltage = V_L)
Applications	Transmission, distribution systems	Motors and low-voltage loads

3 - Phase AC Circuits

Balanced Three-Phase Circuits:

A **balanced system** means:

All three impedances are **equal in magnitude and phase angle**, i.e., $Z_R = Z_Y = Z_B = Z_{ph}$.

Phase voltages are equal and 120° apart.

Then, the **line currents** are also **equal in magnitude** and **120° apart**.

Power in a Balanced 3-Phase System

Total power (average):

$$P = Power\ per\ phase \times 3 = 3V_{ph}I_{ph}\cos\phi$$

Using line quantities:

For **Star**: $V_{ph} = V_L/\sqrt{3}$, $I_{ph} = I_L$

For **Delta**: $V_{ph} = V_L$, $I_{ph} = I_L/\sqrt{3}$

Hence in both cases,

$$\boxed{P = \sqrt{3}V_L I_L \cos\phi}$$

Reactive Power

$$Q = \sqrt{3}V_L I_L \sin\phi$$

Apparent Power

$$S = \sqrt{3}V_L I_L$$

Power Factor

$$\cos\phi = \frac{P}{S}$$

3 - Phase AC Circuits

Problem 1 — Balanced Star (Y) connected load

Given: Line voltage $V_L = 400\ \mathrm{V}$. Balanced Y-connected load with per-phase impedance $Z_{ph} = 10\angle 30^\circ\ \Omega$ (i.e. magnitude 10 Ω, phase angle $+30^\circ$ — inductive).

Find: Phase voltage V_{ph}, phase current I_{ph} (magnitude & angle), total active power P, reactive power Q, and apparent power S.

Solution

1. $V_{ph} = \dfrac{V_L}{\sqrt{3}} = \dfrac{400}{\sqrt{3}} = 230.9401\ \mathrm{V}.$
2. Phase current:

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.9401}{10\angle 30^\circ} = 23.0940\angle(-30^\circ)\ \mathrm{A}.$$

(Angle negative because voltage divided by $+30^\circ$ gives current lagging by 30°.)

3. Power per phase (active and reactive):

$$P_{ph} = V_{ph} I_{ph} \cos 30^\circ = 230.9401 \times 23.0940 \times 0.8660254 \approx 4618.802\ \mathrm{W}.$$

$$Q_{ph} = V_{ph} I_{ph} \sin 30^\circ = 230.9401 \times 23.0940 \times 0.5 \approx 2666.667\ \mathrm{VAR}.$$

3 - Phase AC Circuits

Problem 2 — Balanced Delta (Δ) connected load

Given: Line voltage $V_L = 415\ \mathrm{V}$. Balanced Δ-connected load with per-phase impedance $Z_{ph} = 20\angle 36.87^\circ\ \Omega$ (power factor $\cos 36.87^\circ = 0.8$, lagging).

Find: Phase current I_{ph}, line current I_L (magnitude), and total active power P.

Solution

1. For Δ: $V_{ph} = V_L = 415\ \mathrm{V}$.
2. Phase current (through each phase winding):

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{415}{20\angle 36.87^\circ} = 20.75\angle(-36.87^\circ)\ \mathrm{A}.$$

3. Line current magnitude (for Δ): $I_L = \sqrt{3}\, I_{ph}$ (magnitude):

$$I_L = \sqrt{3} \times 20.75 \approx 35.940\ \mathrm{A}.$$

4. Total active power:

$$P = 3V_{ph}I_{ph}\cos\phi = 3 \times 415 \times 20.75 \times 0.8 \approx \boxed{20\,666.97\ \mathrm{W}}.$$

Summary (rounded):

$$I_{ph} = 20.75\angle - 36.87^\circ\ \mathrm{A},\ I_L \approx 35.94\ \mathrm{A},\ P \approx 20.667\ \mathrm{kW}.$$

3 - Phase AC Circuits

Summary Formulas

Quantity	Formula
Phase voltage (Star)	$V_{ph} = V_L/\sqrt{3}$
Line current (Star)	$I_L = I_{ph}$
Phase voltage (Delta)	$V_{ph} = V_L$
Line current (Delta)	$I_L = \sqrt{3}I_{ph}$
Total Power	$P = \sqrt{3}V_L I_L \cos\phi$
Reactive Power	$Q = \sqrt{3}V_L I_L \sin\phi$
Apparent Power	$S = \sqrt{3}V_L I_L$
PF from 2-wattmeter readings	$\cos\phi = \dfrac{W_1 + W_2}{\sqrt{3}(W_1 - W_2)}$
Total power (2-wattmeter)	$P = W_1 + W_2$

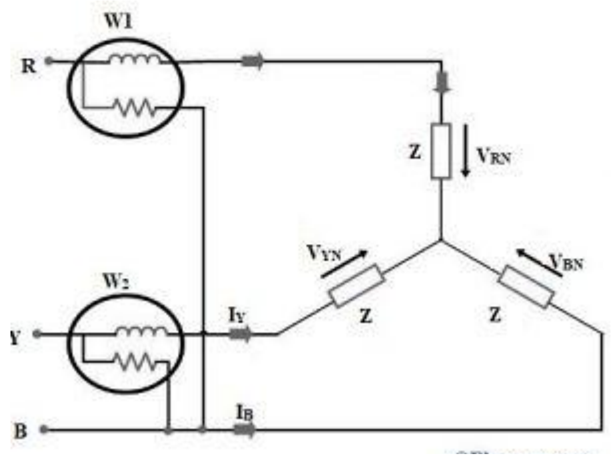
Measurement of Power in a 3-Phase System — Two Wattmeter Method

This is a **practical and standard method** for measuring total power in a **balanced or unbalanced 3-phase load**.

Expressions

For a **balanced load** with phase angle ϕ(between phase voltage and current):

$$W_1 = V_L I_L \cos(30^\circ - \phi)$$
$$W_2 = V_L I_L \cos(30^\circ + \phi)$$



Total Power:

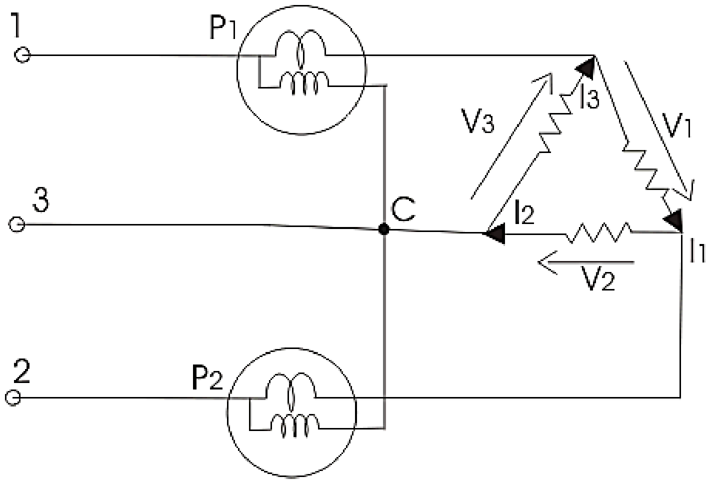
$$P_{total} = W_1 + W_2 = \sqrt{3} V_L I_L \cos\phi$$

Reactive Power:

$$Q = \sqrt{3}(W_1 - W_2) = \sqrt{3} V_L I_L \sin\phi$$

Power Factor from Wattmeter Readings

$$\cos\phi = \frac{W_1 + W_2}{\sqrt{3}(W_1 - W_2)}$$



3 - Phase AC Circuits

Given:

$V_L = 400$ V, $I_L = 10$ A, $\cos\phi = 0.8$ lagging.

Find:

$W_1, W_2, P_{total}, Q, S.$

Solution:

$$W_1 = V_L I_L \cos(30^\circ - \phi) = 400(10)\cos(30^\circ - 36.87^\circ) = 4000\cos(-6.87^\circ) \approx 3976.8 \text{ W}$$

$$W_2 = V_L I_L \cos(30^\circ + \phi) = 400(10)\cos(30^\circ + 36.87^\circ) = 4000\cos(66.87^\circ) \approx 1580.5 \text{ W}$$

$$P_{total} = W_1 + W_2 = 3976.8 + 1580.5 = 5557.3 \text{ W}$$

Check with formula:

$$P = \sqrt{3} V_L I_L \cos\phi = 1.732(400)(10)(0.8) = 5542.4 \text{ W}$$

1 - Phase Transformers

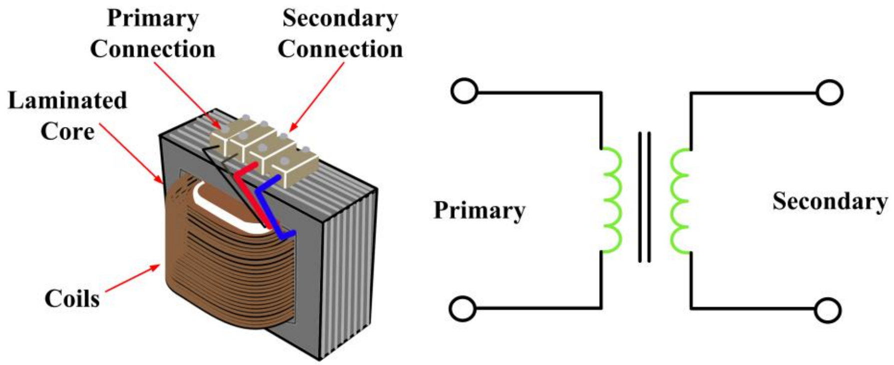
A **single-phase transformer** is a **static electromagnetic device** that transfers electrical energy between two circuits at **constant frequency**, based on **Faraday's law of electromagnetic induction**.

Function: Step-up or step-down AC voltage levels while keeping frequency same.

Construction:

A single-phase transformer has **two windings** wound on a **magnetic core**:

1. **Primary winding (N_1 turns):** connected to the input (supply side).
2. **Secondary winding (N_2 turns):** connected to the load (output side).
3. **Magnetic core:** provides a low-reluctance path for magnetic flux; made of **laminated silicon steel** to reduce eddy current losses.



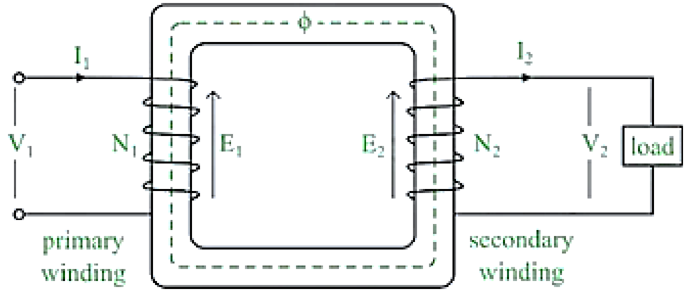
Principle of Operation:

The transformer works on **Faraday's law of electromagnetic induction** — **mutual induction** between two windings.

1. When AC voltage V_1 is applied to the primary, an alternating magnetic flux (ϕ) is produced in the core.
2. This time-varying flux links both primary and secondary windings.
3. According to Faraday's law, an EMF is induced in each winding:

$$e_1 = -N_1 \frac{d\phi}{dt}, e_2 = -N_2 \frac{d\phi}{dt}$$

4. If the secondary circuit is closed, this induced EMF drives current through the load.



1 - Phase Transformers

EMF Equation of Transformer

Let:

ϕ_m =maximum flux in the core (Wb)

f= frequency of AC supply (Hz)

N_1, N_2 =number of turns on primary and secondary windings

For a sinusoidal flux:

$$\phi = \phi_m \sin(\omega t)$$

Instantaneous EMF induced per turn:

$$e = -\frac{d\phi}{dt} = -\omega\phi_m \cos(\omega t)$$

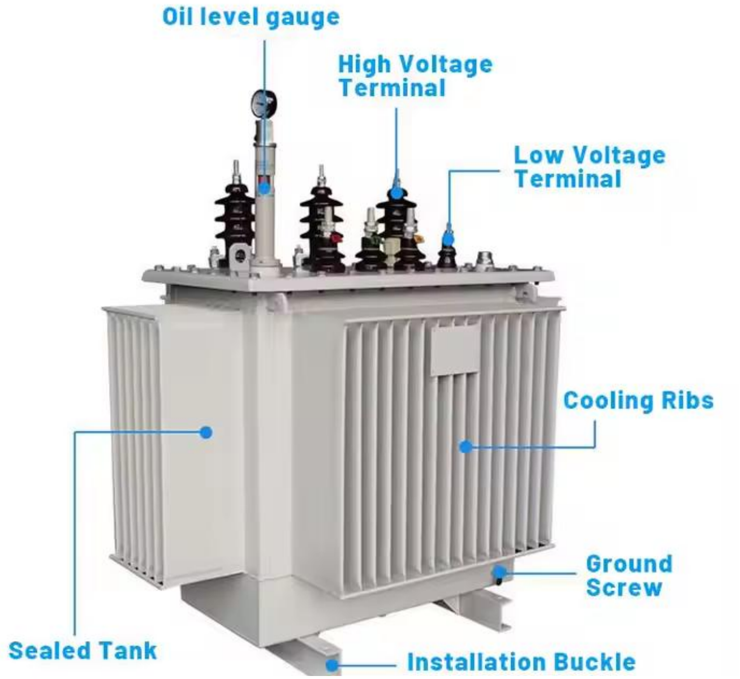
Maximum value of EMF per turn $= \omega\phi_m = 2\pi f \phi_m$.

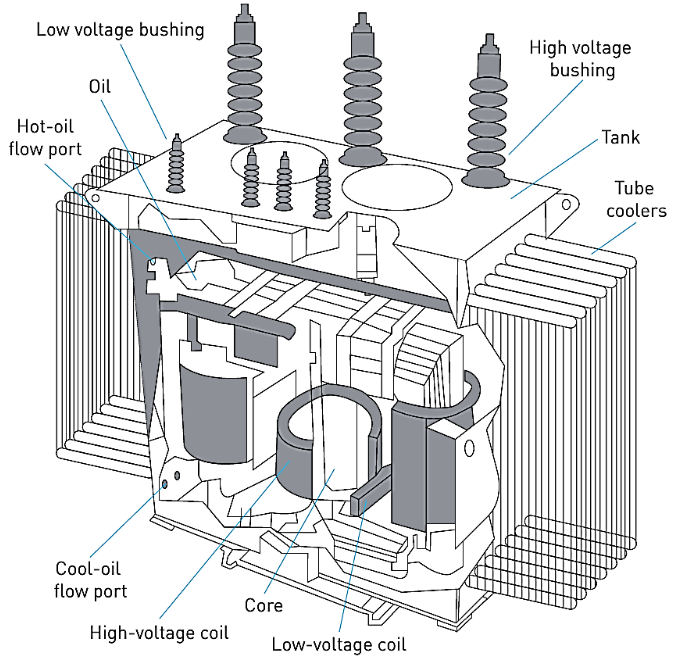
Hence RMS value per turn:

$$E_{turn} = 4.44 f \phi_m$$

Then,

$$\boxed{E_1 = 4.44 f N_1 \phi_m, \qquad E_2 = 4.44 f N_2 \phi_m}$$





1 - Phase Transformers

Voltage (Turn) Ratio:

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

where

K= **turns ratio** (or **transformation ratio**).

If $K > 1$ **Step-up transformer** (voltage increased).

If $K < 1$ **Step-down transformer** (voltage decreased).

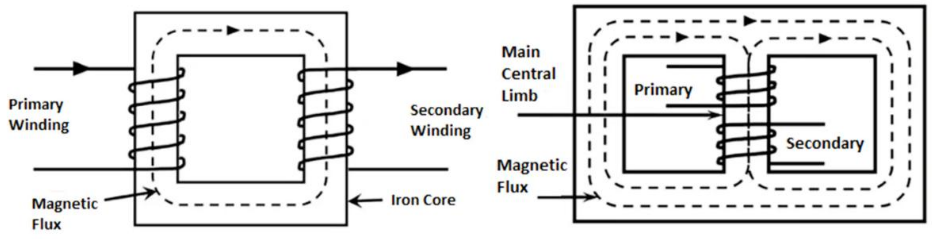
For an **ideal transformer** (no losses):

$$\boxed{\frac{E_1}{E_2} = \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}}$$

1 - Phase Transformers

Comparison: Core-Type vs Shell-Type Transformer

Point	Core-Type Transformer	Shell-Type Transformer
1. Winding Placement	Windings placed on **two separate limbs**.	Windings placed on **central limb** (sandwiched).
2. Leakage Flux	**More leakage flux** (windings farther apart).	**Less leakage flux** due to close coupling of windings.
3. Copper Requirement	Requires **more copper** (longer coil turns).	Requires **less copper** (shorter winding length).
4. Core Requirement	**Less core material**.	Requires **more core material**.
5. Efficiency	Slightly **lower efficiency**.	Usually **higher efficiency**.
6. Mechanical Strength	Lower mechanical strength; less fault withstand capability.	Higher mechanical strength; better short-circuit withstand capacity.
7. Cooling	Cooling is **better** (windings exposed).	Cooling is **less effective** (windings concentrated).
8. Applications	Preferred for **distribution transformers** (small/medium rating).	Preferred for **power transformers** (high rating).



Core Type Transformer   Shell Type Transformer

1 - Phase Transformers

Rating of a Transformer:

The **rating of a transformer** specifies the **maximum load** it can supply **without exceeding temperature limits** and **without causing excessive losses or insulation damage**.

A transformer is always rated in: **kVA (kilo-volt-ampere) not in kW** because:

$$\boxed{\text{kVA} = \frac{V \times I}{1000}}$$

1. Copper losses depend on current (I^2R).

2. Iron losses depend on voltage (V).

Both losses together determine the rating: independent of power factor (pf).

Hence, transformers are rated by **voltage** × **current**, not by power delivered to the load.

Current Based on Transformer Rating

For a single-phase transformer:

$$I = \frac{kVA \times 1000}{V}$$

For a three-phase transformer:

$$I = \frac{kVA \times 1000}{\sqrt{3}\, V_L}$$

Used to calculate primary and secondary full-load currents.

Types of Transformer Losses

Transformer losses are divided into **core (iron) losses**, **copper losses**, and **miscellaneous losses**.
A transformer is designed to minimize these losses to achieve high efficiency.

1. Core (Iron) Losses: Core losses occur in the **iron/steel core** when alternating magnetic flux passes through it.
They are **present even at no-load** and depend only on **voltage and frequency**, not on load current.
Core losses consist of:

(a) Hysteresis Loss : Occurs due to **reversal of magnetization** in the core as AC flux alternates.
Every cycle, the magnetic domains realign, causing **friction-like loss** inside the material.
Depends on:

- Volume of core
- Frequency
- Peak flux density
- Material type

How to reduce it: Use **silicon-steel core** or **CRGO (Cold Rolled Grain Oriented Steel)** which has low hysteresis loss.

(b) Eddy Current Loss: Induced EMF in the core causes **circulating currents** (eddy currents).
These currents cause **I^2R heating** in the metal core.
Eddy current loss depends on:

- Flux density
- Frequency
- Thickness of laminations
- Electrical resistivity of core

How to reduce it: Use **thin laminated cores**, each sheet insulated from the next → reduces path for eddy currents.

1 - Phase Transformers

Copper Losses (Winding Losses):

Losses due to the **resistance of primary and secondary windings** when current flows through them.

Formula:

$$P_{cu} = I_1^2 R_1 + I_2^2 R_2$$

Copper loss varies with **load current**.
At full load, copper loss is maximum.
At no-load, copper loss is almost zero.

Stray Losses: Small losses due to:
Leakage flux interacting with tank walls and structures
Induced eddy currents in mechanical supports
Non-ideal coupling
Usually included within copper and core losses or considered a small additional term.

1 - Phase Transformers

Ideal Transformer:

An **ideal transformer** is a theoretical transformer with the following assumptions:

Assumptions

1. No copper (I^2R) losses, winding resistance = 0
2. No core (iron) losses, hysteresis & eddy losses = 0
3. No leakage flux, all flux links both windings
4. Infinite permeability, negligible magnetizing current
5. Perfect coupling, mutual flux = total flux

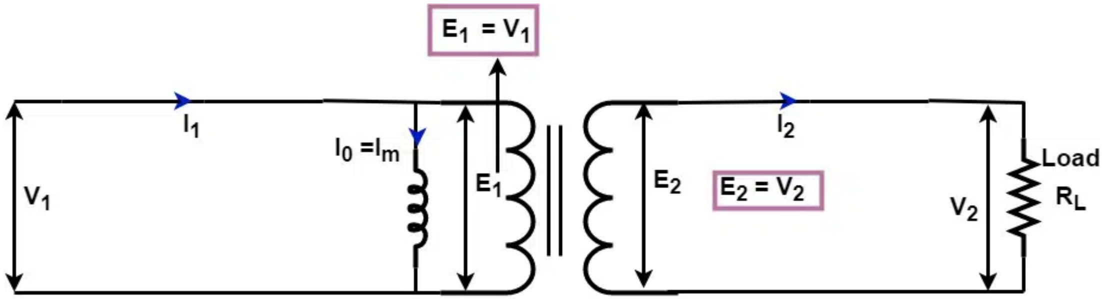
Practical Transformer:

A real transformer has:

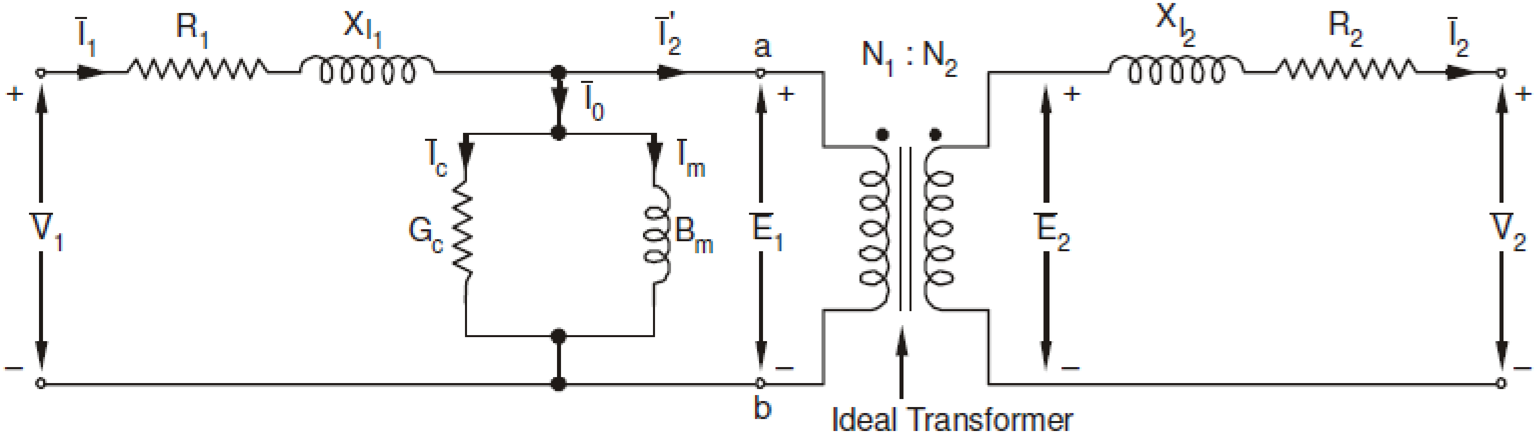
1. **Copper losses** in windings
2. **Core losses** (hysteresis + eddy currents)
3. **Leakage flux**, modeled as leakage reactance
4. **Magnetizing current**, required to establish flux

These imperfections are included in the practical equivalent circuit.

1 - Phase Transformers



Equivalent Circuit of an Ideal Transformer



Equivalent Circuit of Practical Transformer

R1, R2 : Resistance of Primary and Secondary windings
X1, X2 :Leakage reactance of primary and secondary windings
Io :primary input current due to iron losses and copper losses
Im : Magnetising or reactive component of current
Ic : Active component of current

1 - Phase Transformers: Efficiency of a Transformer

Transformer efficiency (η):

$$\eta = \frac{Output\ Power}{Input\ Power}$$

For a load with power factor $\cos\phi$:

$$\boxed{\eta = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + P_i + P_{cu}}}$$

Where:

P_i =**Iron loss** = constant

P_{cu} =**Copper loss** = $I^2 R$,varies with load

Condition for Maximum Efficiency

Transformer efficiency is maximum when:

$$\boxed{P_{cu} = P_i}$$

At this point:

$$\eta_{max} = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + 2P_i}$$

All-Day (Energy) Efficiency

Used for **distribution transformers** where load varies during the day.

$$\eta_{all-day} = \frac{kWh\ Output}{kWh\ Input\ (24\ hrs)}$$

DC MOTORS

A DC Motor converts electrical energy to mechanical energy based on the principle of electromagnetic torque. When a current-carrying conductor is placed in a magnetic field, it experiences a force (Lorentz force), producing rotation.

Construction of a DC Motor: A DC motor has the same basic construction as a DC generator.

1.Stator (Field System)

1. Provides **stationary magnetic field**.
2. Poles with field windings (electromagnets) or permanent magnets.

2.Rotor (Armature)

1. Cylindrical core with slots carrying **armature conductors**.
2. Conductors carry current and experience **force** to produces torque.

3.Commutator

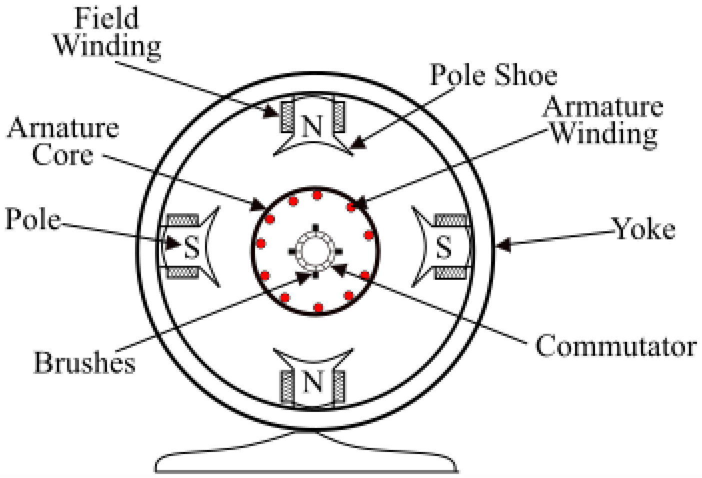
1. Mechanical rectifier.
2. Ensures **unidirectional torque** even though rotor conductors move in alternating magnetic fields.

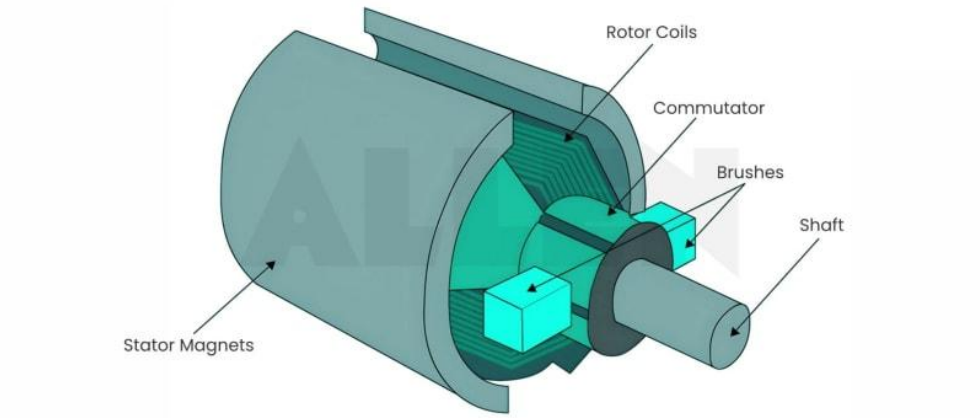
4.Brushes

1. Made of carbon/graphite.
2. Provide electrical contact between rotating commutator and external supply.

5.Frame/Yoke

1. Outer body; supports poles and provides mechanical strength.





DC Motors

Electromagnetic Force on a Conductor:

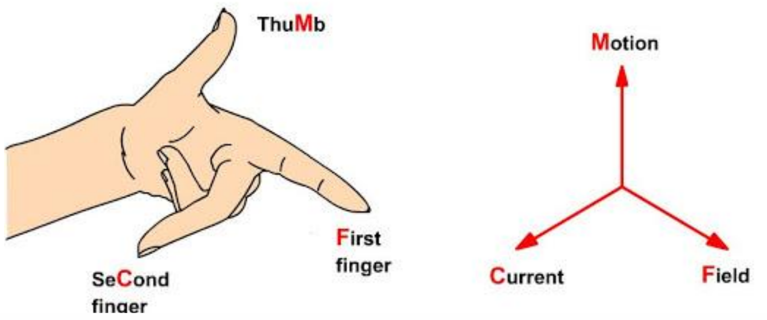
When current I flows through a conductor placed in magnetic field B:

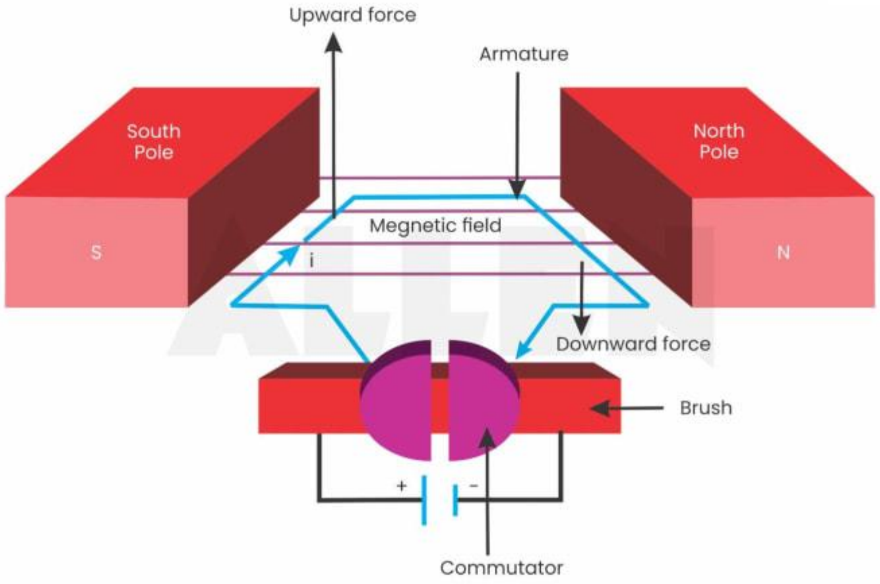
$$F = BIL\sin\theta$$

B =flux density
I= current through conductor
L= length of conductor
θ= angle between magnetic field and conductor
In a DC motor, conductors are arranged such that $\theta = 90°$.
So force becomes:

$$\boxed{F = BIL}$$

This force acts **perpendicular** to both the magnetic field and the current direction (Fleming's Left Hand Rule).





In a DC motor, when the armature rotates under the influence of the driving torque, the armature conductors move through the magnetic field, and therefore an EMF is induced in them by the generator action. This induced EMF in the armature conductors acts in opposite direction to the applied voltage Vs and is known as the **back EMF or counter EMF.**

Role of Commutator: Ensures Unidirectional Torque

In a DC motor:

Armature rotates inside a stationary magnetic field
Direction of conductor motion changes continuously (left side to right side)
But torque must always act **in the same direction** to keep rotation smooth

This is achieved by the commutator:

Reverses the **direction of current** in a conductor every half-turn
Ensures that the **force direction** on each conductor remains the *same*
Therefore, torque remains unchanged and smooth continuous rotation
Without the commutator, torque would reverse every 180°, and the motor would **vibrate instead of rotating**.

DC Motors

Types of DC Motors: Based on how field winding is connected:

1. DC Shunt Motor

Field winding connected **in parallel** (shunt) with armature.
Thick wire, many turns
Constant field flux

2. DC Series Motor

Field winding connected **in series** with armature.
Thick wire, few turns
Field depends on load current

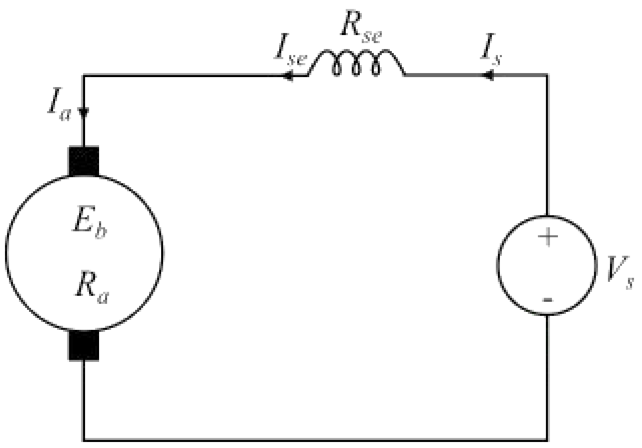
3. DC Compound Motor

Combination of shunt + series field windings
Cumulative compound (series aiding shunt)
Differential compound (series opposing shunt)

DC Motors

Series DC Motor:

A DC motor whose field winding is connected in series with the armature winding so that whole armature current passes through the field winding is called a **series DC motor**



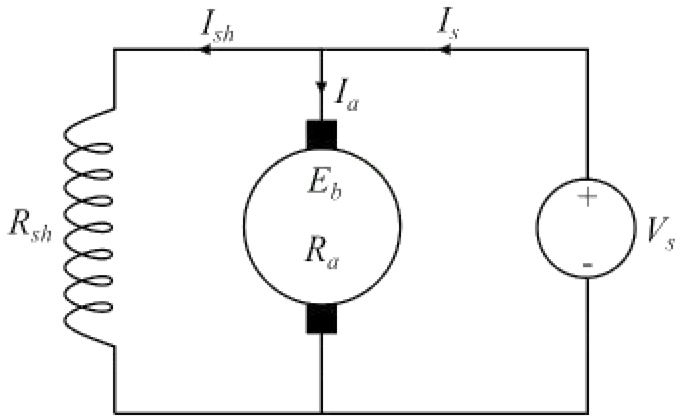
Series DC Motor

It is made up of thick wire with less number of turns so that it possesses minimum resistance.
The following are some important expressions for the series DC motor:
Armature current, Ia = Ise = Is
Where,
Ise is the series field current and
Is is the supply current.

DC Motors

Shunt DC Motor:

A DC motor whose field winding is connected in parallel with the armature winding so that total supply voltage is applied across it, is known as a shunt DC motor



Shunt DC Motor

In a shunt DC motor, the shunt field winding has a large number of turns of thin wire so that it has high resistance, and therefore only a part of supply current flows through it and the rest flows through the armature winding.

Following are the important expressions of a shunt DC motor:

Armature current, $I_a = I_s - I_{sh}$

Shunt field current, $I_{sh} = V_s / R_{sh}$

Supply Voltage, $V_s = E_b + I_a R_a$

Compound DC Motor:

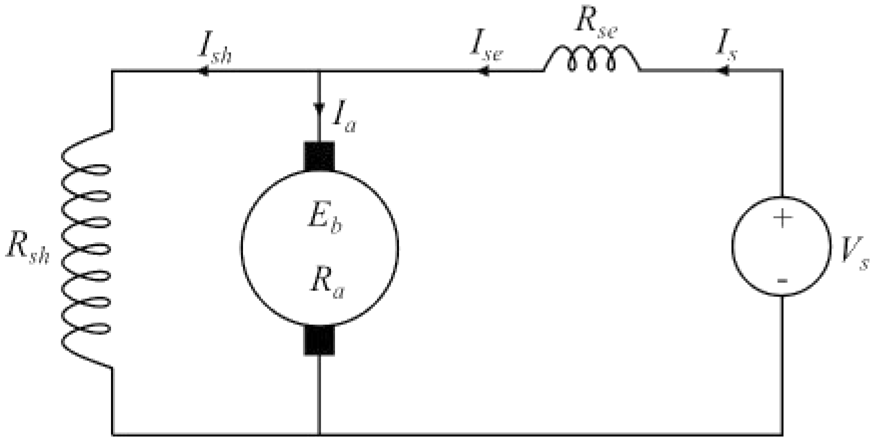
A compound DC motor is one which has two sets of field windings on each magnetic pole one is in series and the other is in parallel with the armature winding.

Compound DC motors are sub-divided into the following two types:

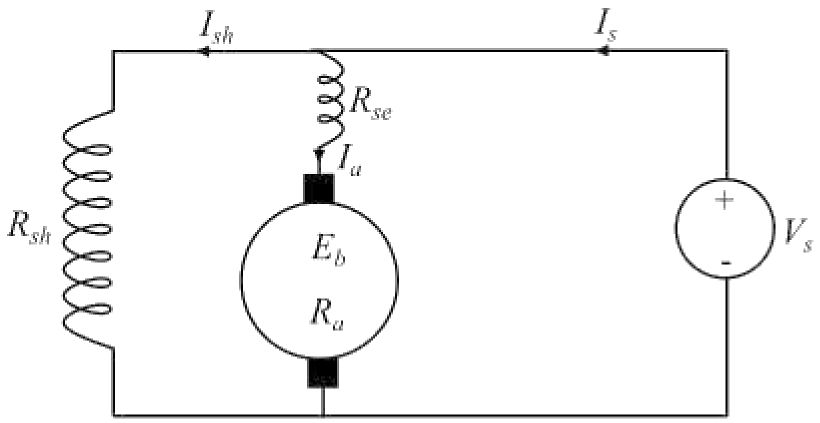
Short-shunt compound DC motor

Long-shunt compound DC motor

A short-shunt compound DC motor is one in which only shunt field winding is in parallel with the armature winding



Short Shunt DC Motor



Long Shunt DC Motor

A long-shunt compound DC motor is one in which shunt field winding is in parallel with both series field winding and armature winding as shown

Torque–Speed Characteristics:

1. DC Shunt Motor

Speed is **constant** (little change with load)
Moderate starting torque
Speed–torque curve: almost **flat**

2. DC Series Motor

Speed is **very high at no-load** (NEVER run without load)
Excellent **high starting torque**
Speed decreases sharply with load

3. DC Compound Motor

Characteristics between shunt and series
Cumulative compound:
- High starting torque
- Better speed regulation than series

Speed and Torque Characteristics

As the torque in a DC motor increases, flux (ϕ) decreases. This is because when torque is increased, armature current increases resulting in reduction of air gap flux ϕ i.e., due to armature reaction and saturation. Therefore, for increased torque causing a drop in the speed of the motor.

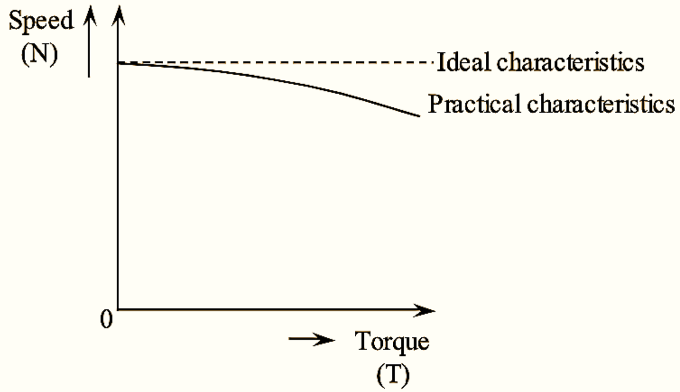


Figure : Speed-torque Characteristics of D.C Shunt Motor

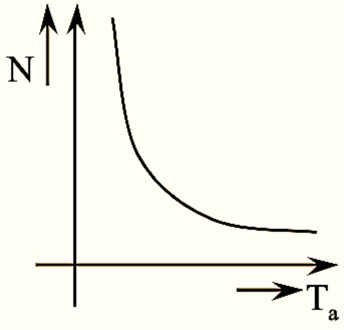


Figure : Speed-torque Characteristics of Series Motor

Series motor has high torque at low speed and vice-versa. Thus, the series DC motor is used where high starting torque is required.

Important: At no-load, the armature current is very small and so is the flux. Hence, the speed increases to a dangerously high value which can damage the machine. Therefore, a series motor should never be started on no-load.

DC Motor

Applications of DC Motors: DC Shunt Motor

(*Constant speed applications*)

Lathes

Fans, blowers

Centrifugal pumps

Machine tools

Conveyors

DC Series Motor:

(*High starting torque applications*)

Traction (electric trains)

Hoists and cranes

Elevators

Electric vehicles

Air compressors

DC Compound Motor:

(*High starting torque + good speed regulation*)

Rolling mills

Presses

Shears

Heavy planers

Industrial lifts

DC Motors

Selection Criteria for DC Motors:

While selecting a motor for an application, consider:

1. Required Starting Torque

High starting torque choose **series** or **cumulative compound**

Constant-torque loads choose **shunt**

2. Speed Regulation

Good speed regulation **shunt motor**

Variable speed + high torque **series or compound**

3. Load Characteristics

Constant load **shunt motor**

Heavy intermittent load **series or compound**

4. Safety

Avoid series motor for no-load applications (unsafe overspeed).

5. Control Requirements

Shunt motors: easy speed control

Series motors: speed control less stable

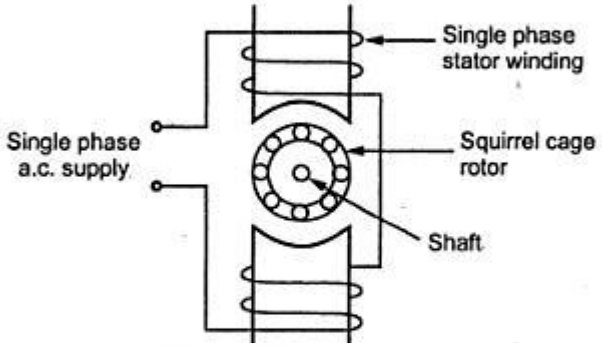
6. Cost & Maintenance

Shunt motors: simpler

Compound motors: more expensive but robust

Single-Phase Induction Motor

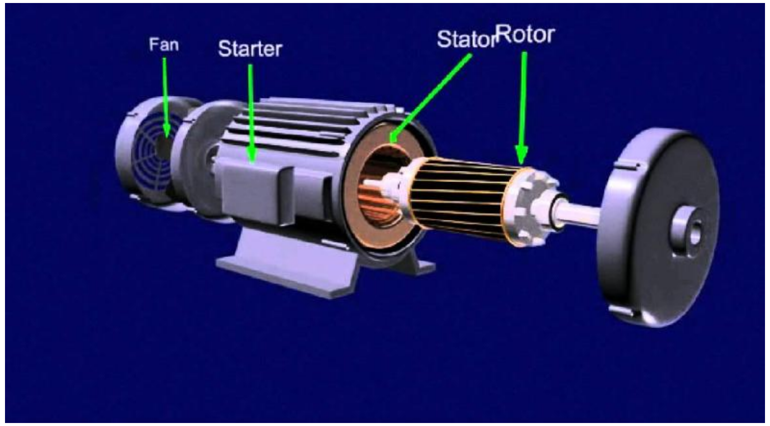
A single-phase induction motor is an alternating-current motor designed to operate on a **single-phase supply**. It converts electrical energy into mechanical energy by the process of **electromagnetic induction**.
Due to its simple construction, ruggedness, and ease of operation, it is widely used in domestic and commercial appliances of fractional horsepower ratings.



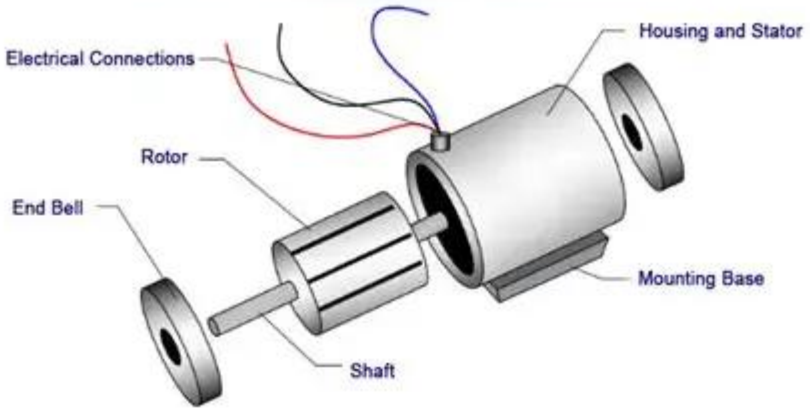
Applications of Single-Phase Induction Motor:

These motors are used where:
Power rating is small (fractional horsepower), Cost is low,
Simplicity and reliability are required

Fans (ceiling fans, table fans)
Refrigerators
Air conditioners
Washing machines
Mixers, grinders, juicers
Blowers & exhaust fans
Pump sets
Drilling machines
Small machine tools
Vacuum cleaners
Shaded-pole motors: clocks, toys, record players



Parts of Single Phase Induction Motor



A single-phase induction motor has **stator**, **rotor**, **main winding**, **starting winding**, and **capacitor/switch**.

Working Principle:

When a single-phase AC voltage is applied to the stator main winding, an alternating current flows through it, producing a **pulsating magnetic field**.
This field alternates in magnitude but **does not rotate**; hence, it **cannot produce a starting torque**.
The pulsating flux induces an EMF in the rotor conductors, causing rotor currents to flow. However, since the flux is not rotating, the induced rotor currents produce **equal and opposite torques** at standstill. The net torque is therefore zero.
A single-phase induction motor is not self-starting.
Once the rotor is given an initial rotation (by auxiliary winding or manual push), the motor continues to run due to the induction principle

Auxiliary starting methods (split-phase, capacitor-start, capacitor-run, shaded-pole) are used to make the motor self-starting.

Double-Field Revolving Theory

To explain the behavior of the single-phase induction motor, the **double-field revolving theory** is most commonly used.

Decomposition of Pulsating Field:

A pulsating magnetic field produced by the single-phase stator winding can be resolved into **two equal rotating magnetic fields**:

One rotating **forward**

One rotating **backward**

Both rotating at **synchronous speed (Ns = 120f/P)**

Each field having magnitude **ϕm/2**

Operation at Standstill:

The forward rotating field produces a torque T_f

The backward rotating field produces a torque T_b

Because the motor is stationary, these torques are equal:

$$T_f = T_b$$

Hence the **net starting torque is zero**, and the motor fails to start on its own.

Operation Once Rotating:

If the rotor is given an initial rotation in the forward direction (through auxiliary winding starting methods):

Slip with respect to the forward field becomes **small**, producing **large forward torque**

Slip with respect to the backward field becomes **large**, producing **small backward torque**

Thus:

$$T_{net} = T_f - T_b > 0$$

The motor accelerates and continues to run as a normal induction motor with the forward rotating field dominating.